

CLOUD-BASED INTEGRATED MANAGEMENT SYSTEM FOR ABR DIAGNOSTIC CENTER

A CAPSTONE PROJECT

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for the Degree of Bachelor of Science in Information Technology

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It does not contain any material previously published or written by another person, nor any material to a substantial extent entitled “**Cloud-Based Integrated Management System for ABR Diagnostic Center**”, except where due acknowledgement is made in the paper. Any contribution made to the research by others is explicitly acknowledged.

I further certify that the signatures appearing herein are genuine and that the work and ideas presented in this study are the original work of the researchers, developed under the supervision of their adviser, **Mr. Eduardo S. Rodrigo**.

Signed this 1st day of April in the year of our Lord, 2026.

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ABSTRACT

Title: Cloud-based Integrated Management System for ABR Diagnostics Center

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The Cloud-Based Integrated Management System for ABR Diagnostic Center was developed to modernize the clinic's operations by replacing outdated paper-based processes with a single, efficient online platform. Built using the Agile development approach, the system solves key problems such as manual appointment scheduling, easily lost or damaged paper records, long patient waiting times, billing errors, and limited communication options. Patients can now conveniently browse available services and book appointments online from anywhere, without needing to visit the clinic in person.

Clinic staff benefit from an interactive real-time dashboard that displays patient flow, laboratory test progress, daily revenue, and upcoming appointments. A smart queueing module organizes patients according to priority, provides live status updates, and greatly reduces waiting times. Dedicated laboratory and imaging modules allow staff to manage test results, generate digital reports, and securely store all records in the cloud.

Overall, this system streamlines daily workflows, enhances data security, improves the patient experience through faster service and easier access, and gives management better insights into operations proving that a well-designed digital solution can significantly raise the quality and efficiency of healthcare services at ABR Diagnostic Center

DEDICATION

This study is dedicated to the following persons:

To our almighty God who gives us wisdom, knowledge, and understanding.

To our parents who give us financial and moral support.

To our friends who are always giving us moral support.

To the staff of ABR Diagnostic Center who gives us their trusts and lends us their information that we need to accomplish the system.

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

1.1 Project Context

A web system can be an application or system that run through the internet without being installed on the user's computer it became famous specially during the pandemic because people are not allowed to go outside therefore the web-based system helps them to do more such as; online shopping, browsing their favorite social media sites or even making an order to make a food deliveries. On the business side, it can help the company to continue operating remotely. The web-based system operation kept running and became a long-term trend even after the pandemic.

The proponent's client, ABR Diagnostic Center, envisioned a healthier community that has easy access to essential diagnostic services, contributing to preventive health care and early detection of disease. To make the client's vision possible, their company mission is to provide an accessible and affordable laboratory test to promote better health and well-being in the community. To help the client on their mission, the proponents want to help them by creating a Cloud-Based Integrated Management System for ABR Diagnostic Center.

1.2 Background of the Study

ABR Diagnostic Center was established in 2001 by Arnel Romeo and Annaliza Romeo and it is located at Pakil, Laguna. The clinic operates daily from 6:00 AM to 8:00 PM and is staffed by medical professionals including a pathologist, medical technologists, and

administrative personnel. The center offers multiple diagnostic services and accommodates an estimated 70 to 80 patients per day.

In its current setup, the clinic relies heavily on manual processes. Patients may either schedule appointments or walk in, and most transactions are handled using paper-based records and logbooks. During the process, patient details, laboratory results, and billing information are manually written or encoded using templates. The release of results also requires physical presence, and duplicate requests depend on retrieving information from logbooks.

1.3 Statement of the Problem

These manual procedures present several challenges. Records are vulnerable to damage or loss, retrieving patient data can be time-consuming, and report generation is prone to inaccuracies. In addition, manual billing computations may lead to errors that could affect the clinic's financial records. With the increasing number of patients and demand for efficient services, these limitations highlight the need for a more reliable and automated system.

1.4 Objectives of the Study

General Objective

The study aims to develop a Cloud-based Integrated Management System for ABR Diagnostic Center. The general objective is to enhance the clinic's transaction management

through a centralized digital platform that makes appointments, queuing, patient record management, billing, and report generation transactions more efficient.

Specific Objectives

- To create a cloud-based management system that will guarantee the availability of clinic records
- To develop a billing system that computes payments automatically, minimizing errors, and ensuring proper financial transactions
- To develop a queuing system that follows the patient priority lane to help manage transactions more efficiently.
- To provide patients with online access to their previous records and enable them to schedule appointments conveniently through the system
- To implement automated email notifications for appointment confirmations, reminders, and test result availability
- To generate automated reports for laboratory transactions and patient records to enhance accuracy, efficiency, and decision-making in laboratory operations.

1.5 Scope and Delimitation

Scope

- Admission Staff: This user handles front office transactions such as appointments, creating transactions, queuing, patient information, and releasing results. The

admission staff verifies patient identities, creates patient records, manage appointments, encodes transactions, and updates patient status across different stations.

- Appointments - Full access (create, view, edit, delete): Can manage all appointment requests, approve or decline appointments, verify appointments when patients arrive, and send appointment reminders
- Patient Information - Full access (create, view, edit, delete): Can search, register, edit, view, and delete patient records, as well as create patient website accounts with login credentials.
- Transactions - Full access (create, view, edit, delete): Can gather transaction services, create new transactions, edit transaction details, update queue checklists, and search transactions for verification purposes.
- Invoices - Full access (create, view, edit, delete): Can access all billing information, search billing records for verification, and view invoice details.
- Discounts - View only: Can view available discount options when processing transactions but cannot modify discount settings.
- Employees - View only: Can view employee information, particularly when adding signatories to medical reports
- Users - View, create, and update access: Can view user accounts, create new user accounts for patients and staff, and update existing user information (cannot delete users).

- Patient: This user only sees their own medical and transaction history. The patient can create appointments, browse services on the system, receive email notifications, view test results, and access their profile information.
 - Appointments - Personal access only: Can browse available services, create appointment requests, view their own appointment history and status, and receive appointment confirmations and reminders via email.
 - Transactions - Personal view only: Can view their own transaction history, see the status of their laboratory tests, access and download their medical reports, and track their billing and payment information.
- Cashier: This user is responsible for managing all financial transactions and billing processes within the system. The cashier computes service fees, processes payments, generates invoices, and updates billing information.
 - Appointments - View only: Can view appointment details to verify scheduled services and prepare billing information.
 - Patient Information - Create, view, update, and restore access: Can search patient records, view patient details for billing purposes, create new patient records if needed, update patient information, and restore deleted patient records (cannot delete).
 - Queues - Full access (create, view, edit, delete): Can monitor patient queues to track patient flow and coordinate billing processes accordingly

- Transactions - View only: Can search and view transaction details to verify services rendered for billing purposes but cannot modify transaction information
- Invoices - Full access (create, view, edit, delete): Can calculate service fees, add discounts, encode payment details, generate and print invoices, and manage all billing records
- Users - View only: Can view user account information but cannot create, modify, or delete user accounts.
- MedTech: The primary role of this user is to encode conducted tests findings from patients. MedTech can access patient test checklists, conduct medical tests, add signatory information, and save test results.
 - Appointments - Full access (create, view, edit, delete): Can view appointment schedules and patient appointment details to prepare for testing procedures
 - Patient Information - Full access (create, view, edit, delete): Can access complete patient records to review medical history and patient details relevant to laboratory testing
 - Queues - Full access (create, view, edit, delete): Can view queue lists filtered by their station, call patients from the queue, and update queue status during the testing process.
 - Transactions - Full access (create, view, edit, delete): Can search and view transaction details to identify which tests need to be conducted for each patient.

- Invoices - Full access (create, view, edit, delete): Can access billing information to verify which services have been paid for and need to be performed.
- Discounts - View only: Can view discount information applied to patient transactions.
- Employees - View only: Can view the list of medical doctors and professionals to add as signatories on medical reports.
- Users - View, create, and update access: Can view user information, create accounts if need, and update user details (cannot delete users).
- Chief MedTech: Handles the diagnostic center's changes on details. The primary role of this role is to conduct maintenance such as, employee details, service details, role details, permission details, log activity, etc.
- Employees - Can view and update employee details to add as signatories on medical reports.
- Service - Can view and update the diagnostic center's services (add or remove new services).
- Roles - Can view and update role details of users according to the diagnostic center's new changes.
- Permission - Can change permissions of each user on the system.
- Users - Can update the system's users (not limited to employees as per request) details.

- Payment Method -The chief MedTech can view and update the payment methods (add or remove new payment methods).
- Billing Setting - Can update the billing setting's details and information.
- Discount Setting - Can update the discount details according to the diagnostic center's changes.
- Log Activity - Can view the log activities of the users of the system.

Delimitation

The system is limited to in-person payments and cannot process digital transactions (e.g., credit cards, e-wallets) due to the system not integrating with any payment service provider or API, and the billing will only be transacted on the clinic premises. Patient identity verification is conducted manually by clinic staff during physical visits without automated tools like biometric scanners or AI-based authentication. The services listed in the study are updated only up to the company's services as of March 2025. Any additional services after this date will not be included in the study.

1.6 Significance of the Study

This part of the study explained the importance of the development of the system to the following individuals:

ABR Diagnostic Center

The main beneficiary of this study is the clinic ABR Diagnostic Center. This study will help the clinic manage its transactions more efficiently by automating appointments,

queuing, billing, and reports. This improves workflow, reduces manual errors, and supports better service delivery.

Clinic's Staff

The system will streamline their workflow by automating tasks like patient admission, billing, and test result management, reducing manual effort and errors. It also allows faster and more organized access to patient records and lab results, improving the quality and efficiency of healthcare services.

Patients

Patients can conveniently book appointments online, view their medical records, and receive updates without needing to go to the clinic physically. This saves time and improves their overall experience at the clinic.

Proponents

The creation of this system enables the proponents to put into practice their technical expertise and skills in actual problem-solving. It also provides them with exposure to working with live clients and project development.

Future Proponents

The proposed study can be used as a reference or guide for the future proponents that have similar systems. It may serve as a learning tool for system design, research documentation, and project management.

1.7 Definition of Terms

Upon implementation, the following terms are clearly defined both conceptually and operationally to facilitate better understanding:

ABR. An acronym that stands for Arnel B. Romeo, referring to the clinic's owner. They used that acronym as the name for their clinic.

Appointment System. A system feature that lets patients schedule their visit to the clinic in advance.

Audit Trail. A system record that tracks all actions taken by users, mostly when they create, update, or delete records. It is used for monitoring and security.

Billing System. A component of the system responsible for calculating service fees, generating invoices, and managing patient payments.

Invoice. A document that shows the amount a patient needs to pay for the services received.

MedTech. Short for medical technologist, refers to a staff member or licensed professional responsible for conducting medical laboratory tests and procedures.

Patient. an individual who avails services from the clinic and whose personal and medical information is recorded and managed in the system.

Patient Information. Refers to the personal, medical, and contact details collected and stored in the system for identification and healthcare service purposes.

Queueing System. A system that manages the order in which patients are served based on their appointments and patient priority type.

Roles and Permissions. Refers to access controls in the system that define what specific actions a user can perform based on their assigned position or responsibility.

Services. A type refers to medical procedures or laboratory tests offered by the clinic that patients can book or avail themselves of through the system.

Test Results. Refers to a medical report that the patient may get after undergoing various medical tests.

CHAPTER II**REVIEW OF RELATED LITERATURE AND STUDIES****2.1 Introduction**

This chapter examines existing research and reviews related literature on the ABR Integrated Management System. It focuses on four main areas: online appointment system, patient queue management system, patient information management system, and billing system. These areas help build the foundation for creating a digital solution that meets the needs of healthcare centers like ABR Diagnostic Center.

2.2 Review of Related Literature**Local Literature**

Book Title: Online Scheduling System for Doctors and Patients in a Hospital]

Authors: Marion Rylan Q. De Guzman, Jonn Louis N. Ordoñez, Ronald O. Somocierra, and Gloren S. Fuentes.

Year of Publication: 2021

Source: The outpatient department (OPD) of a hospital delivers diagnostic and consultative services to non-emergency patients. This study developed an online appointment scheduling system for the OPD to efficiently manage patient consultations. The proposed system allows patients to view physician availability and book time slots in advance. It reduces patient wait times by minimizing unnecessary hospital visits and

provides clinicians with real-time access to their daily schedules, thereby enhancing operational efficiency and service accessibility.

Book Title: PHIrecord: A Medical Record Management System for Rural Health Facilities in the Philippines

Authors: Allen G. Diaz, Adrienne Denise Gumtang, Cristine Joy A. Orpiada, Angelika S. Balagot, Evelyn A. Villanueva and Marycris A. Manalang

Year of Publication: 2024

Source: Traditionally, physicians relied on paper-based methods for managing patient medical histories and treatments. This study proposed a centralized medical record management system to address issues such as errors, lost records, and retrieval difficulties. The system provides quick access to patient records for physicians and nurses, enables seamless sharing of medical records across healthcare facilities, and includes referral features. Patient confidentiality is maintained through permission-based access, while patients can view their own records and request official copies.

Book Title: Web-based billing and support system for simplified water management

Authors: Chinley Mencias, Cathlyn L. Ylagan and Sheryl May Lainez

Year of Publication: 2024

Source: This study developed a web-based billing system that automatically computes usage and generates billing statements. The system incorporates SMS notifications for bills, due dates, and disconnection notices. Evaluated using ISO 25010 standards, the

system demonstrated high levels of functional suitability, performance efficiency, usability, compatibility, and security. It significantly improves accuracy, efficiency, and convenience for both management and customers.

Foreign Literature

Book Title: Applications of Artificial Intelligence in Business, Education and Healthcare

Authors: Khulood Khalid AlHajeri, Mukhtar AL-Hashimi, Sherine Badawi & Allam Hamdan

Year of Publication: 2021

Source: The online patient appointment system significantly improves the quality of healthcare services. Patient acceptance of the system does not differ significantly by gender, age, nationality, education level, or disease type. The study recommends upgrading the Management Information System (MIS), including infrastructure, software, personnel, and data management, to enhance service quality and healthcare performance.

Book Title: Effect of Queue Management System on Patient Satisfaction in Emergency Department

Authors: Ali Bidari, Shabahang Jafarnejad, Nazanin Alaei Faradonbeh

Year of Publication: 2021

Source: This study investigated the impact of a queue management system on patient satisfaction in emergency department waiting areas using a prospective randomized single-blinded design with 236 patients. Patients exposed to the queue management system

reported higher satisfaction levels and more accurate perceptions of waiting time compared to those under standard waiting procedures.

Book Title: Interoperability of heterogeneous health information systems

Authors: Amir Torab-Miandoab, Taha Samad-Soltani, Ahmadreza Jodati & Peyman Rezaei-Hachesu

Year of Publication: 2023

Source: Incompatibility among health information systems wastes resources and lowers care quality. This review of 36 studies highlighted the growing importance of interoperability since 2003, particularly in national electronic health record initiatives. Key standards include HL7 FHIR, CDA, HIPAA, and SNOMED-CT, while technologies such as SOA, APIs, XML, and SQL are essential for implementation. Semantic interoperability was emphasized as critical for consistent data exchange.

Book Title: Security and Privacy of Patient Information in Medical Systems Based on Blockchain Technology

Authors: Hongjiao Wu, Ashutosh Dhar Dwivedi and Gautam Srivastava

Year of Publication: 2021

Source: This study proposed a blockchain-based Patient-oriented Privacy Preserving Access Control model to secure private medical information. The system uses cryptographic algorithms and smart contracts to prevent data theft. Simulation results

showed strong performance with a maximum information throughput of 550 kbit/s and storage response time below 1,000ms.

Book Title: Protection Degree and Migration in the Stochastic SIRS Model: A Queueing System Perspective

Authors: Yuhan Li, Ziyang Zeng, Minyu Feng and Jürgen Kurths

Year of Publication: 2021

Source: This paper presented individual and population models to analyze epidemic propagation using a queueing system perspective, incorporating individual protection degree and human migration. The models highlight how individual behaviors and population-level dynamics influence system outcomes.

2.3 Review of Related Studies

Local Studies

Book Title: A Quantitative Evaluation of Online Appointment System at Mindanao State University–Main Campus: Employing the System Usability Scale (SUS) and Technology Acceptance Model (TAM)

Authors: Edison M. Esberto, Rose Ann C. Malaborbor and Alliana M. Ablan

Year of Publication: 2023

Source: This study developed and evaluated a web-based appointment system using SUS and TAM. The results showed a high overall client satisfaction rate of 90.2%,

demonstrating the system’s effectiveness in reducing wait times and improving appointment scheduling efficiency during the pandemic.

Book Title: Efficiency and Effectiveness of Utility Billing Management System (UBMS) in the Collection of Electric Services

Authors: Roselyn Resurreccion and Ma. Teresa Alvarez

Year of Publication: 2025

Source: This study assessed the Utility Billing Management System (UBMS) in a rural electric cooperative. The system showed strengths in usability, portability, efficiency, and reliability. However, challenges such as inaccurate billing statements and inadequate user training were identified as major implementation issues.

Foreign Studies

Book Title: Evaluation of Patient Satisfaction with the New Web-Based Medical Appointment Systems “Mawid” at Primary Health Care Level in Southwest Saudi Arabia: A Cross-Sectional Study (New)

Authors: MS Mahfouz et. al.

Year of Publication: 2023

Source: This cross-sectional study assessed patients’ satisfaction and perception toward the web-based medical appointment system “Mawid” implemented in primary health care centers in Jazan, Saudi Arabia. The findings revealed generally positive patient perceptions

and satisfaction with the online appointment system, highlighting its role in improving accessibility and reducing waiting times in primary care settings.

Book Title: The Association Between Wait Times and Patient Satisfaction in Primary Health Care Centers (New)

Authors: KF Alrasheedi et al.

Year of Publication: 2019

Source: This study examined the relationship between waiting times and patient satisfaction in selected primary health care centers in the Al Qassim region of Saudi Arabia. It found a significant association between longer wait times and lower patient satisfaction, underscoring the importance of effective queue management and appointment systems in enhancing the overall patient experience in healthcare facilities.

2.4 Synthesis

According to the literature review, digital technologies can improve critical healthcare operations such as appointment scheduling, queue management, patient information handling, and invoicing. These studies regularly demonstrate that automation may reduce wait times, improve service efficiency, boost data security, and increase patient happiness. However, the majority of existing systems address only one or two functions at a time, limiting their total operational effectiveness.

For ABR Diagnostic Center, these findings support the development of an integrated management system that combines online appointment scheduling, queue management,

patient information management, and billing automation in one platform. Such a system can help address the center's current manual processes, improve workflow efficiency, reduce administrative burden, and provide better patient service. By adopting a unified digital solution, ABR Diagnostic Center can modernize its operations and respond more effectively to the needs of both staff and patients.

2.5 Summary Table of Related Literature and Studies

Table 1. Comparison of Existing Applications for Healthcare Management and the Proposed ABR Diagnostic Center System

Authors	Year	Title	Country	Key Focus	Key Findings
De Guzman et al	2021	Online Scheduling System for Doctors and Patients	Philippines	Appointment System	Reduced waiting time and improved scheduling efficiency
Diaz et al.	2024	PHIrecord: Medical Record	Philippines	Patient Information System	Improved record accessibility, security,

		Manageme nt System			and data sharing
Mencias et al.	2024	eBillH20: Web-based Billing System	Philippines	Billing System	Automated billing improved accuracy and efficiency
AlHajeri et al.	2021	AI in Healthcare Systems	Kuwait	Appointme nt & MIS	Improved healthcare quality and system acceptanc e
Bidari et al.	2021	Queue Manageme nt System Study	Iran	Queue System	Increased patient satisfaction and accurate

					wait perception
Torab-Miandoab et al.	2023	Interoperability in Health Systems	Iran	System Integration	Highlighted need for standards like HL7 FHIR and HIPAA
Wu et al.	2021	Blockchain in Medical Data Security	China	Data Security	Enhanced privacy and protection using blockchain
Li et al.	2021	Queueing System in Epidemic Models	Germany/China	Queue Theory	Showed impact of behavior and

					system dynamics
Esberto et al.	2023	Online Appointment System Evaluation	Philippines	Appointment System	Achieved 90.2% user satisfaction
Resurreccion & Alvarez	2025	Utility Billing Management System	Philippines	Billing System	Efficient but requires proper user training
Mahfouz et al.	2023	Mawid Appointment System Study	Saudi Arabia	Appointment System	Improved accessibility and reduced waiting time

Alrasheedi et al.	2019	Wait Time vs Patient Satisfaction	Saudi Arabia	Queue System	Longer wait times reduce patient satisfaction
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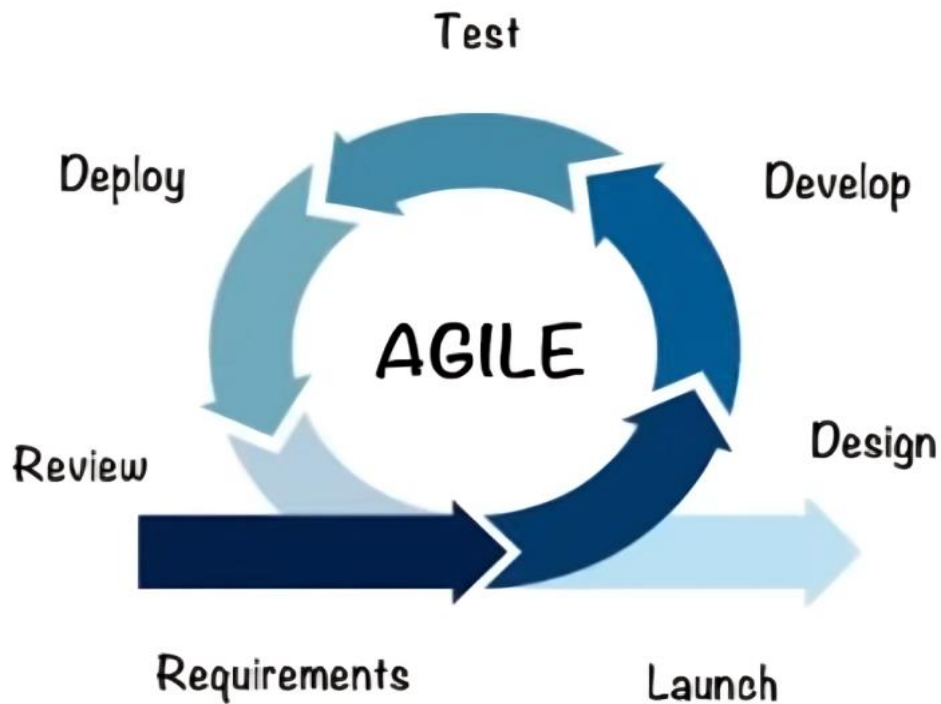
CHAPTER III

METHODOLOGY AND RESEARCH DESIGN

This chapter discusses the research design of the proposed system. It also contains the methodology used, processes, information, and diagrams that represent the proposed system.

3.1 Research Method

Figure 1. Research Method and Design (Agile Methodology)



Note: Developed by the researchers based on Agile SDLC models (ChartDiagram, 2025)

Figure 1 illustrates that the methodology used in this study is the Agile methodology. Agile emphasizes team collaboration, flexibility, iterative development, and rapid delivery of functional system components. It focuses on developing a project into small pieces called sprints. Each sprint results in a working version of the system that can be tested and reviewed. Allowing continuous feedback from the ABR Diagnostic Center that can be used for adjustments based on user input and changing requirements throughout the development process. Its emphasis is on handling evolving requirements, frequent client feedback being crucial for a student project, and delivering functional increments.

- **Requirements:** This phase includes gathering and defining the system's needs in every sprint that the proponents plan. This includes interviewing the client for the proponents to gather information about the business process that relates to the sprint target. This also includes the requesting of client samples, like if the next sprint is creating features for billing, the proponents may gather copies of the receipt or invoice. These requirements evolve or may increase as client feedback is received. In this phase, the proponents can set a timeline for the upcoming sprint.
- **Design:** After the requirements are gathered, the next phase is to design. In this phase, the user interface layouts, data flow diagram, and database architecture are designed. The design is kept flexible and minimal at first to support Agile's adaptive nature.
- **Development:** After designing, the proponents begin coding and implementing features in small, manageable parts. Agile promotes development in small increments so that

each feature can be developed in a short cycle. Allowing for continuous and early feedback.

- **Test:** After the development of the feature, the proponents who are assigned to testing immediately start to test the developed feature. This will ensure the feature functionalities and detect bugs. This may include user acceptance testing (UAT), integration testing, and usability testing. In this phase, the proponent's goal is to identify as early as possible.
- **Deploy:** Once tested, the study has a staging environment for real-world validation. This enables the proponents and the client to observe how the system performs and to gather insights for improvement.
- **Review:** The deployed feature is evaluated by the proponent's capstone adviser and the ABR Diagnostic Center. Their feedback is collected and logged, which helps the proponents make necessary adjustments and refinements for the next development cycle.
- **Launch:** After all major and critical modules are completed, tested, and approved. The full system is launched for live use. In Agile, the launching phase is allowed, which means the earlier module can go live while the other module is still being developed.

Context Diagram

Figure 2. Proposed Context Diagram of ABR Diagnostic Center

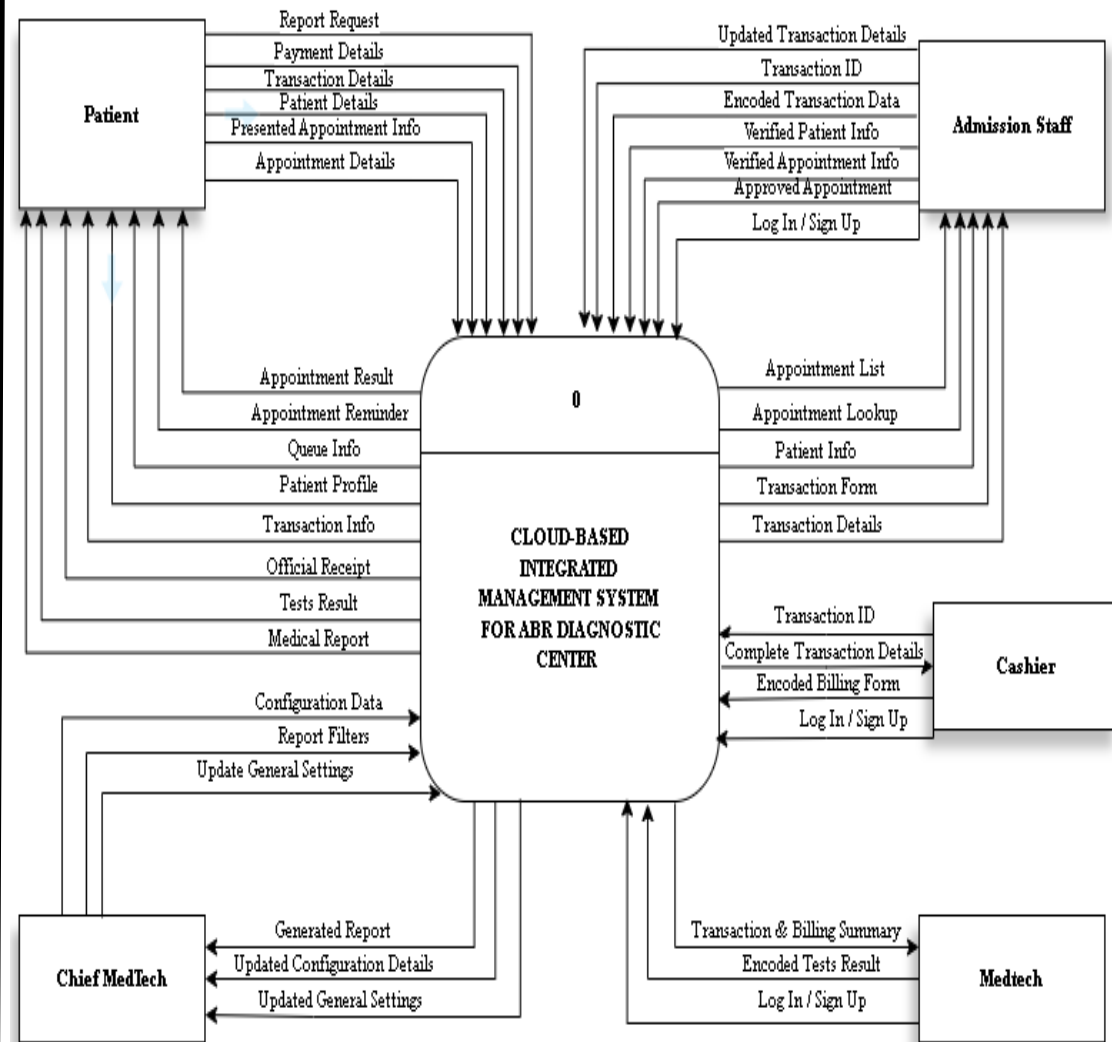


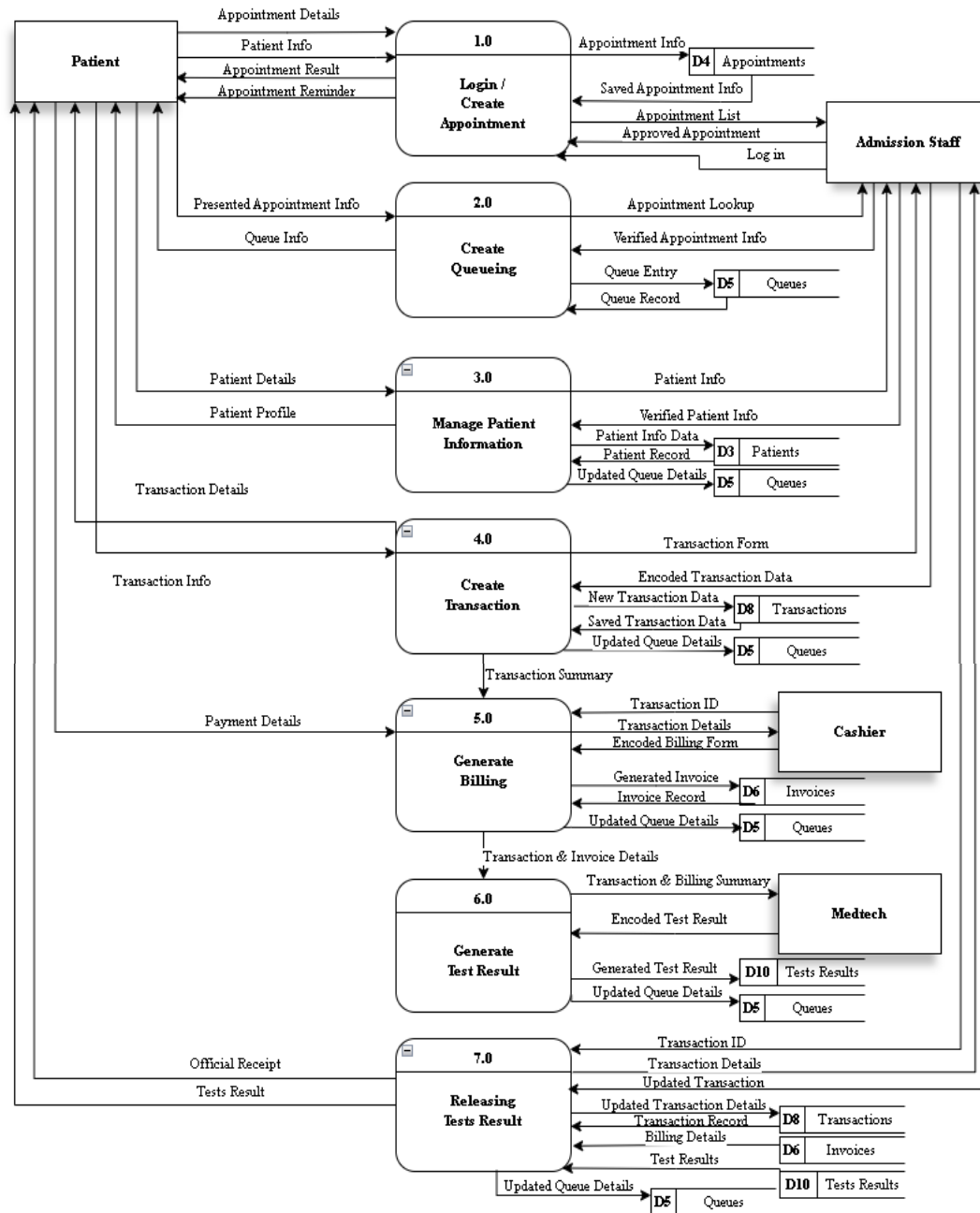
Figure 2 illustrates the enhanced data flow of the proposed ABR Diagnostic Center Integrated Management System, where the process starts with the patient submitting essential information such as personal details, appointment preferences, transaction records, and payment data. The system responds by validating the input and providing the

patient with key outputs: confirmed appointment results, reminders for upcoming appointments, real-time queue updates, and a complete patient profile for reference.

Once the appointment details are in place, the admission staff enters the picture by verifying the patient's identity and the accuracy of their appointment. They encode the verified transaction and appointment details into the system. Once approved, this information becomes part of the centralized records, ensuring data consistency and operational transparency. The cashier then accesses the system to receive the necessary transaction identifiers and billing forms. After processing the payment details and validating the transaction, the cashier updates the system with finalized billing records, which are also reflected in the patient's profile and overall transaction history. Simultaneously, the medtech conducts the required diagnostic tests based on the services the patient requested. The test outcomes, such as lab results or vital health data, are entered back into the system. These results are processed and delivered to the patient in the form of comprehensive medical reports and test findings. Meanwhile, the chief medtech oversees the system's configuration. They input data such as report filters and configuration settings to get customized reporting and to have full control over necessary settings. In return, they receive automatically generated reports and updated configuration details, giving them insights into operational metrics, patient statistics, or financial summaries.

Data Flow Diagram

Figure 3. Proposed Data Flow Diagram Level 1 of the ABR Diagnostic Center



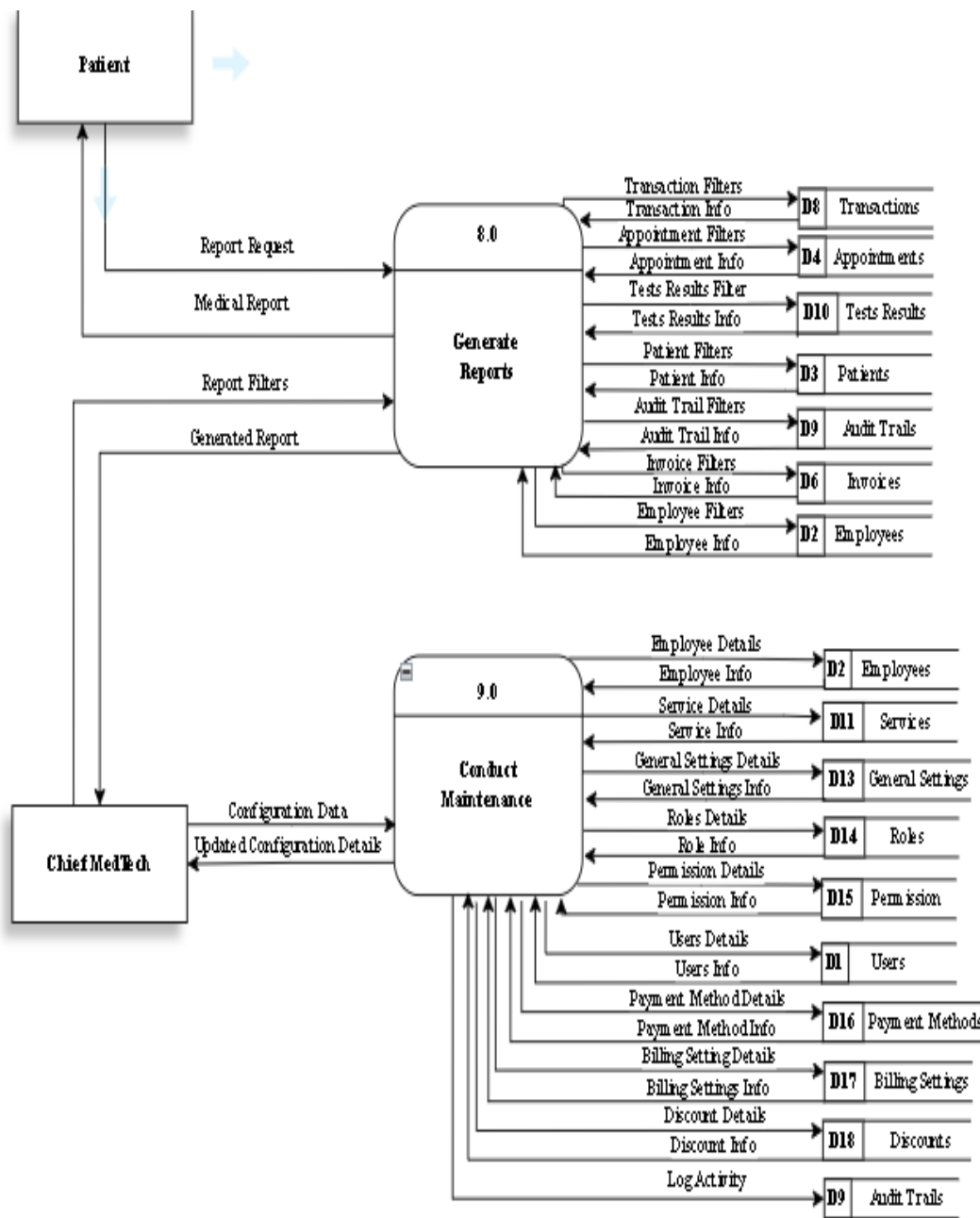


Figure 3 represents the data flow diagram of the proposed system for the ABR Diagnostic Center. It consists of 9 main processes of the proposed system, namely appointment, queueing, managing patient information, creating transactions, generating billing,

generating test results, releasing test results, generating reports, and conducting maintenance.

The appointment process will be that the patient will create an appointment by providing appointment details that consist of selected services, appointment dates, and their basic info like name, email, and contact number. In this process, admission staff will be included by providing approval of the appointment. After the approval, the patient will receive an appointment reminder when the appointment day comes.

The queueing process is when a patient is in the clinic premises; the patient will provide their appointment details that the admission staff will need to verify. When verified, the admission staff will create a queue entry to the Queue (D5) datastore. After saving, the patient will receive their queue info.

The next process, named Manage Patient Information, is the next station where the patient is called. In this process, a new patient will need to provide their details, and the admission staff will verify them. When verified, the new patient information will be created together with their account credentials. In this process, the patient will receive their patient profile, which consists of their information and credentials.

The Create Transaction process allows the patient to provide key details such as the purpose of the lab tests and the requesting company, which will be used in the test result display. During this process, selected services are gathered and added to form the complete

transaction. The system also updates the patient's checklist and queue status. Once completed, the patient receives their transaction summary.

The Generate Billing process handles the computation of service fees. It begins when the cashier retrieves the patient's transaction summary by entering the transaction ID into the system. This provides a list of services along with their prices, allowing the system to calculate the total cost. The patient then provides payment details such as receipt name and payment method, which the cashier encodes into the system. Once completed, the system generates an invoice and updates the patient's queue status.

The Generate Test Result process begins when the medtech retrieves the patient's transaction summary from the system. After reviewing the details, the medtech conducts the necessary medical tests on the patient. Once the testing is completed, the test outcomes are encoded into the test result form. The system then updates the test result record and the patient's queue status accordingly.

Releasing test results is the final step for the patient. Once called, the admission staff searches for the patient's transaction in the system and updates its status to indicate completion. The system retrieves records from the invoice, test results, and transaction data stores to prepare the medical report and official receipt for printing. These documents are then given to the patient through email and hard copy, and the patient's queue status is marked as completed.

This process, Generate Reports, allows patients and chief medtech s to request various reports, such as medical reports, transaction summaries, test results, and audit logs. The system filters and gathers data from multiple sources, including transactions, appointments, patients, invoices, tests, employees, and audit trails, based on the selected report filters. The requested reports are then compiled and presented in a structured format for viewing or printing. This process, conduct maintenance, is managed by the chief medtech and involves configuring and updating essential system data. It includes managing employee records, services, roles, permissions, users, billing settings, discounts, payment methods, general settings, and system logs. All updates are stored in their respective data stores to ensure the system operates with the latest configuration and data integrity is maintained.

Figure 4. Proposed Data Flow Diagram Level 2 of Process 1

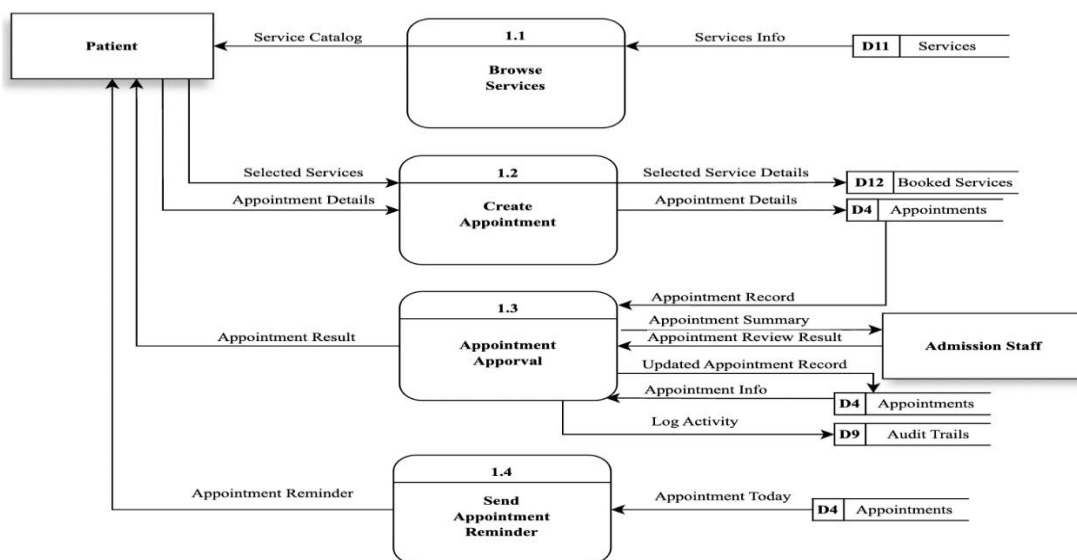


Figure 4 illustrates the data flow of Process 1 in the proposed system for ABR Diagnostic Center, detailing how appointment scheduling is handled. The process begins with the

patient browsing available services, which the system retrieves from the Services (D11) datastore and presents as a service catalog. After selecting desired services, the patient proceeds to create an appointment by submitting the appointment details and selected services. These details are then stored in the booked services and appointments data stores.

The system generates an appointment summary for the admission staff to review. Once approved, the system updates the appointment record and logs the activity into the audit trail for tracking purposes. The approved appointment result is then sent back to the patient. Finally, the system automatically identifies all appointments scheduled for the current day and sends timely reminders to the corresponding patients.

Figure 5. Proposed Data Flow Diagram Level 2 of Process 2

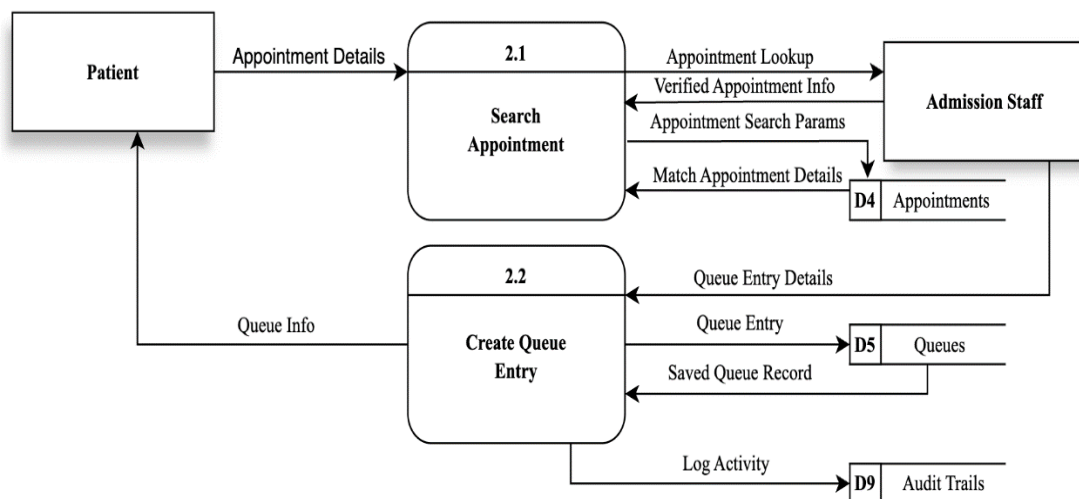


Figure 5 illustrates the data flow of Process 2 in the proposed ABR Diagnostic Center system, which focuses on creating a queue entry for the patient. When the patient arrives at the clinic, they present their appointment details to the admission staff. The staff then

performs an appointment lookup using search parameters to verify the appointment against the Appointments (D4) datastore. Once a valid appointment is found, the admission staff encodes the necessary queue entry details, which the system saves into the Queues (D5) datastore.

After successfully saving the record, the system provides the queue information back to the patient and logs all related activities into the audit trail for monitoring purposes.

Figure 6. Proposed Data Flow Diagram Level 2 of Process 3

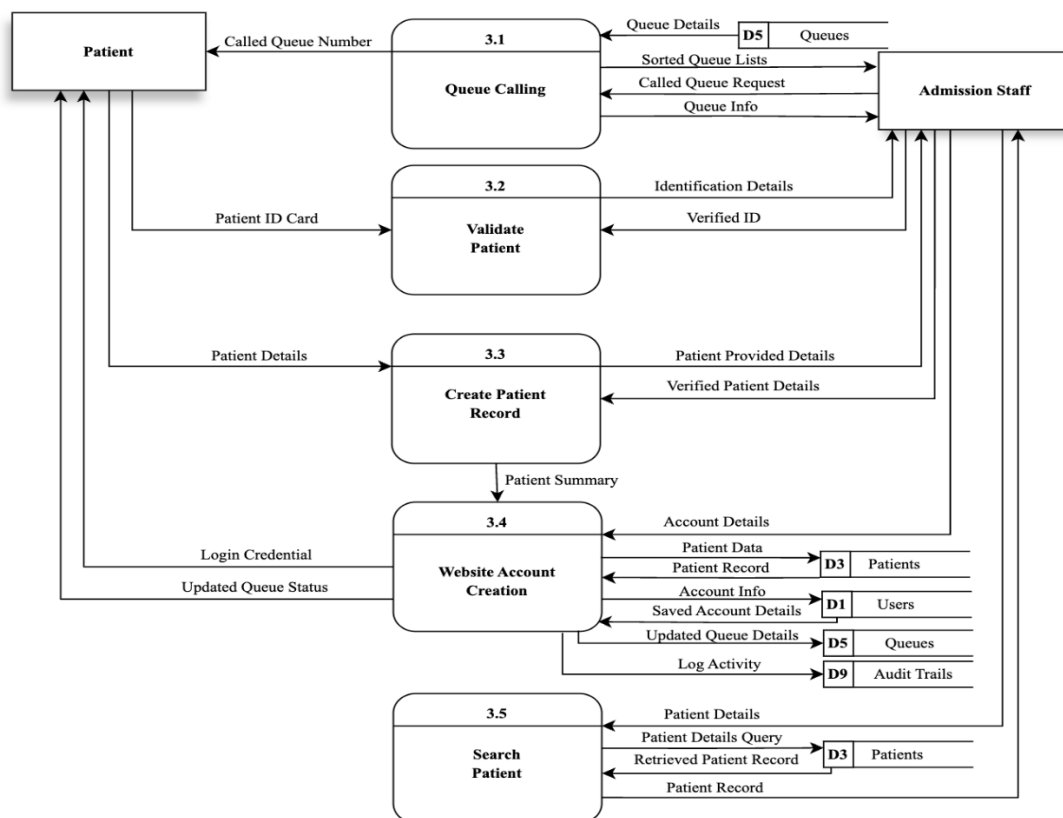


Figure 6 illustrates the data flow of Process 3 in the proposed ABR Diagnostic Center system, which focuses on creating a patient record for the patient. The process begins with

the system displaying all sorted queue entries from the Queues (D5) datastore. Once a queue number is called, the system notifies both the admission staff and the patient by displaying the queue information. When the patient approaches, they present a valid ID for verification. If the patient is new, the admission staff proceeds to create a new patient record by collecting the patient's personal details. Next, the patient provides their login credentials, such as email and password, which are encoded to create their website account.

The system then stores the patient's information in the Patients (D3) datastore and the account credentials in the Users (D1) datastore, both of which return confirmation records. Finally, the updated queue details are saved in the Queues (D5) datastore, and the entire activity is logged in the audit trail. At the end of the process, the patient receives their login credentials, profile details, and updated queue status. If the patient record exists, or already has an account, the admission staff will only need to search the patient record in the system.

Figure 7. Proposed Data Flow Diagram Level 2 of Process 4

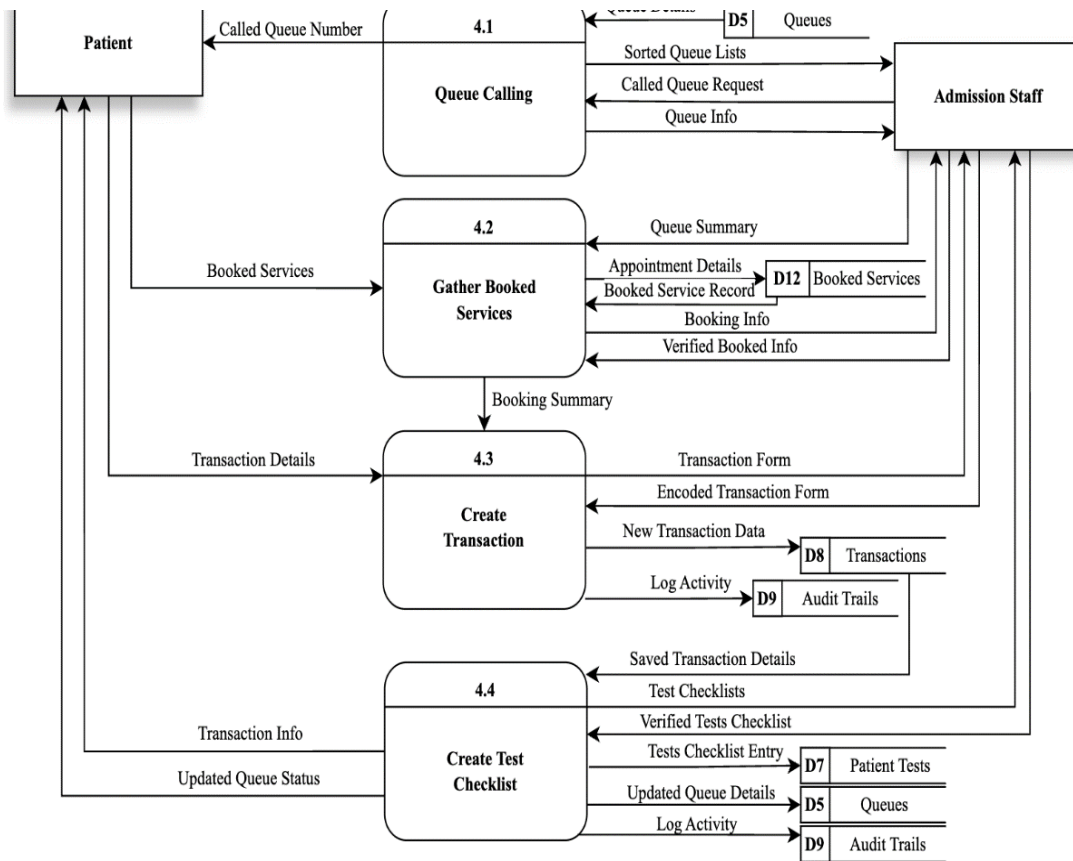


Figure 7 illustrates the data flow of Process 4 in the proposed ABR Diagnostic Center system, which handles the creation of a transaction record for the patient. The process starts with the system displaying the sorted queue entries from the Queues (D5) datastore. Once a queue number is called, both the admission staff and the patient are notified. When the patient arrives, they may provide new or additional service selections if none were chosen during the appointment. If preselected services exist, the admission staff retrieves them from the Booked services (D12) datastore. The system then displays the service booking information for review and encoding.

Next, the patient provides transaction details, such as the purpose of the lab tests or the requesting company. The admission staff encodes these into the transaction form, which is saved in the Transactions (D8) datastore, with corresponding activity logs recorded in the Audit Trail (D9) datastore. After saving the transaction, the system generates a checklist for the required tests. Once verified, the checklist is saved in the Patient Tests (D7) datastore. The queue status is also updated in the Queues (D5) datastore, and the system logs the activity again for tracking purposes. At the end of this process, the patient receives their transaction summary and updated queue information.

Figure 8. Proposed Data Flow Diagram Level 2 of Process 5

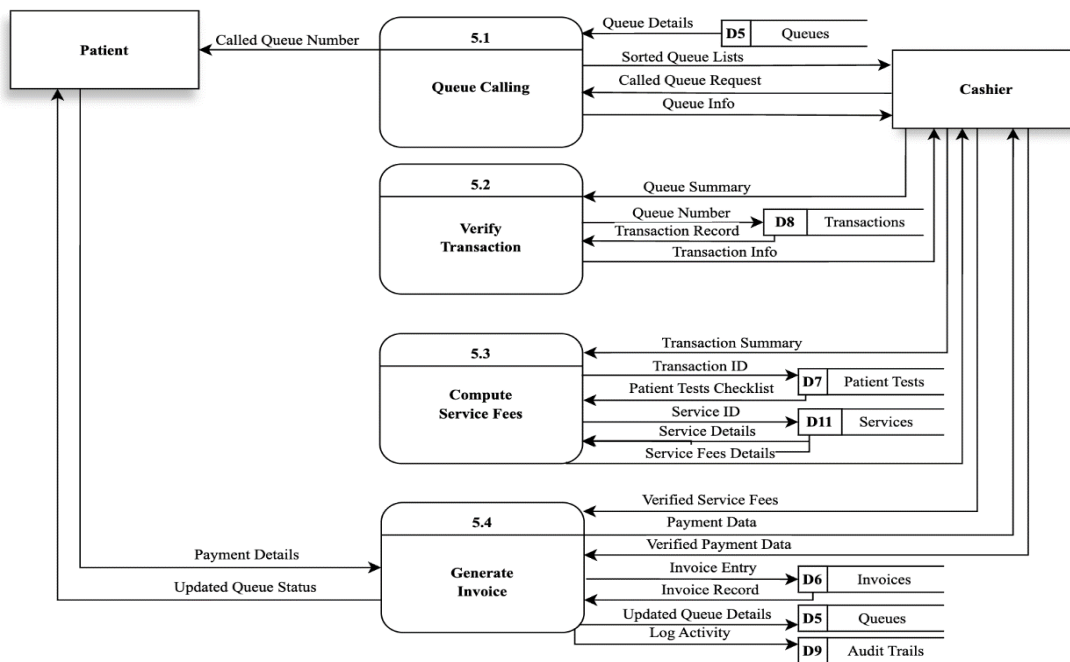


Figure 8 illustrates the data flow of Process 5 in the proposed ABR Diagnostic Center system, which focuses on computing transaction fees and generating the patient's invoice.

The process begins with the system displaying the sorted queue entries from the Queues (D5) datastore. When a queue number is called, both the cashier and the patient are notified. The cashier then verifies the transaction by providing the queue summary, which allows the system to retrieve the related transaction record from the Transaction (D8) datastore. This transaction information is then displayed to the cashier. To compute the service fees, the cashier provides the transaction summary to the system. The system retrieves the patient's checklist from the Patient Tests (D7) datastore and matches each service to the services (D11) datastore to get the corresponding prices and details. The system automatically calculates the total fees and displays them for the cashier to review and confirm. Once verified, the patient provides payment details such as the receipt name and payment method. After confirming the payment, the system saves the invoice to the Invoices (D6) datastore, updates the queue status in the Queues (D5) datastore, and logs the activity in the Audit Trail (D9) datastore. Finally, the patient receives the updated queue status as confirmation that the billing process is complete.

Figure 9. Proposed Data Flow Diagram Level 2 of Process 6

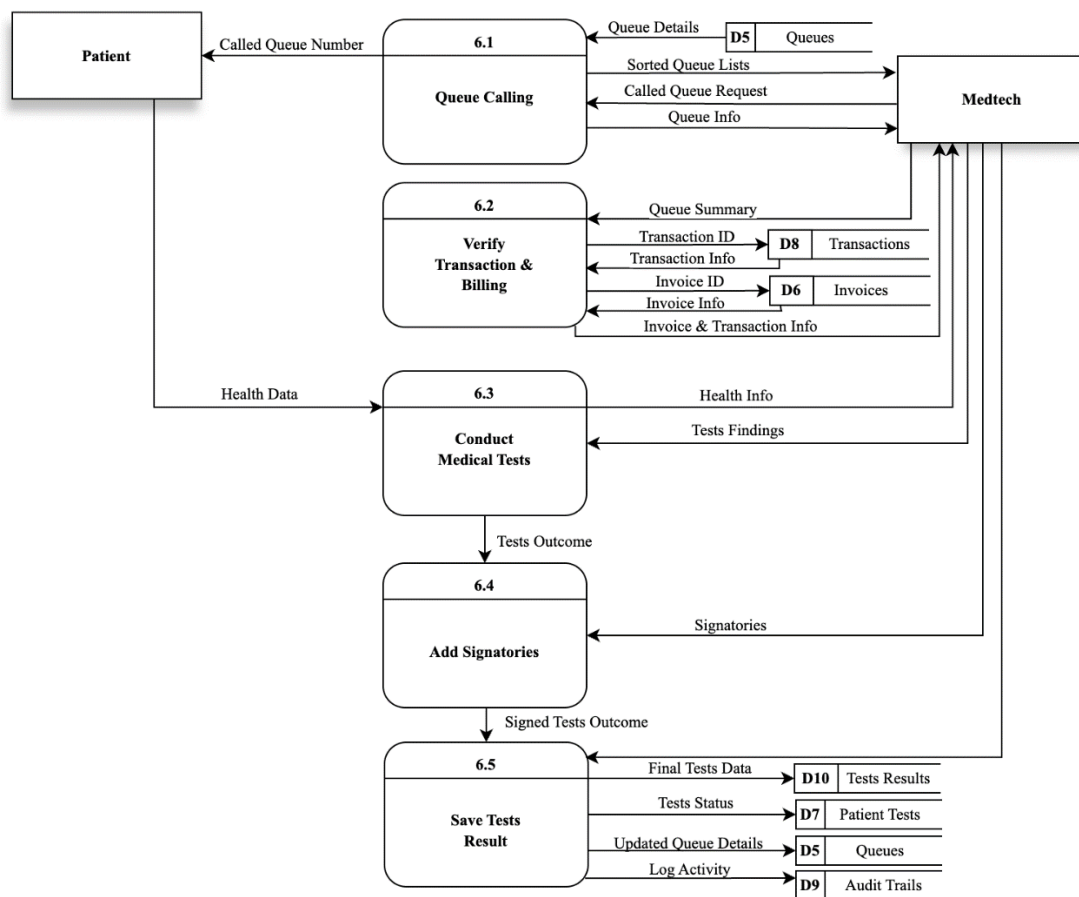


Figure 9 illustrates the data flow of Process 6 in the proposed ABR Diagnostic Center system, which covers the conduct and processing of medical tests. The process begins when a queue number is called by the system from the sorted queue list retrieved from the Queues (D5) datastore. Both the medtech and the patient are notified of the call queue number. The medtech verifies the transaction and billing information by accessing the transaction and invoice records from the respective datastores. Once verified, the patient provides their health data to proceed with the medical test. The medtech then conducts the tests, and the findings are recorded. The test outcomes are forwarded to the next step, where authorized

signatories are added. Once the results are signed, the system proceeds to save the final test results. This includes storing the signed test data into the Test Results (D10) and Patient Tests (D7) datastores, updating the patient's queue status, and logging the activity in the Audit Trail (D9) datastore. At the end of this process, both the test results and the updated queue information are completed and recorded within the system.

Figure 10. Proposed Data Flow Diagram Level 2 of Process 7

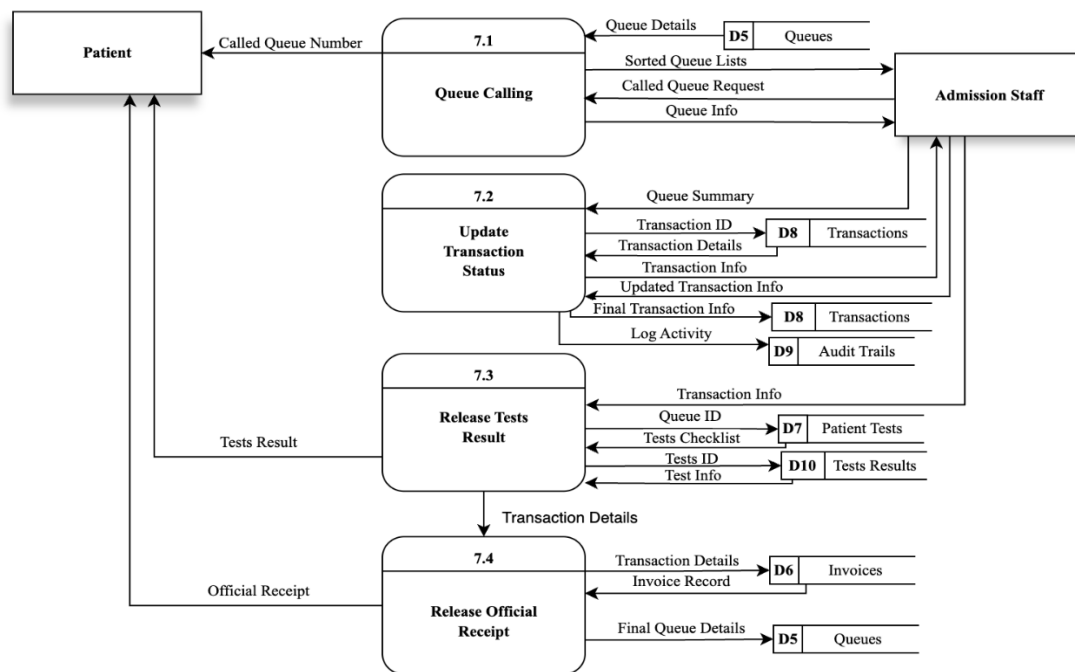


Figure 10 illustrates the data flow of Process 7 in the proposed ABR Diagnostic Center system, which focuses on the releasing of patient documents. The process begins when the system calls a queue number from the sorted list retrieved from the Queues (D5) datastore, notifying both the patient and the medtech. Using the queue summary provided by the admission staff, the system retrieves the corresponding transaction record from the

Transactions (D8) datastore via the transaction ID. The admission staff then updates the final transaction information, which is saved back into the Transactions (D8) datastore, and the system logs this activity in the audit trail. To release the medical test results, the system uses the transaction info to gather the patient's test checklist, and each test item is matched against the Test Results (D10) datastore to compile the necessary test information or medical report for printing and release. For the issuance of the official receipt, the transaction details are used to retrieve the invoice from the Invoices (D6) datastore. Finally, the system updates the final queue status in the Queues(D5) datastore and issues the receipt to the patient.

Figure 11. Proposed Data Flow Diagram Level 2 of Process 9

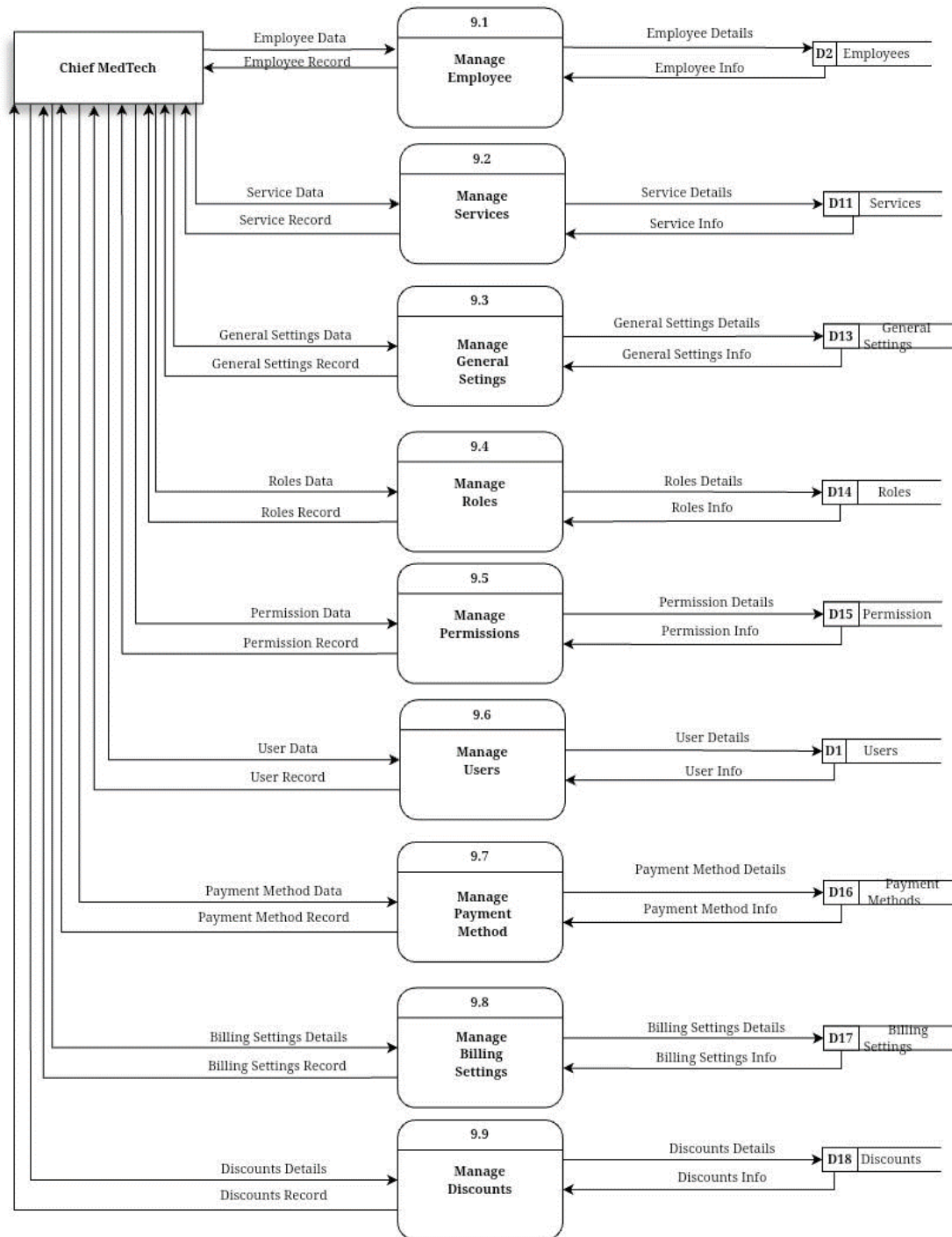


Figure 11 illustrates the data flow of Process 9 in the proposed ABR Diagnostic Center system, which handles the system maintenance and configuration management tasks. This process includes several administrative functions such as managing employees, services, general settings, roles, permissions, users, payment methods, billing settings, and discounts.

These processes are accessed by the system chief medtech to configure and maintain the respective records in their associated datastores. Upon providing new configuration settings, the system updates the targeted datastore, which in turn responds with the updated record information. The system then displays the updated data to inform the chief medtech that the changes have been successfully applied, and the records have been managed accordingly.

Activity Diagram

An activity diagram is a type of Unified Modeling Language (UML) diagram used to model the flow of activities or actions within a system or process. It is similar to a flowchart but offers additional features to represent complex workflows, including concurrency and decision-making processes.

Figure 12. Activity Diagram for Patient - Browse Services

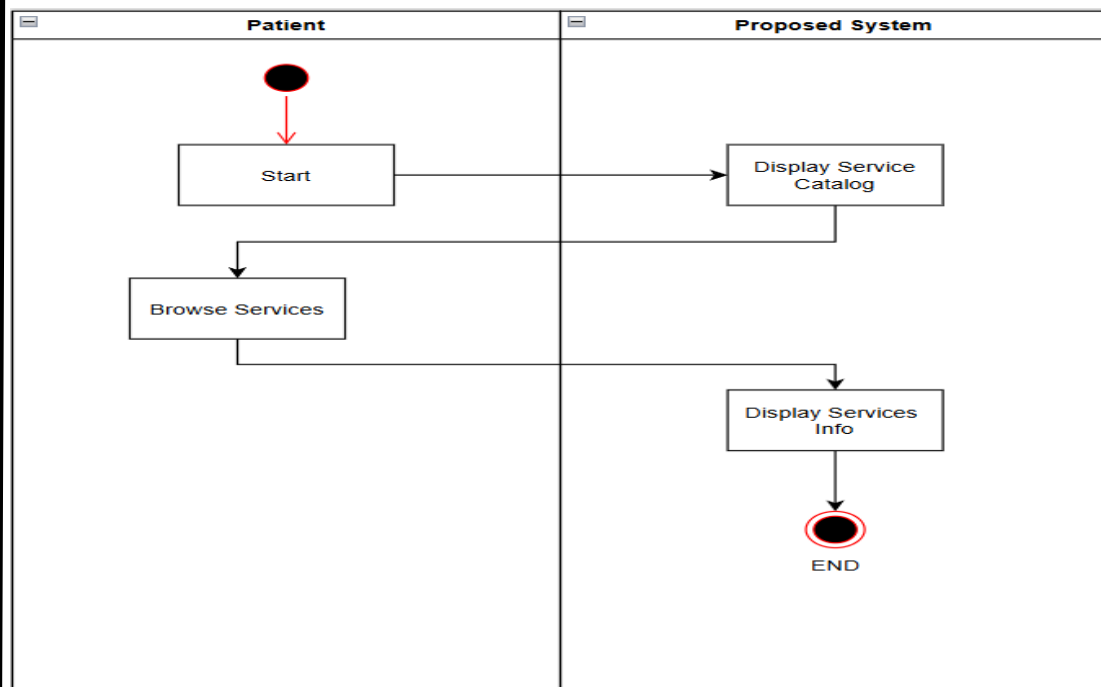


Figure 13. Activity Diagram for Patient - Create Appointment

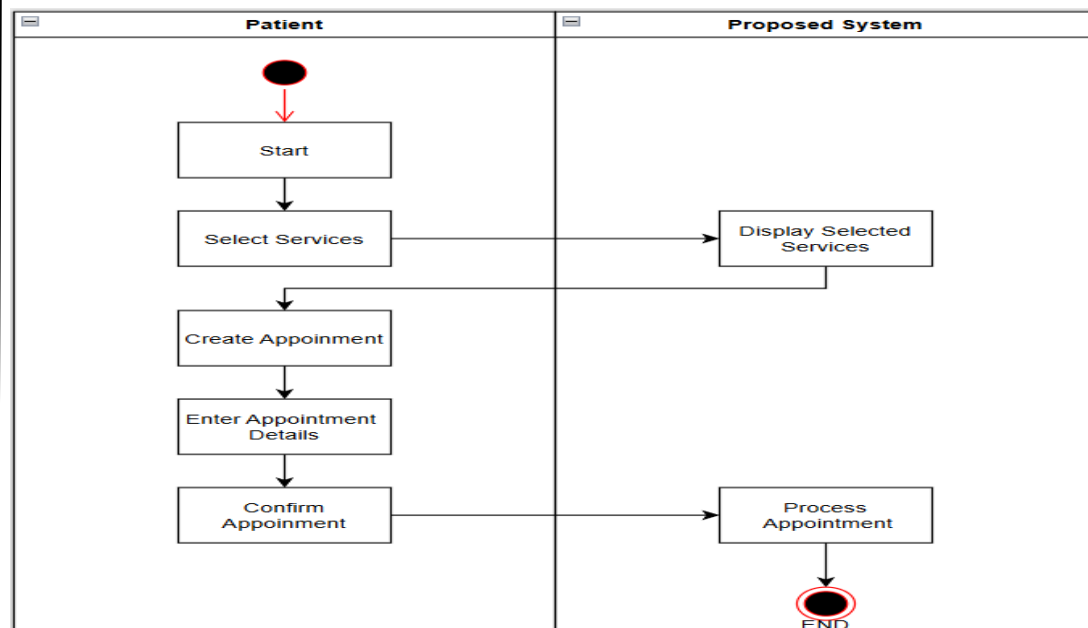


Figure 14. Activity Diagram for Patient & Admission - Appointment Approval

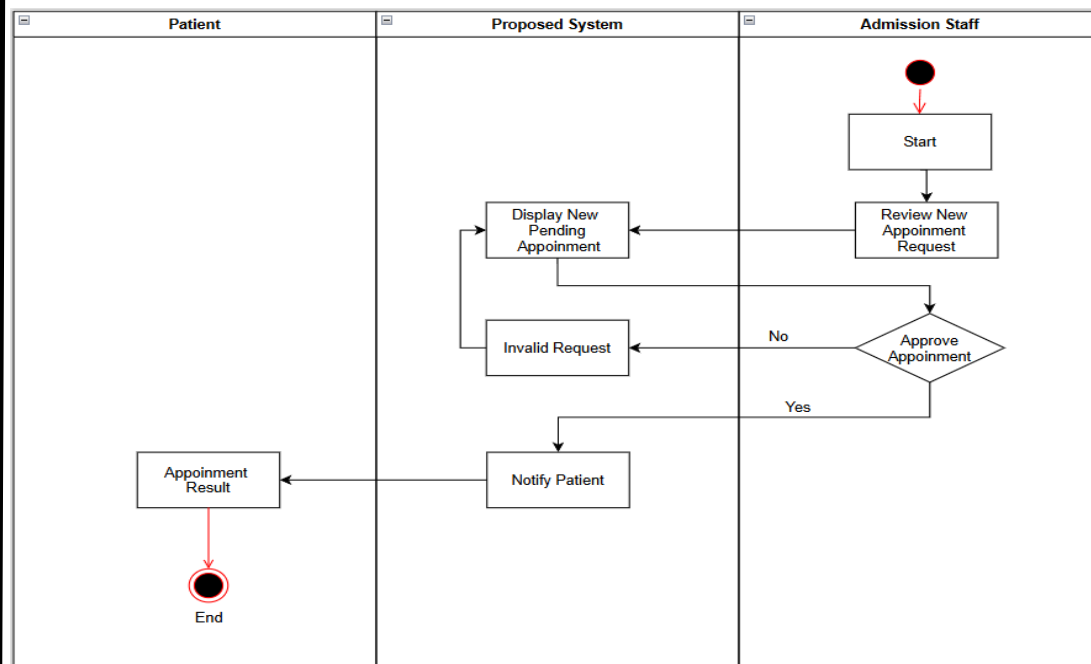


Figure 15. Activity Diagram for Patient & Admission - Search Appointment

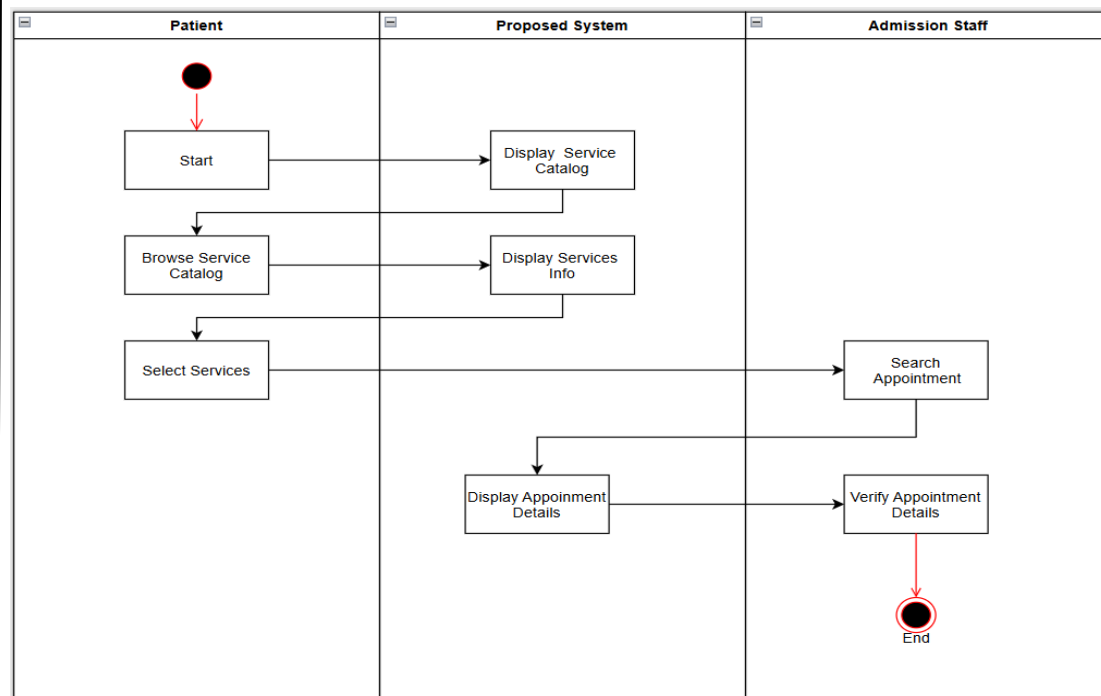


Figure 16. Activity Diagram for Patient & Admission - Create Queue Entry

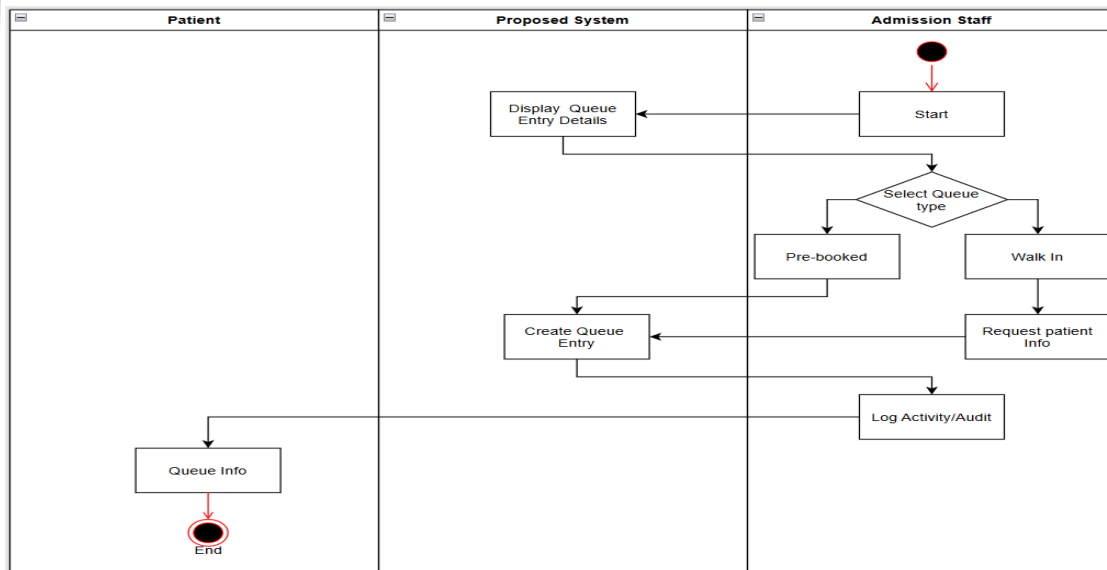


Figure 17. Activity Diagram for Patient & Admission - Queue Calling

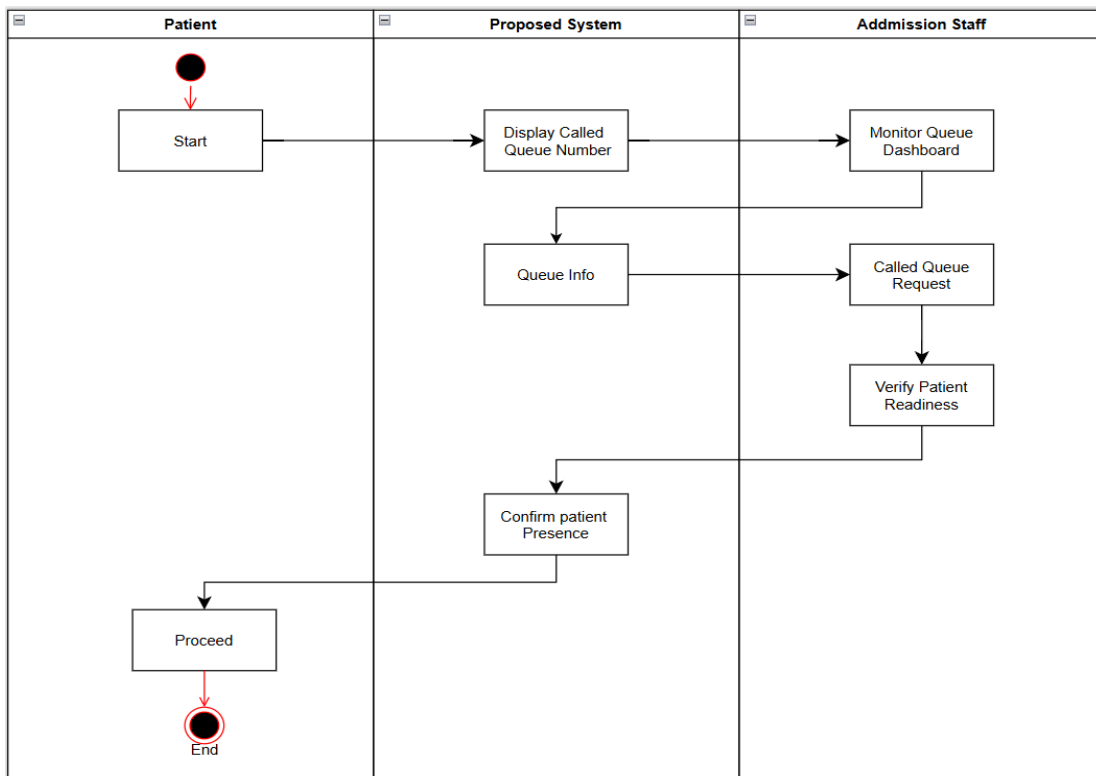


Figure 18. Activity Diagram for Patient & Admission - Validate Patient

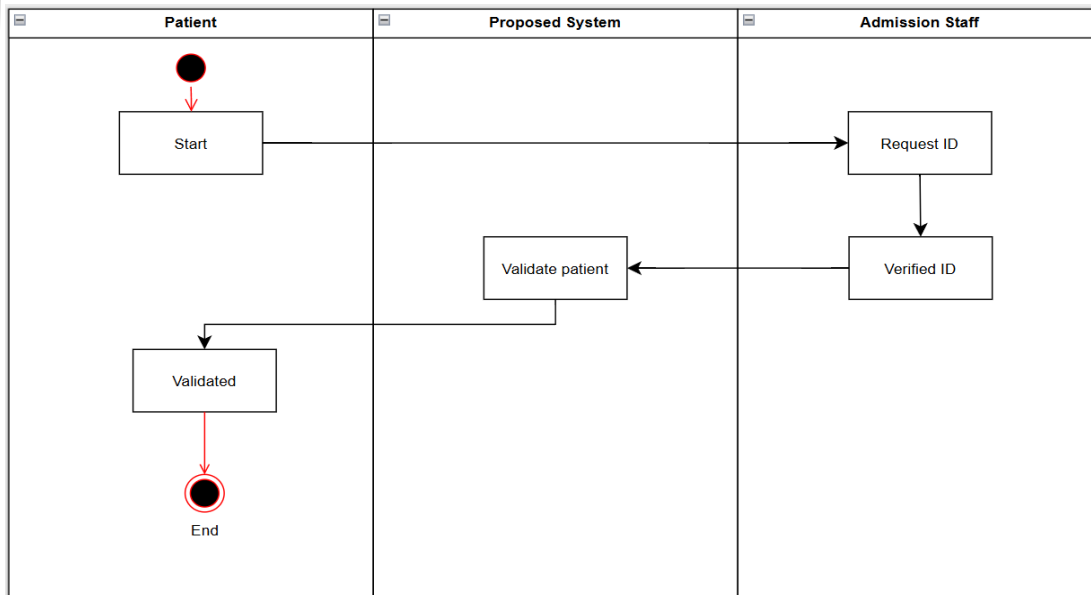


Figure 19. Activity Diagram for Patient & Admission - Create Patient Record

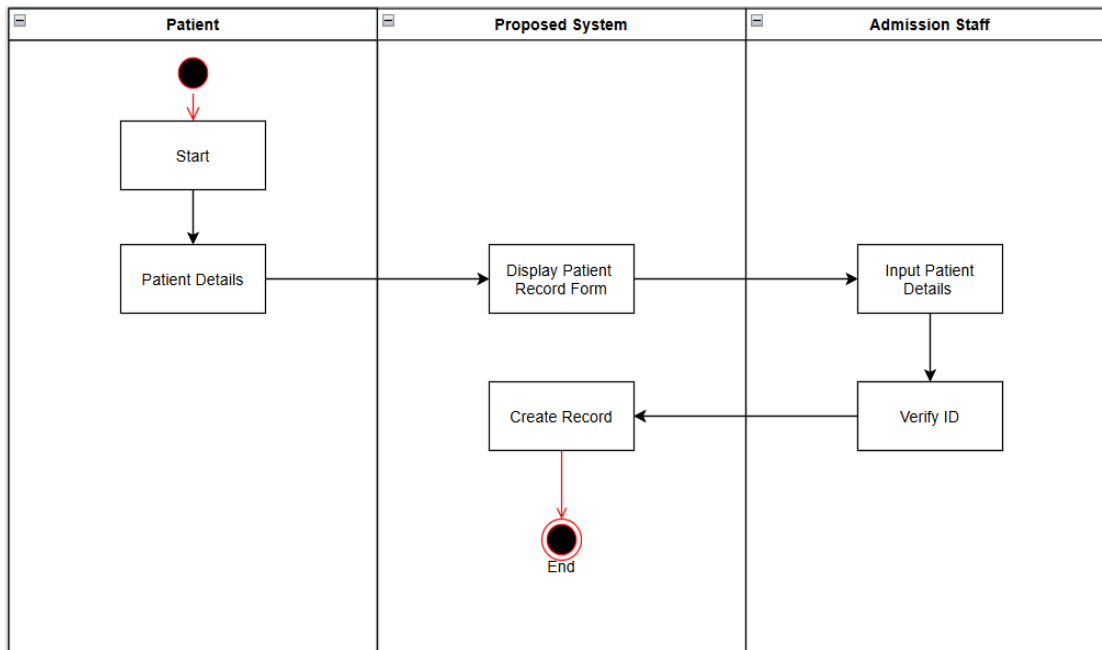


Figure 20. Activity Diagram for Patient & Admission - Website Account Creation

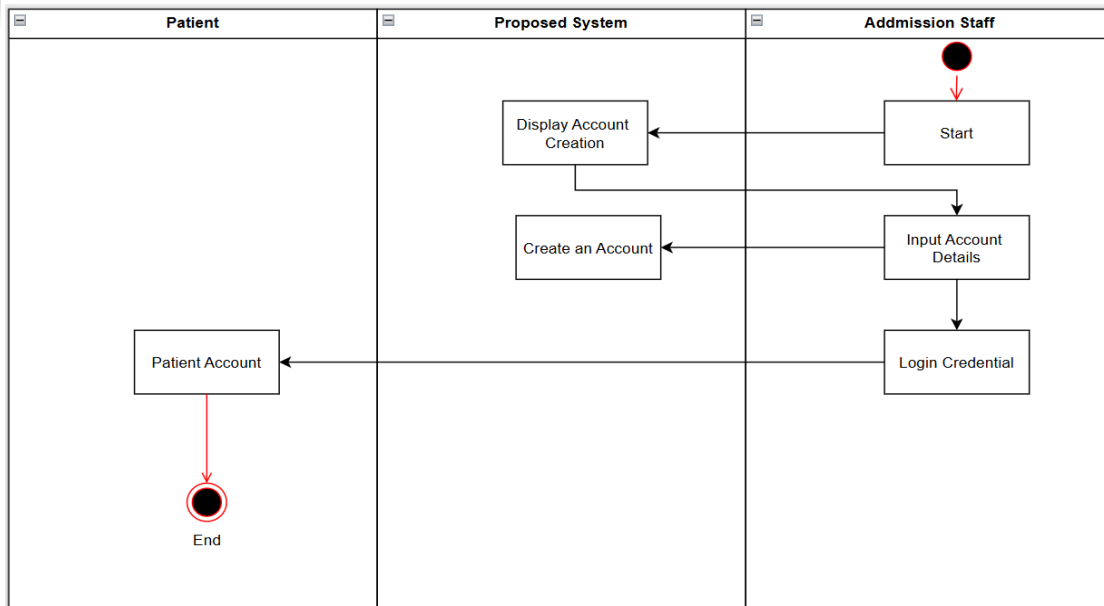


Figure 21. Activity Diagram for Admission - Search Patient

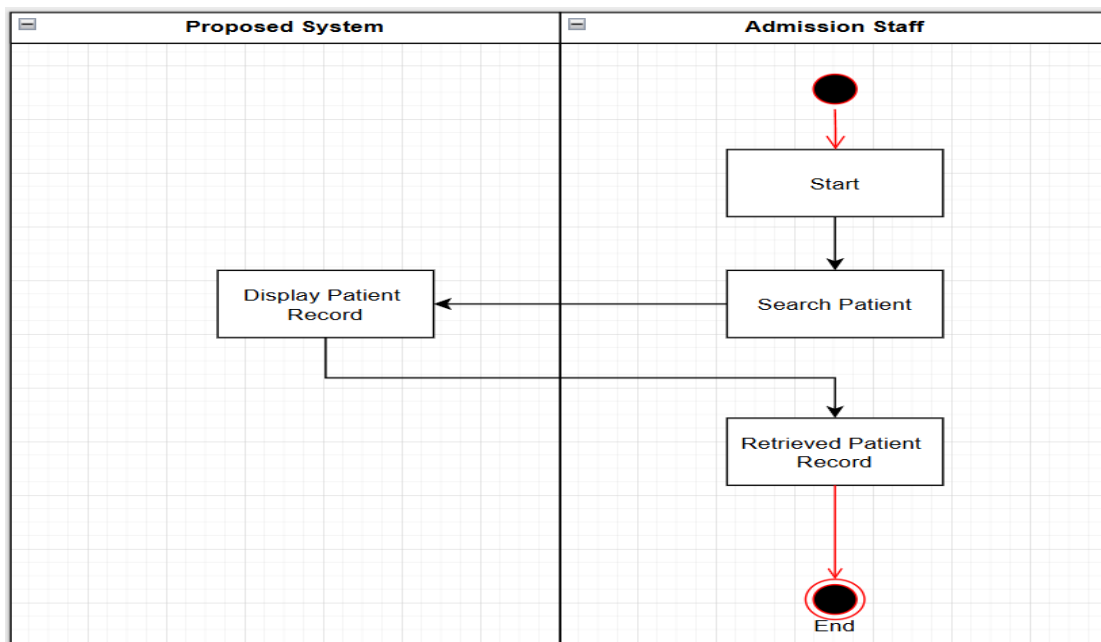


Figure 22. Activity Diagram for Patient & Admission - Queue Calling

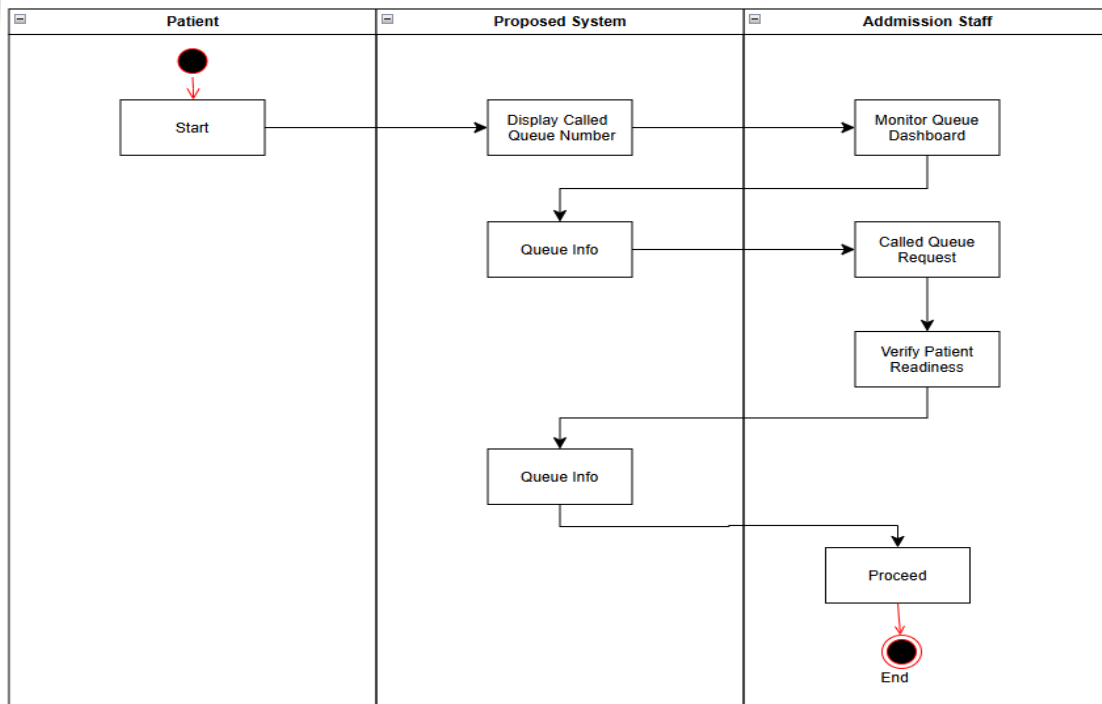


Figure 23. Activity Diagram for Patient & Admission - Gather Booked Services

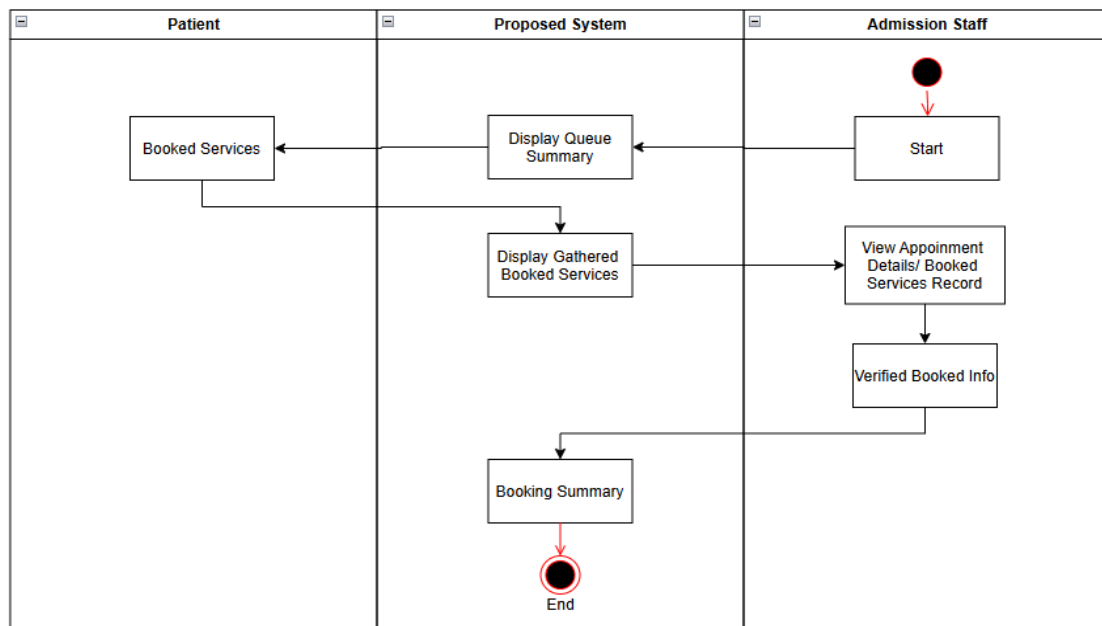


Figure 24. Activity Diagram for Patient & Admission - Create Transaction

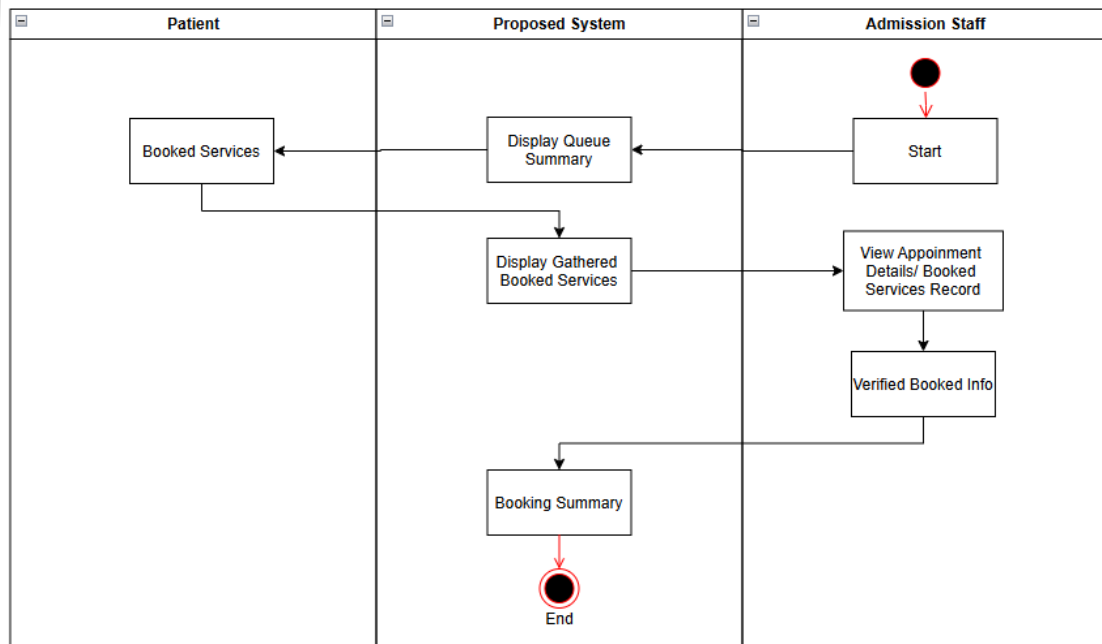


Figure 25. Activity Diagram for Patient & Admission - Create Test Checklist

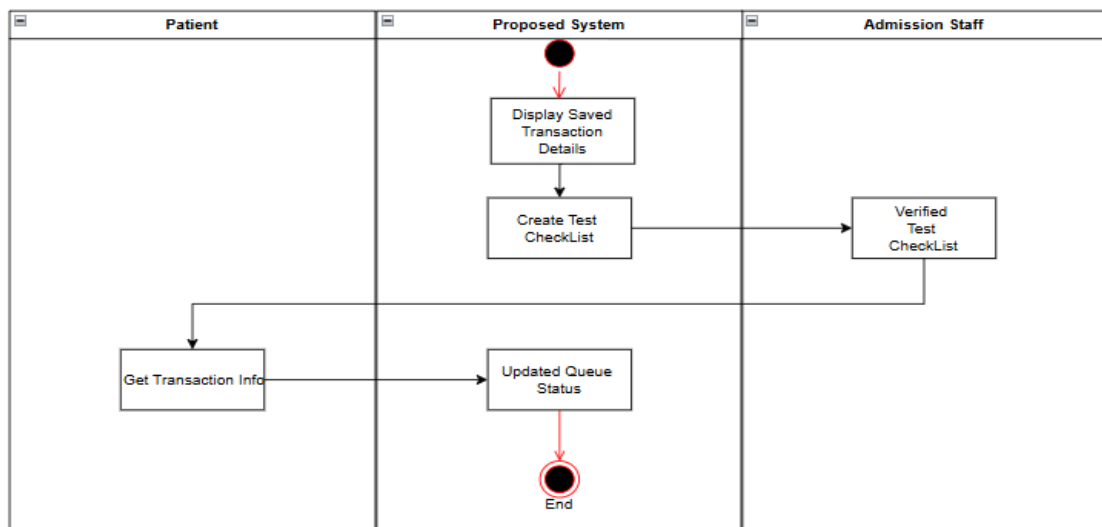


Figure 26. Activity Diagram for Patient & Cashier - Queue Calling

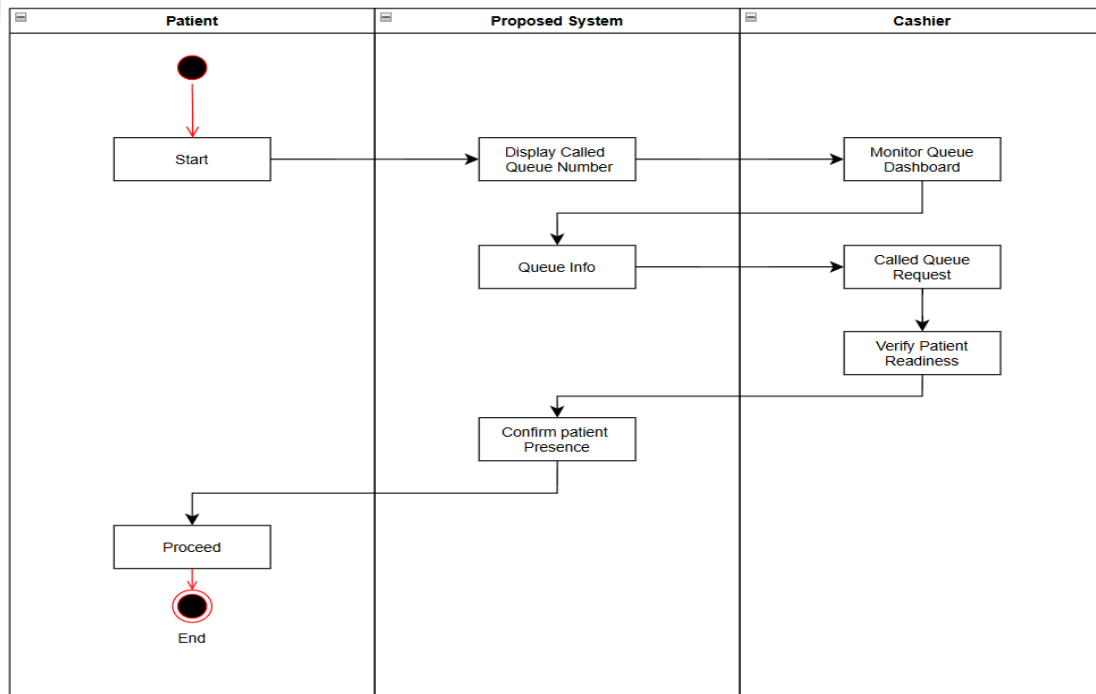


Figure 27. Activity Diagram for Cashier - Verify Transaction

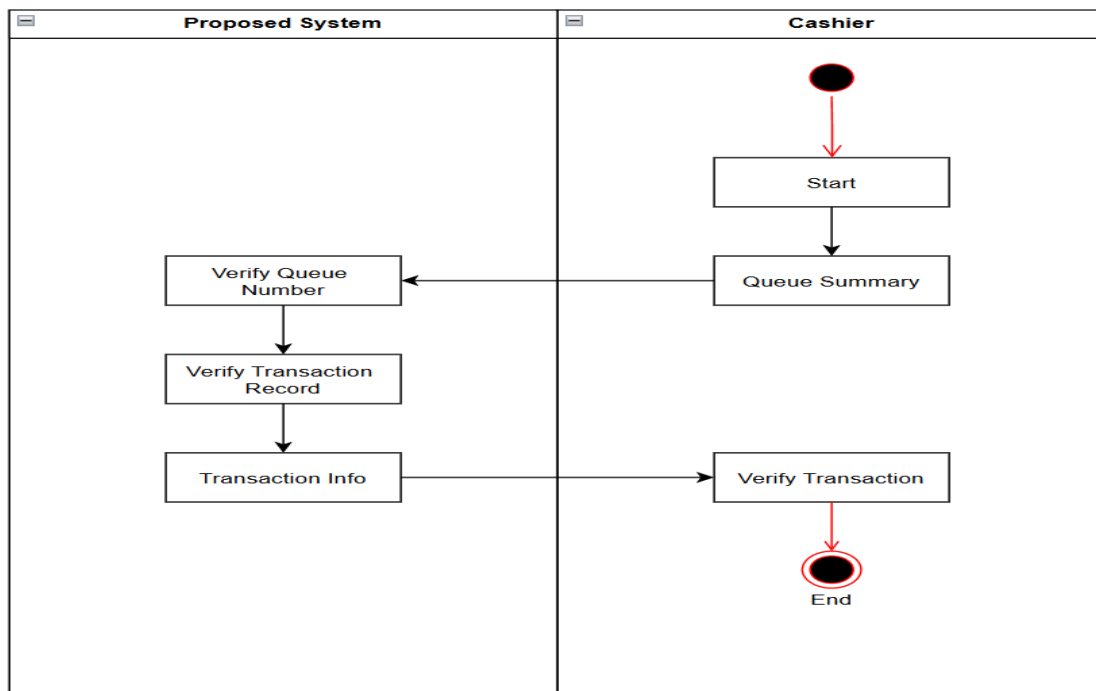


Figure 28. Activity Diagram for Cashier - Compute Service Fees

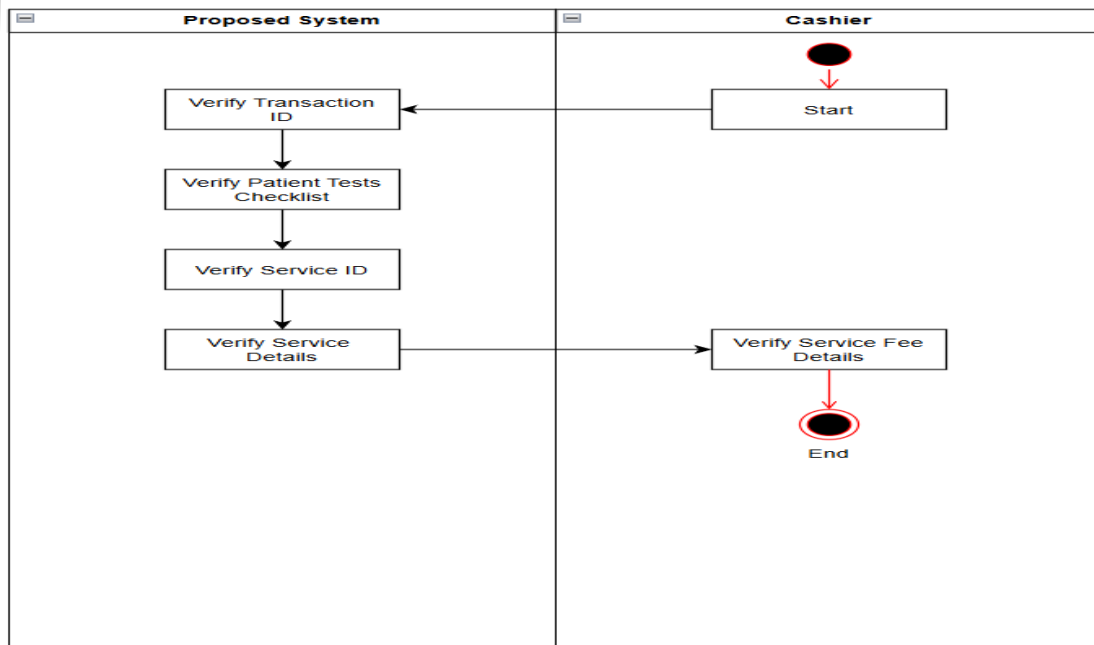


Figure 29. Activity Diagram for Patient & Cashier - Generate Invoice

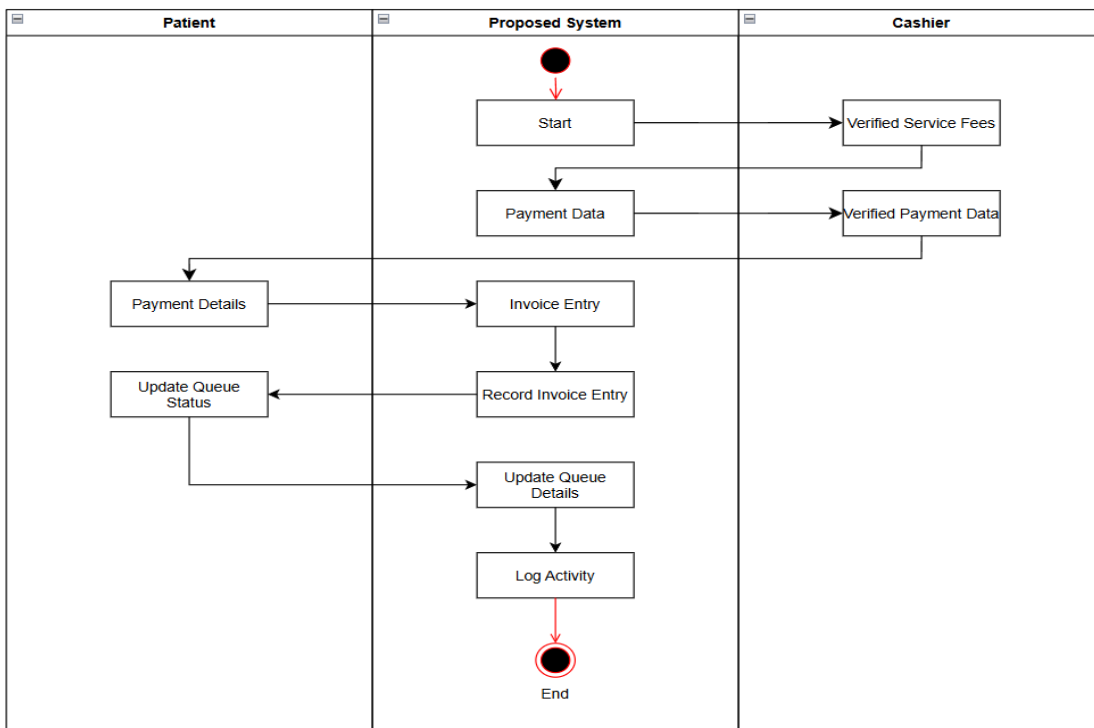


Figure 30. Activity Diagram for Patient & Medtech - Queue Calling

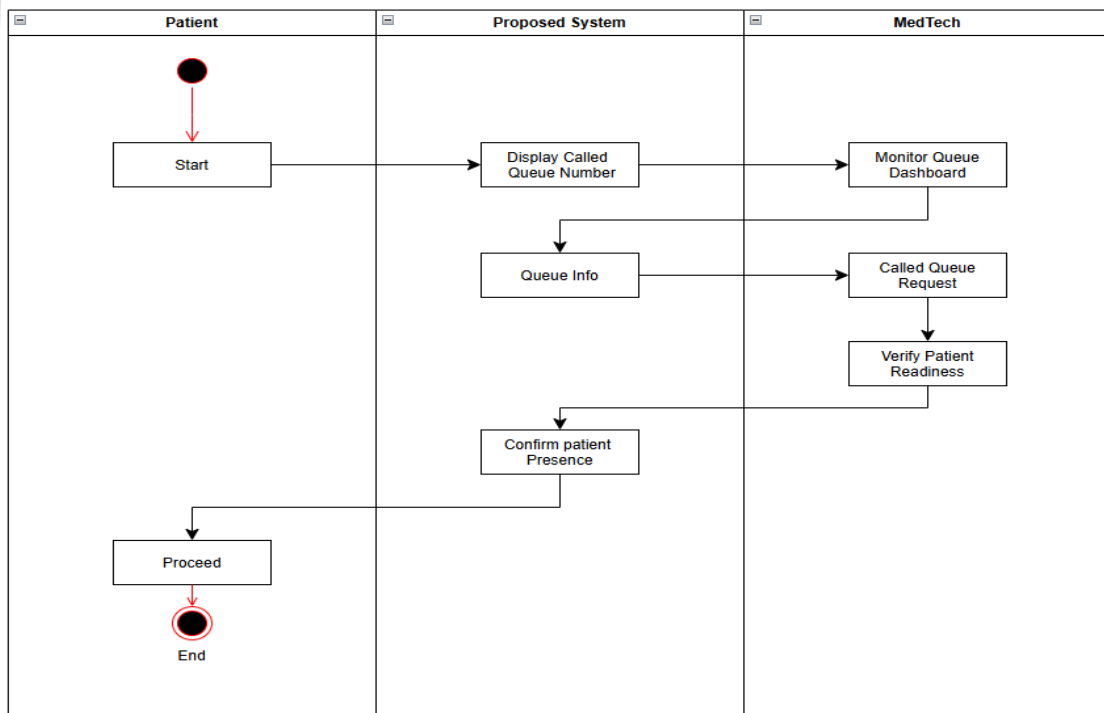


Figure 31. Activity Diagram for Medtech - Verify Transaction & Billing

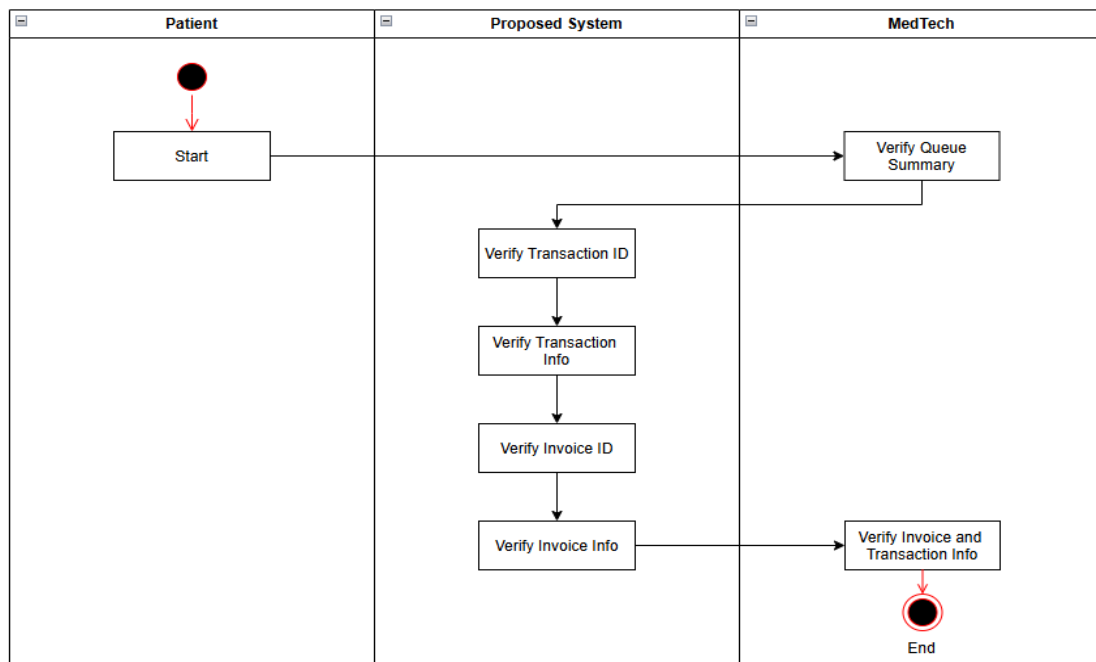


Figure 32. Activity Diagram for Patient & Medtech - Conduct Medical Test

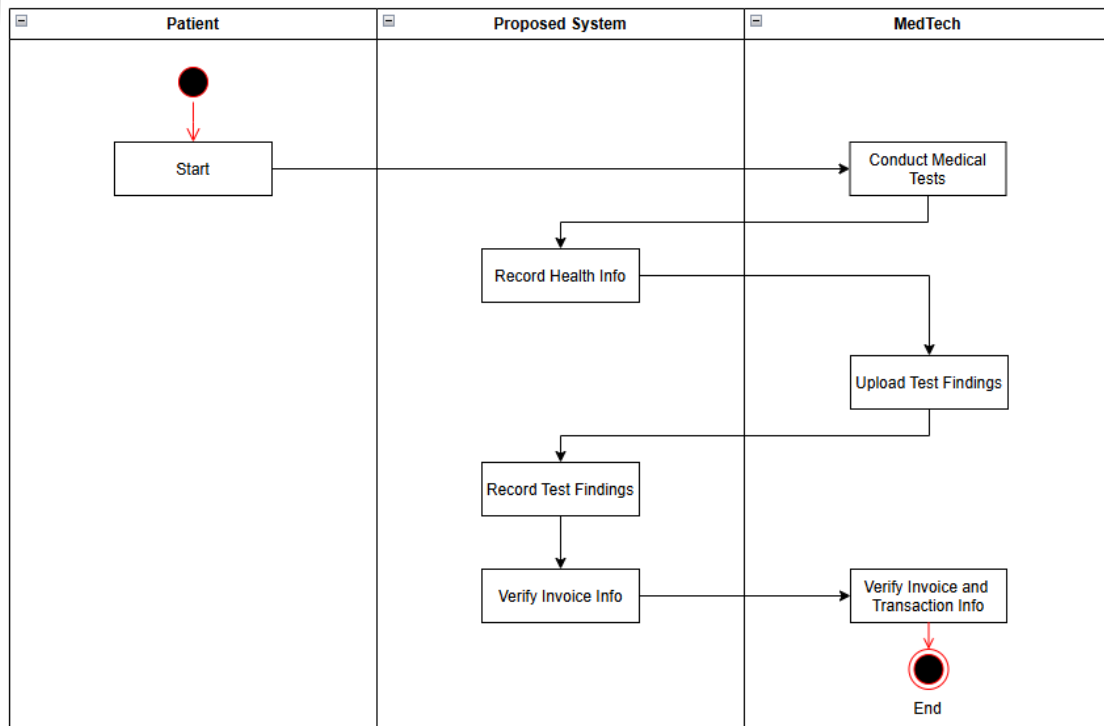


Figure 33. Activity Diagram for Medtech - Add Signatories

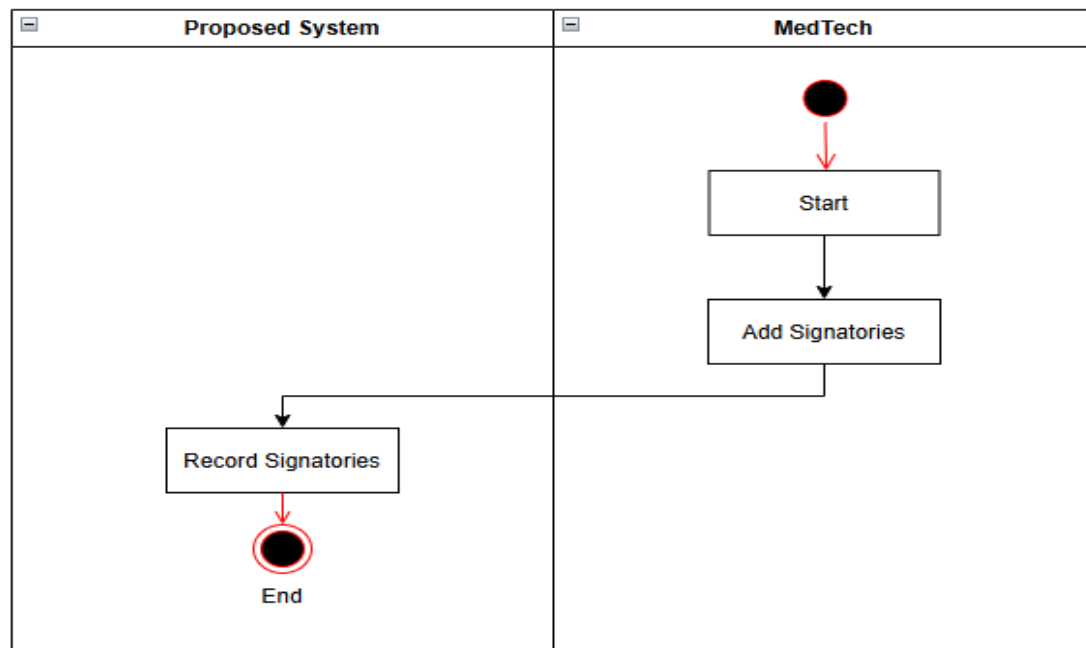


Figure 34. Activity Diagram for Medtech - Save Test Results

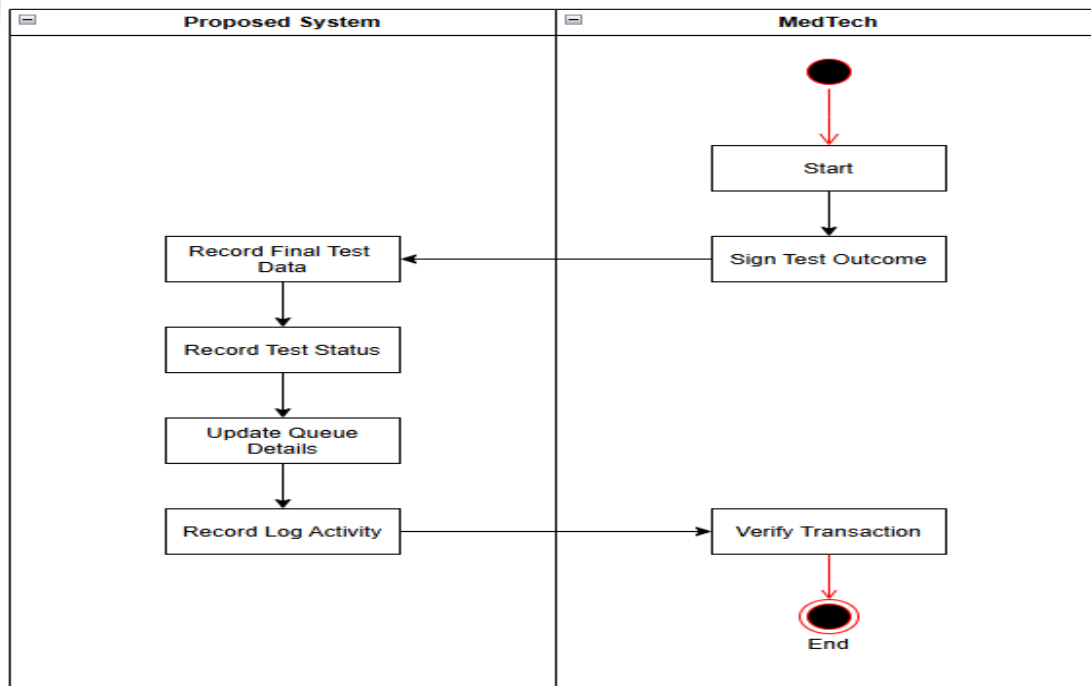


Figure 35. Activity Diagram for Patient & Admission Staff - Queue Calling

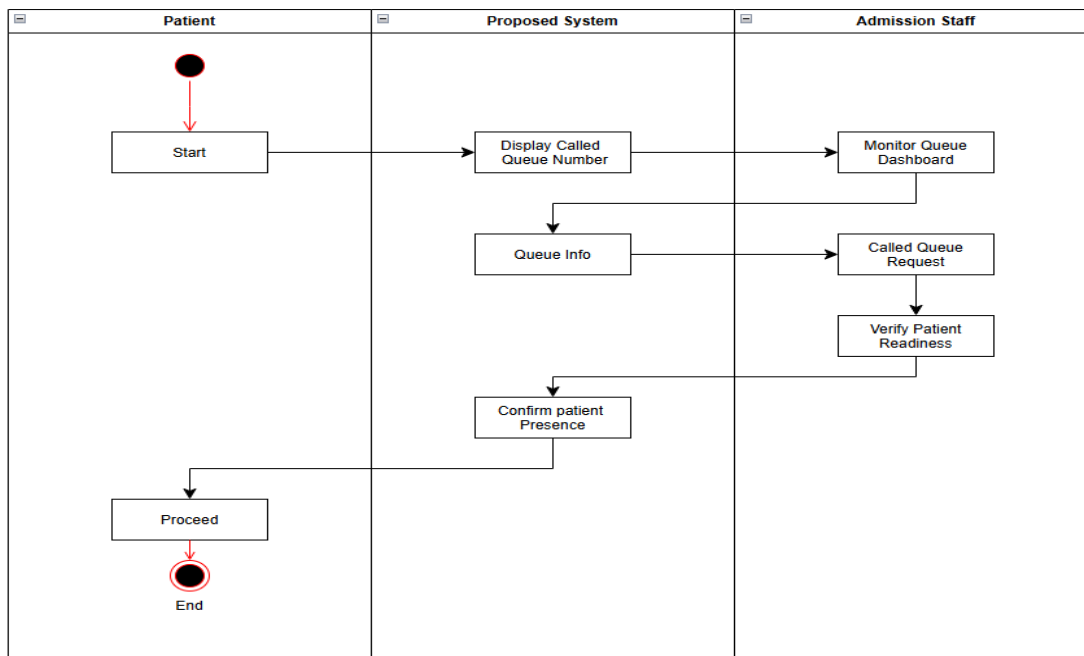


Figure 36. Activity Diagram for Admission Staff - Update Transaction Status

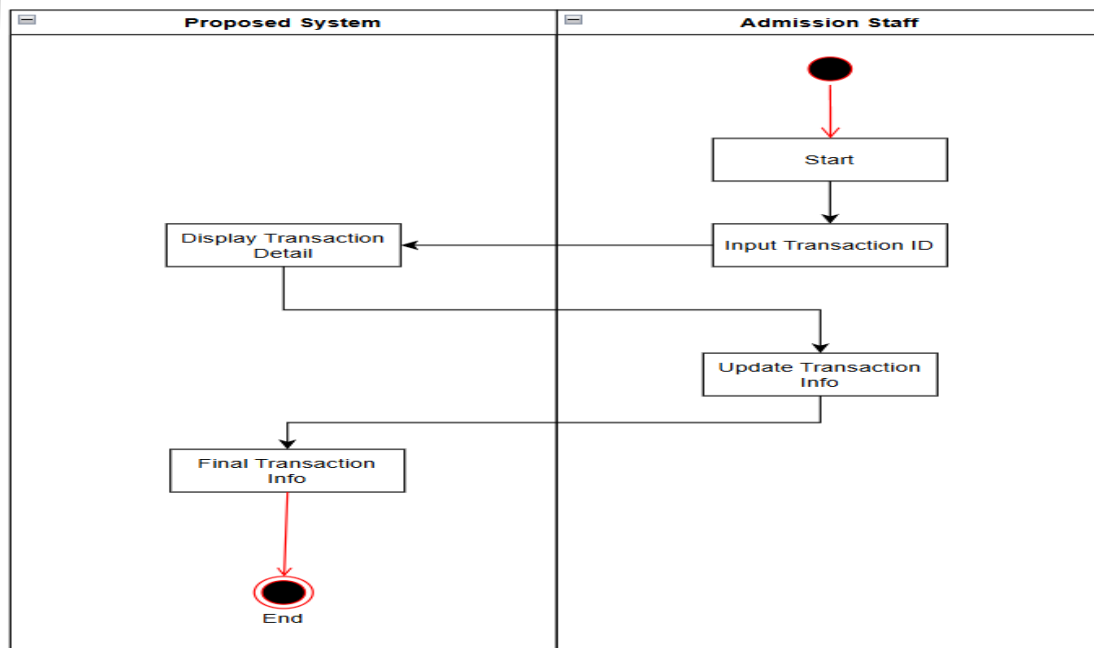


Figure 37. Activity Diagram for Patient & Admission Staff - Release Test Result

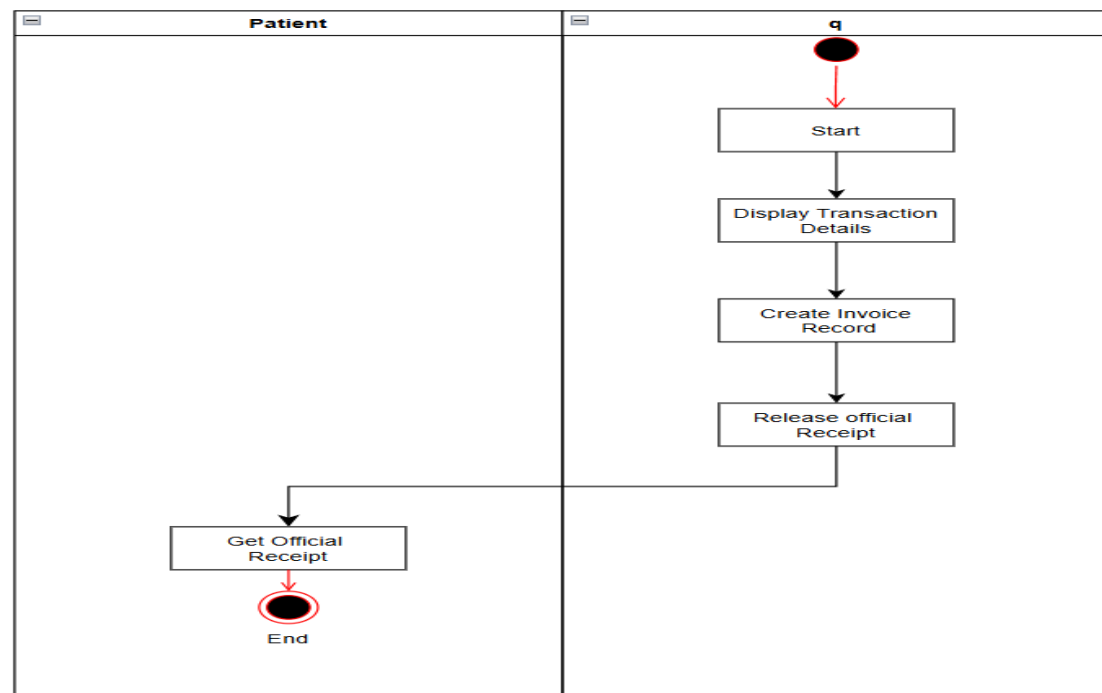


Figure 38. Activity Diagram for Patient & Admission Staff - Release Official Receipt

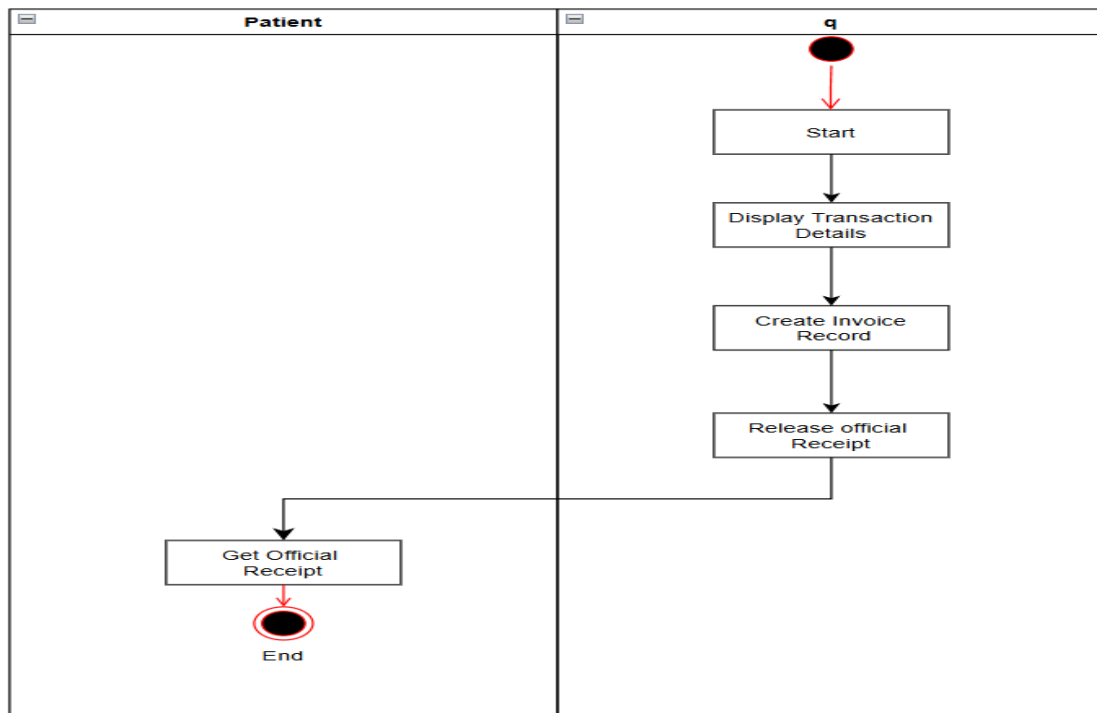


Figure 39. Activity Diagram for chief Medtech - Generate Reports

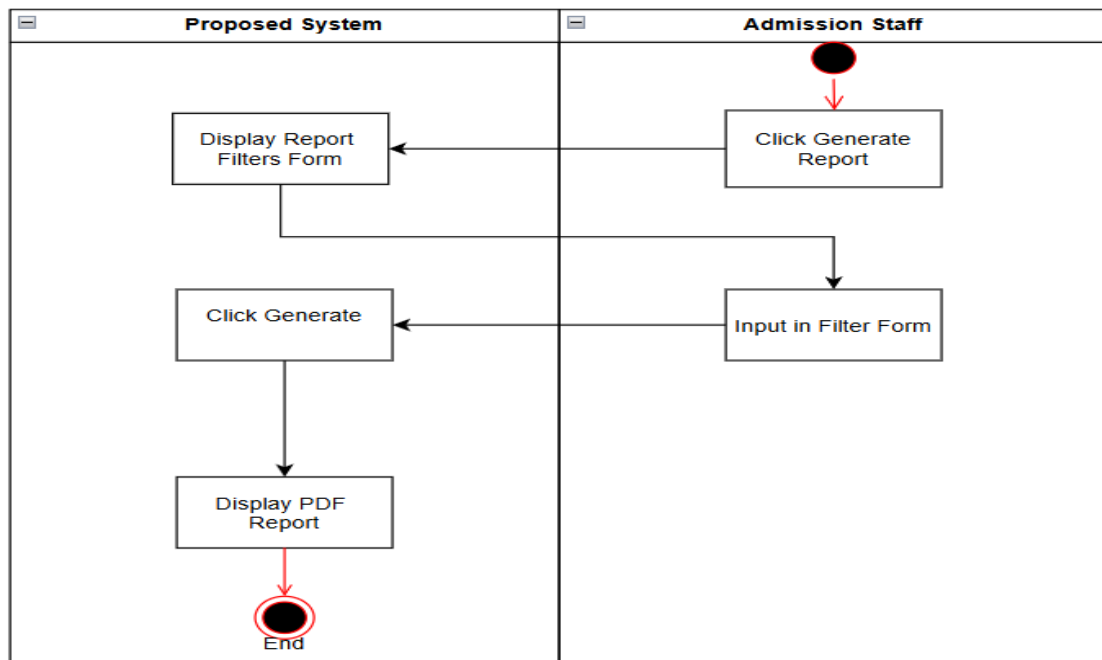


Figure 40. Activity Diagram for Patient - Reports Request

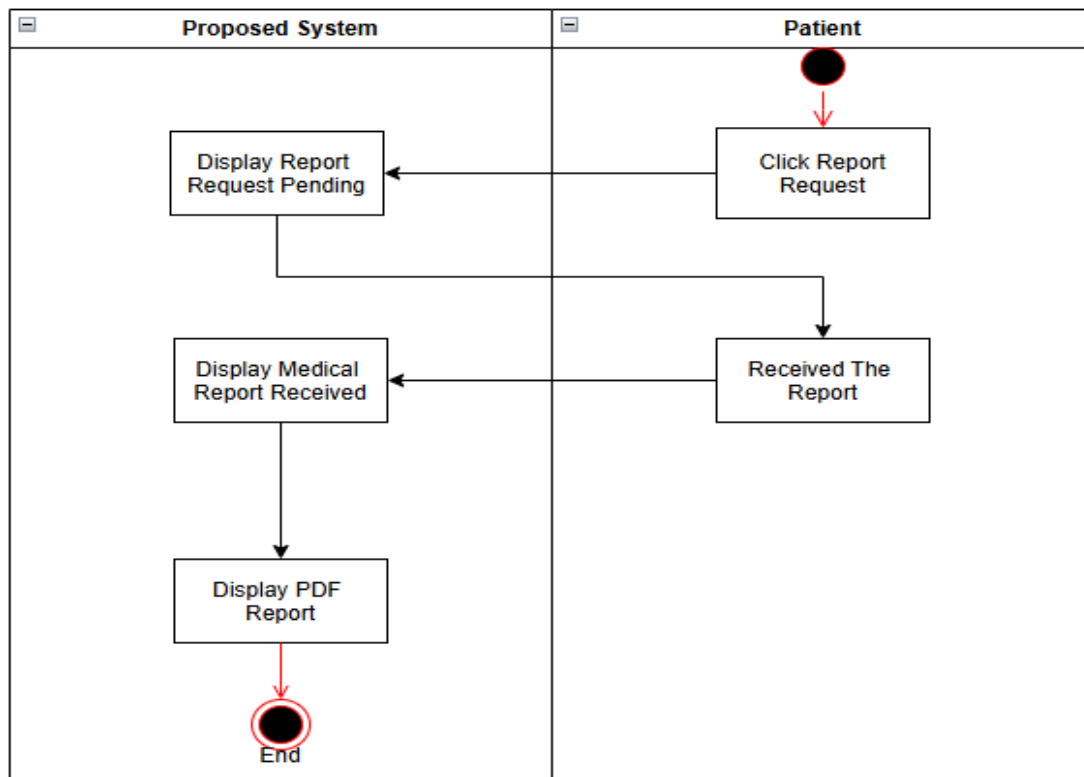


Figure 41. Activity Diagram for chief Medtech - Manage Employee

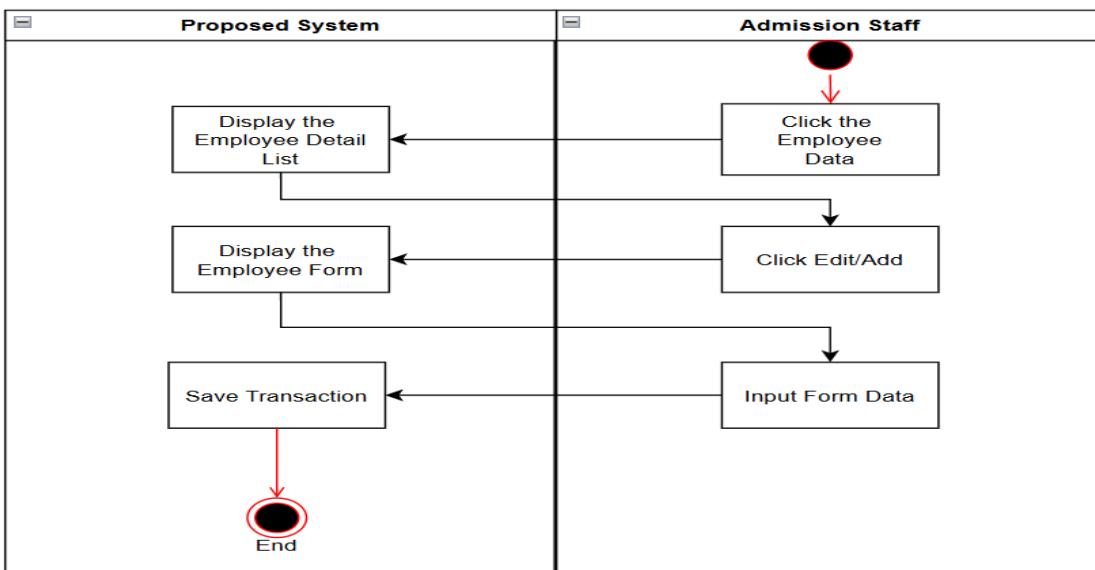


Figure 42. Activity Diagram for chief medtech - Manage Services

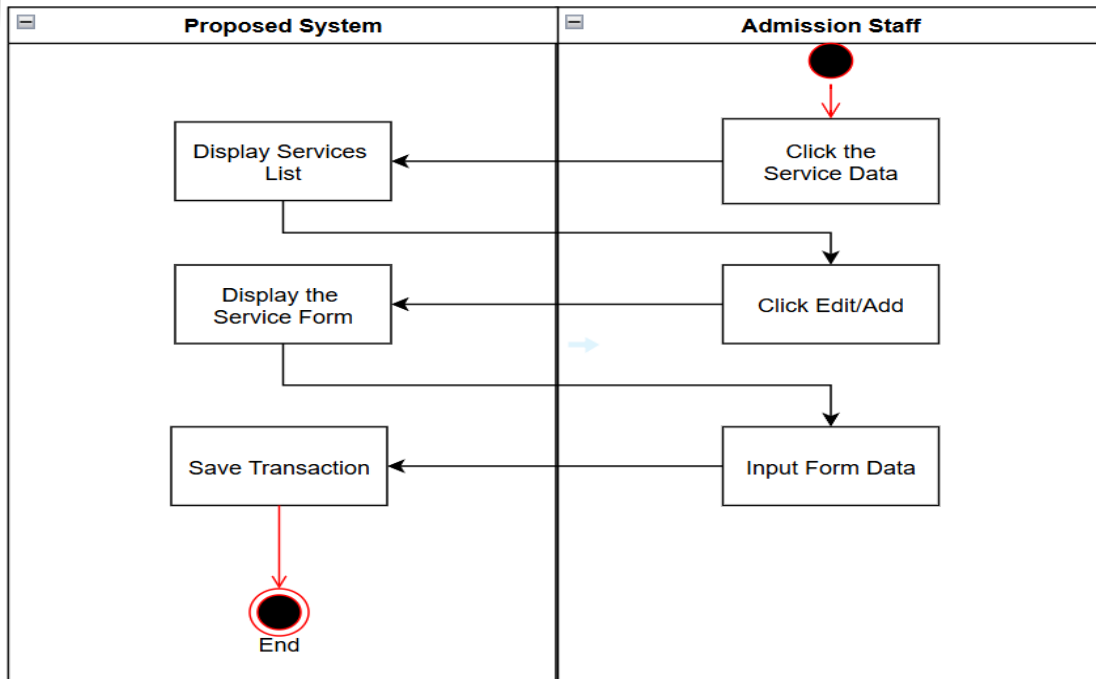


Figure 43. Activity Diagram for chief medtech - Manage General Settings

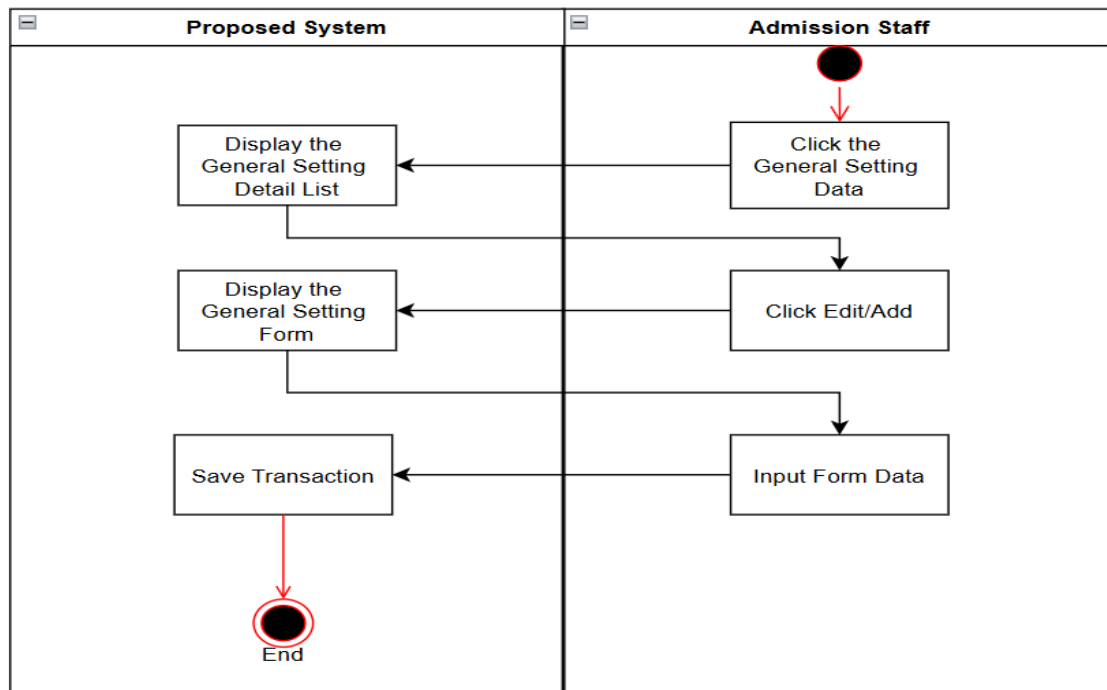


Figure 44. Activity Diagram for chief medtech - Manage Roles

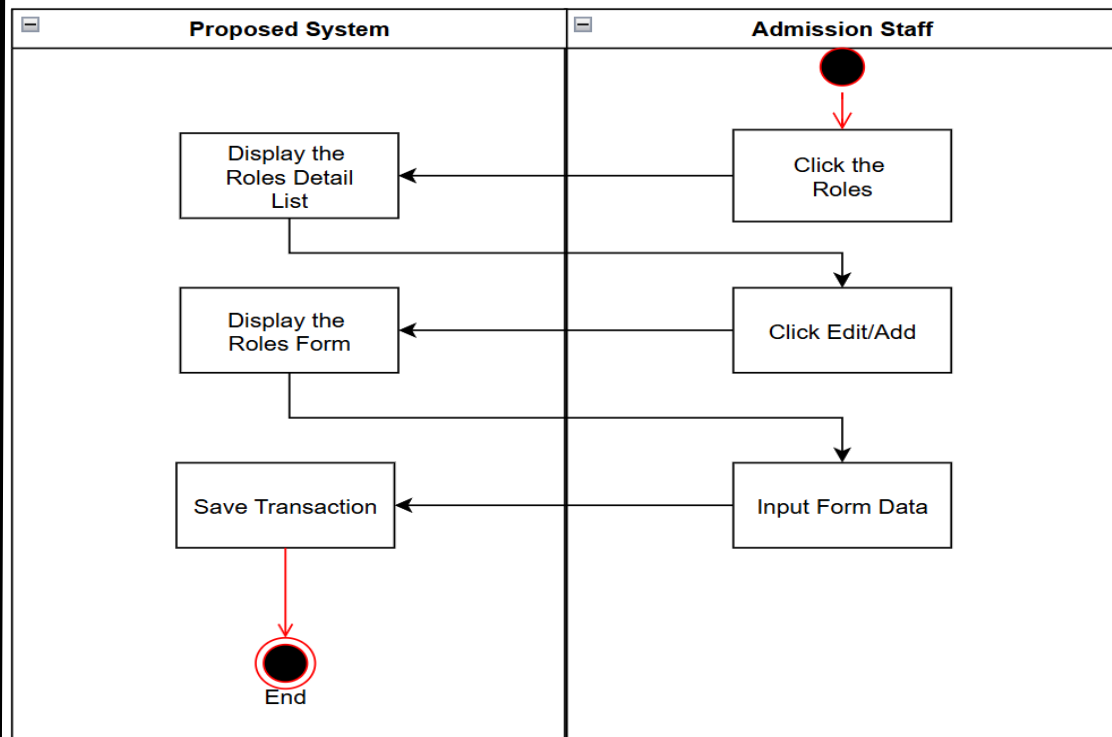


Figure 45. Activity Diagram for chief medtech - Manage Permissions

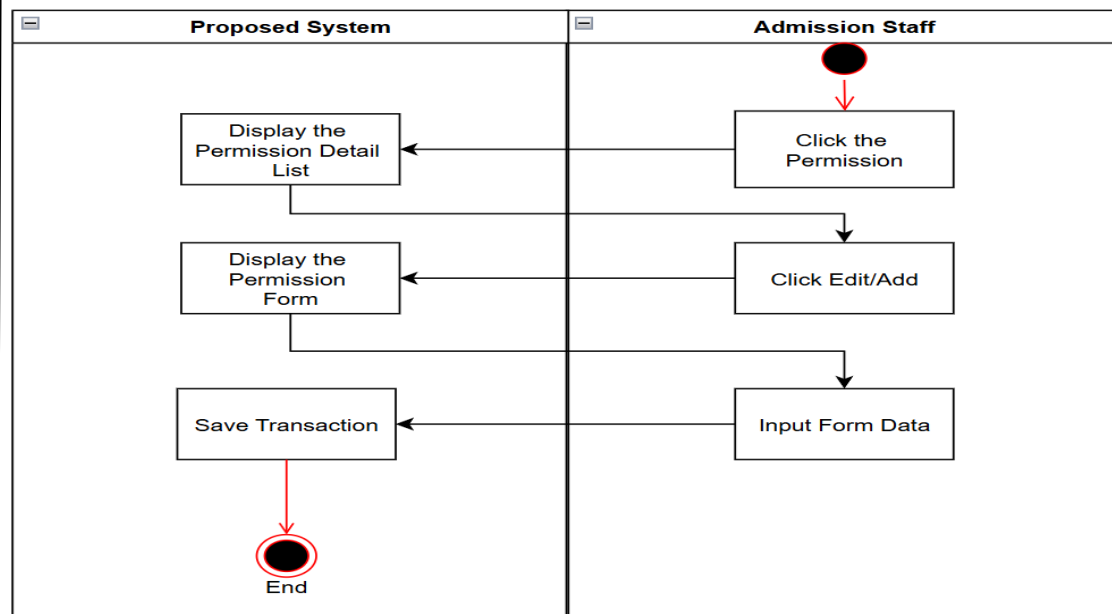


Figure 46. Activity Diagram for chief medtech - Manage Users

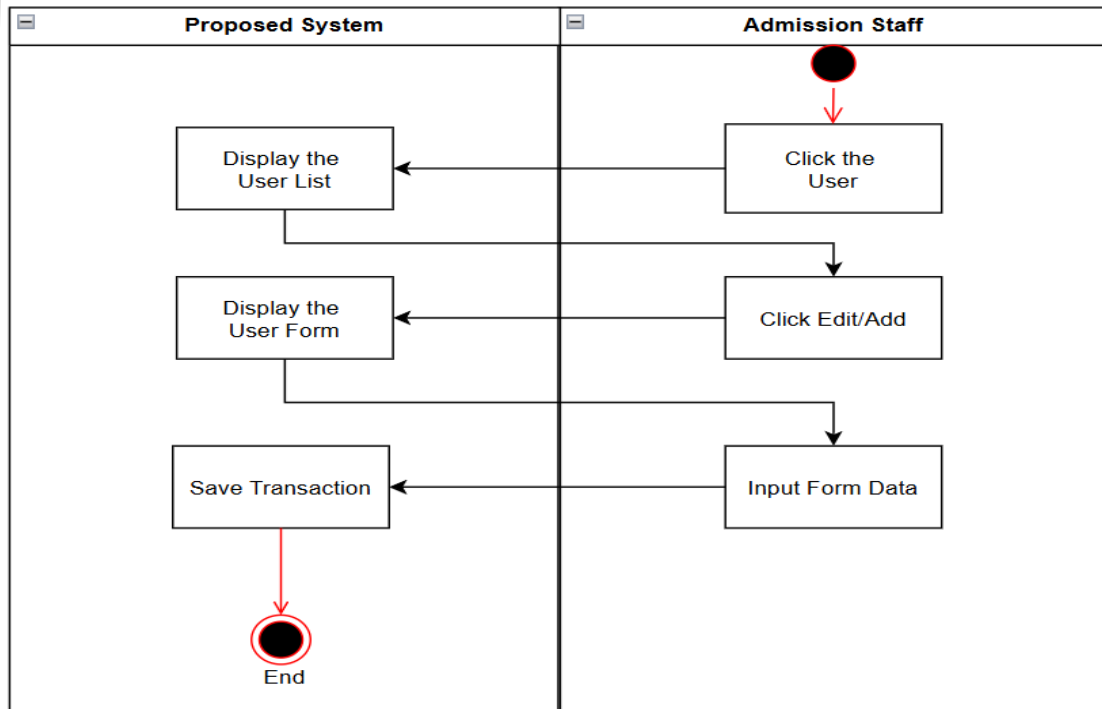


Figure 47. Activity Diagram for chief medtech - Manage Payment Method

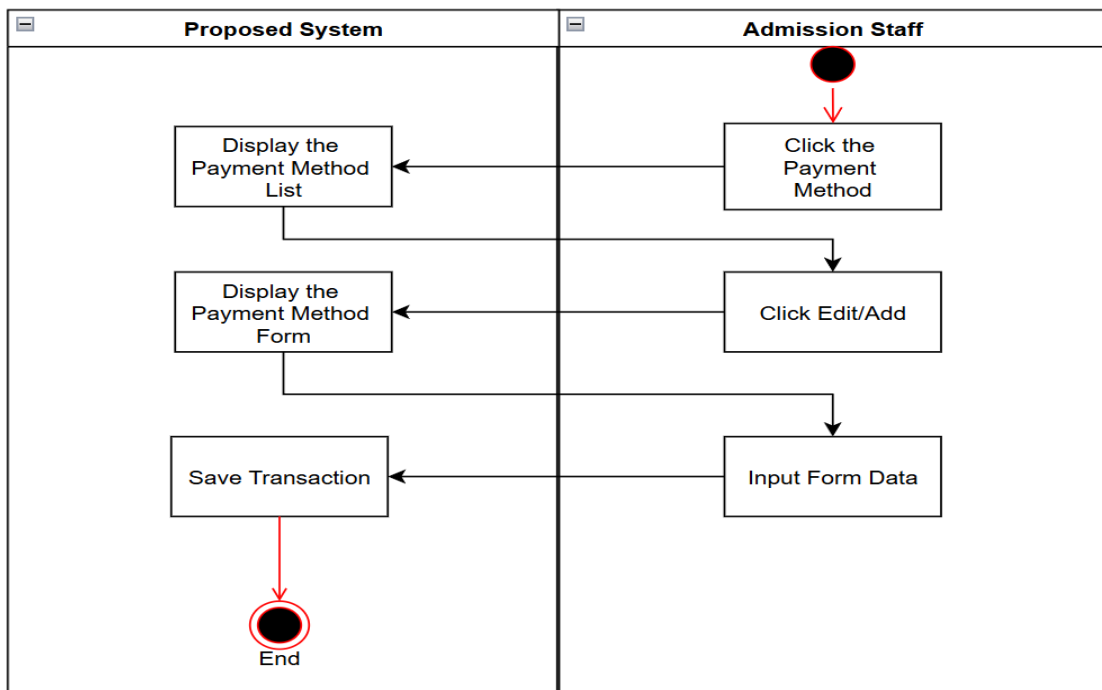


Figure 48. Activity Diagram for chief medtech - Billing Settings

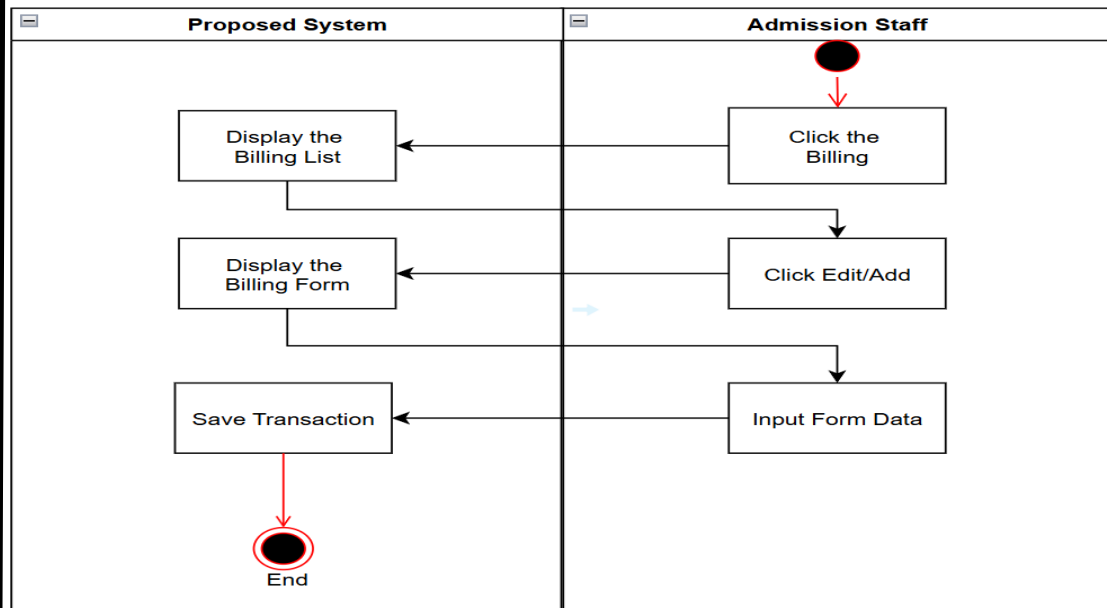
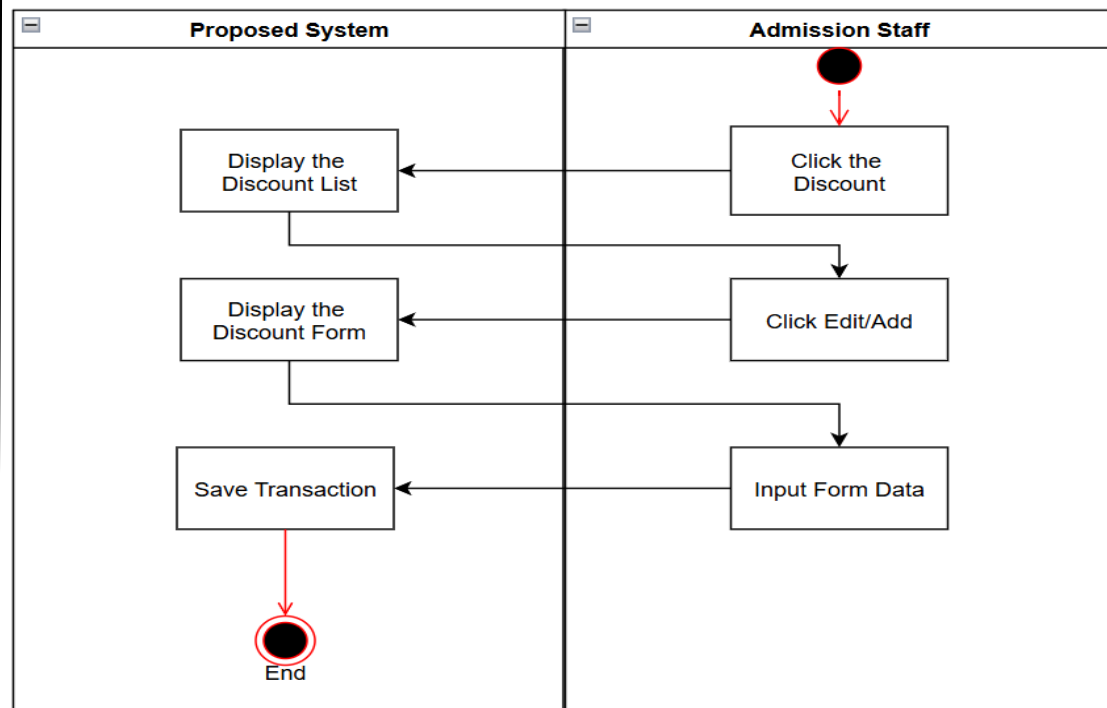


Figure 49. Activity Diagram for chief medtech - Manage Discounts



Data Dictionary

Table 2. Users

Column	Type	Null	Default	Comments
user_id (Primary)	int	No	AUTO_INCR EMENT	Unique user identifier
username	varchar(35)	No		Login username
email	varchar(50)	No		User email address
password_hash	varchar(255)	No		Encrypted password
role_id	int	No		Foreign key to roles table
is_active	tinyint(1)	No	1	Account active status
last_login	datetime	Yes	NULL	Last login timestamp
created_at	timestamp	No	CURRENT_TI MESTAMP	Record creation time
updated_at	timestamp	No	CURRENT_TI MESTAMP	Last update time

Description:

Stores user account information for system authentication and access control. This table manages login credentials, user roles, and account status for all system users including chief medtech s, staff, and other personnel who need access to the system.

Stores user account information for system authentication and access control. This table manages login credentials, user roles, and account status for all system users including chief medtech s, staff, and other personnel who need access to the system.

Table 3. Employee

Column	Type	Null	Default	Comments
employee_id (Primary)	int	No	AUTO_INCREMENT	Unique employee identifier
user_id	int	No		Foreign key to users table
first_name	varchar(30)	No		Employee first name
last_name	varchar(30)	No		Employee last name

middle_name	varchar(30)	Yes	NULL	Employee middle name
employee_code	varchar(20)	No		Unique employee code
department	varchar(50)	Yes	NULL	Department name
position	varchar(50)	No		Job position
contact_number	varchar(20)	Yes	NULL	Contact phone number
address	text	Yes	NULL	Residential address
date_hired	date	No		Date of employment

date_separated	date	Yes	NULL	Date of separation
is_active	tinyint(1)	No	1	Employment status
created_at	timestamp	No	CURRENT_TIMESTAMP	Record creation time
updated_at	timestamp	No	CURRENT_TIMESTAMP	Last update time

Description:

Contains detailed information about employees working in the healthcare facility. This table links to the Users table and stores personal information, employment details, contact information, and employment status. It tracks both active employees and separated staff members.

Table 4. Patients

Column	Type	Null	Default	Comments
patient_id (Primary)	int	No	AUTO_INCREMENT	Unique patient identifier

patient_code	varchar(20)	No		Unique patient code
first_name	varchar(30)	No		Patient first name
last_name	varchar(30)	No		Patient last name
middle_name	varchar(30)	Yes	NULL	Patient middle name
date_of_birth	date	No		Patient birth date
gender	enum('Male','Female','Other')	No		Patient gender
contact_number	varchar(20)	Yes	NULL	Contact phone number
email	varchar(50)	Yes	NULL	Patient email address

address	text	Yes	NULL	Residential address
emergency_co ntact_name	varchar(100)	Yes	NULL	Emergency contact person
emergency_co ntact_number	varchar(20)	Yes	NULL	Emergency contact phone
blood_type	varchar(5)	Yes	NULL	Patient blood type
allergies	text	Yes	NULL	Known allergies
medical_histor y	text	Yes	NULL	Medical history notes
is_active	tinyint(1)	No	1	Patient active status
created_at	timestamp	No	CURRENT_TI MESTAMP	Record creation time

updated_at	timestamp	No	CURRENT_TIMESTAMP	Last update time
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Description:

Maintains comprehensive patient records including personal information, contact details, emergency contacts, and medical information such as blood type, allergies, and medical history. This is the central table for managing all patient data within the healthcare system.

Table 5. Appointments

Column	Type	Null	Default	Comments
appointment_id (Primary)	int	No	AUTO_INCREMENT	Unique appointment identifier
patient_id	int	No		Foreign key to patients table
employee_id	int	Yes	NULL	Assigned staff/doctor
appointment_date	date	No		Scheduled appointment date

appointment_time	time	No		Scheduled appointment time
appointment_type	varchar(50)	Yes	NULL	Type of appointment
status	enum('Scheduled', 'Confirmed', 'In Progress', 'Completed', 'Cancelled', 'No Show')	No	'Scheduled'	Appointment status
reason	text	Yes	NULL	Reason for appointment
notes	text	Yes	NULL	Additional notes
created_by	int	No		User who created the appointment

created_at	timestamp	No	CURRENT_TIMESTAMP	Record creation time
updated_at	timestamp	No	CURRENT_TIMESTAMP	Last update time

Description:

Manages all patient appointments including scheduling information, appointment types, status tracking, and associated staff members. This table handles the complete appointment lifecycle from scheduling through completion or cancellation.

Table 6. Queues

Column	Type	Null	Default	Comments
queue_id (Primary)	int	No	AUTO_INCREMENT	Unique queue identifier
patient_id	int	No		Foreign key to patients table
appointment_id	int	Yes	NULL	Related appointment if any

queue_number	varchar(20)	No		Queue ticket number
queue_date	date	No		Queue date
queue_time	time	No		Queue entry time
service_type	varchar(50)	Yes	NULL	Type of service
priority	enum('Normal', 'Urgent', 'Emergency')	No	'Normal'	Queue priority level
status	enum('Waiting', 'Called', 'Serving', 'Completed', 'Cancelled')	No	'Waiting'	Queue status
called_time	datetime	Yes	NULL	Time patient was called

serving_time	datetime	Yes	NULL	Time service started
completed_time	datetime	Yes	NULL	Time service completed
served_by	int	Yes	NULL	Employee who served
created_at	timestamp	No	CURRENT_TIMESTAMP	Record creation time
updated_at	timestamp	No	CURRENT_TIMESTAMP	Last update time

Description:

Tracks the real-time queue management system for patient services. This table monitors patients waiting for service, their queue numbers, priority levels, and tracks the progression from waiting to service completion. It records timestamps for each stage of the queue process.

Table 7. Invoices

Column	Type	Null	Default	Comments
--------	------	------	---------	----------

id (Primary)	bigint	No	AUTO_INCREMENT	Unique invoice identifier
invoice_number	varchar(255)	No		Invoice reference number
transaction_id	bigint	No		Related transaction
total_amount	float	No		Total amount billed
total_discount	float	No	0	Total amount of discount
amount_paid	float	No	0	Amount paid for invoice
grand_total	float	No	0	Final total after discount
change	float	No	0	Change given after payment

discount_id	bigint	Yes	NULL	Linked discount
created_by	bigint	No		User who created invoice
is_paid	tinyint(1)	No	0	Invoice paid status
deleted_at	timestamp	Yes	NULL	Deletion timestamp
created_at	timestamp	Yes	NULL	Record creation time
updated_at	timestamp	Yes	NULL	Last update time

Description:

Manages billing invoices for patient services including itemized charges, discounts, taxes, and payment tracking. This table maintains the financial records of services rendered, tracks payment status, and calculates outstanding balances.

Table 8. Patient Tests

Column	Type	Null	Default	Comments
--------	------	------	---------	----------

patient_test_id (Primary)	int	No	AUTO_INCR EMENT	Unique test record identifier
patient_id	int	No		Foreign key to patients table
appointment_id	int	Yes	NULL	Related appointment
test_code	varchar(20)	No		Test code/identifier
test_name	varchar(100)	No		Name of the test
test_category	varchar(50)	Yes	NULL	Test category/type
ordered_by	int	No		Employee who ordered test
ordered_date	datetime	No	CURRENT_TIMESTAMP	Date test was ordered

sample_collected	tinyint(1)	No	0	Sample collection status
collection_date	datetime	Yes	NULL	Sample collection date
status	enum('Ordered', 'Sample Collected', 'In Progress', 'Completed', 'Cancelled')	No	'Ordered'	Test status
priority	enum('Routine', 'Urgent', 'STAT')	No	'Routine'	Test priority
notes	text	Yes	NULL	Test notes
created_at	timestamp	No	CURRENT_TIMESTAMP	Record creation time

updated_at	timestamp	No	CURRENT_TIMESTAMP	Last update time
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Description:

Records all laboratory tests and diagnostic procedures ordered for patients. This table tracks the complete test workflow from order placement through sample collection to completion, including test priorities and current status.

Table 9. Transactions

Column	Type	Null	Default	Comments
transaction_id (Primary)	int	No	AUTO_INCREMENT	Unique transaction identifier
transaction_number	varchar(50)	No		Unique transaction number
invoice_id	int	No		Foreign key to invoices table
patient_id	int	No		Foreign key to patients table

transaction_date	datetime	No	CURRENT_TIMESTAMP	Transaction date and time
payment_method_id	int	No		Payment method used
amount	decimal(10,2)	No	0.00	Transaction amount
reference_number	varchar(100)	Yes	NULL	Payment reference number
status	enum('Pending', 'Completed', 'Failed', 'Refunded')	No	'Completed'	Transaction status
processed_by	int	No		Employee who processed

notes	text	Yes	NULL	Transaction notes
created_at	timestamp	No	CURRENT_TIMESTAMP	Record creation time
updated_at	timestamp	No	CURRENT_TIMESTAMP	Last update time

Description:

Records all financial transactions and payments made in the system. This table tracks payment methods, transaction amounts, references, and links payments to their corresponding invoices while maintaining transaction status and processing information

Table 10. Audit Trails

Column	Type	Null	Default	Comments
audit_id (Primary)	bigint	No	AUTO_INCREMENT	Unique audit record identifier

user_id	int	Yes	NULL	User who performed action
table_name	varchar(50)	No		Database table affected
record_id	int	Yes	NULL	ID of affected record
action	enum('CREATE','UPDATE','DELETE','LOGIN','LOGOUT','VIEW')	No		Action performed
old_values	json	Yes	NULL	Previous values (for updates)
new_values	json	Yes	NULL	New values
ip_address	varchar(45)	Yes	NULL	User IP address

user_agent	varchar(255)	Yes	NULL	Browser/client information
timestamp	timestamp	No	CURRENT_TIMESTAMP	Action timestamp

Description:

Provides comprehensive logging and tracking of all system activities for security, compliance, and accountability purposes. This table records user actions, data changes, login/logout events, and maintains a complete audit history with IP addresses and timestamps.

Table 11. Tests Results

Column	Type	Null	Default	Comments
result_id (Primary)	int	No	AUTO_INCREMENT	Unique result identifier
patient_test_id	int	No		Foreign key to patient_tests
parameter_name	varchar(100)	No		Test parameter name

result_value	varchar(255)	Yes	NULL	Test result value
unit	varchar(20)	Yes	NULL	Unit of measurement
reference_range	varchar(100)	Yes	NULL	Normal reference range
flag	enum('Normal', 'Low', 'High', 'Critical')	Yes	NULL	Result flag
result_date	datetime	No	CURRENT_TIMESTAMP	Date result was obtained
verified_by	int	Yes	NULL	Employee who verified result
verified_date	datetime	Yes	NULL	Verification date

notes	text	Yes	NULL	Result notes
created_at	timestamp	No	CURRENT_TIMESTAMP	Record creation time
updated_at	timestamp	No	CURRENT_TIMESTAMP	Last update time

Description:

Stores detailed results for laboratory tests and diagnostic procedures. This table maintains test parameters, result values, reference ranges, abnormal flags, and verification information to ensure accurate and validated test reporting.

Table 12. Services

Column	Type	Null	Default	Comments
service_id (Primary)	int	No	AUTO_INCREMENT	Unique service identifier
service_code	varchar(20)	No		Unique service code

service_name	varchar(100)	No		Service name
service_category	varchar(50)	Yes	NULL	Service category
description	text	Yes	NULL	Service description
base_price	decimal(10,2)	No	0.00	Base service price
duration_minutes	int	Yes	NULL	Expected duration
is_active	tinyint(1)	No	1	Service availability
requires_appointment	tinyint(1)	No	1	Appointment required flag

created_at	timestamp	No	CURRENT_TIMESTAMP	Record creation time
updated_at	timestamp	No	CURRENT_TIMESTAMP	Last update time

Description:

Catalogs all medical services, procedures, and treatments offered by the healthcare facility.

This table defines service details, pricing, duration, availability, and whether appointments are required for each service type.

Table 13. Booked Services

Column	Type	Null	Default	Comments
booked_service_id (Primary)	int	No	AUTO_INCREMENT	Unique booking identifier
patient_id	int	No		Foreign key to patients table
appointment_id	int	Yes	NULL	Related appointment

invoice_id	int	Yes	NULL	Related invoice
service_id	int	No		Foreign key to services table
service_date	date	No		Date service was rendered
quantity	int	No	1	Service quantity
unit_price	decimal(10,2)	No	0.00	Price per unit
total_price	decimal(10,2)	No	0.00	Total price
discount_amount	decimal(10,2)	No	0.00	Discount applied
final_price	decimal(10,2)	No	0.00	Final price after discount

status	enum('Pending', 'Completed', 'Cancelled')	No	'Pending'	Service status
performed_by	int	Yes	NULL	Employee who performed service
notes	text	Yes	NULL	Service notes
created_by	int	No		User who created record
created_at	timestamp	No	CURRENT_TIMESTAMP	Record creation time
updated_at	timestamp	No	CURRENT_TIMESTAMP	Last update time

Description:

Records all services that have been booked or rendered to patients. This table links patients, appointments, and invoices while tracking service quantities, pricing, discounts, and completion status. It maintains a comprehensive record of all patient services.

Table 14. General Settings

Column	Type	Null	Default	Comments
setting_id (Primary)	int	No	AUTO_INCREMENT	Unique setting identifier
setting_key	varchar(100)	No		Setting key name
setting_value	text	Yes	NULL	Setting value
setting_type	enum('text','number','boolean','json','date')	No	'text'	Data type of setting
category	varchar(50)	Yes	NULL	Setting category
description	varchar(255)	Yes	NULL	Setting description
is_editable	tinyint(1)	No	1	Can be edited by users

created_at	timestamp	No	CURRENT_TIMESTAMP	Record creation time
updated_at	timestamp	No	CURRENT_TIMESTAMP	Last update time

Description:

Stores system-wide configuration parameters and settings. This flexible table manages various application settings with different data types, allowing chief medtech s to configure system behavior, features, and preferences without code changes.

Table 15. Roles

Column	Type	Null	Default	Comments
role_id (Primary)	int	No	AUTO_INCREMENT	Unique role identifier
role_name	varchar(50)	No		Role name
role_code	varchar(20)	No		Unique role code

description	text	Yes	NULL	Role description
is_active	tinyint(1)	No	1	Role active status
created_at	timestamp	No	CURRENT_TI MESTAMP	Record creation time
updated_at	timestamp	No	CURRENT_TI MESTAMP	Last update time

Description:

Defines user roles and their hierarchical structure within the system. This table manages role-based access control by categorizing users into different roles (e.g., Admin, Medtech, Staff, Patient) with unique permissions and responsibilities.

Table 16. Permissions

Column	Type	Null	Default	Comments

permission_id (Primary)	int	No	AUTO_INCREMENT	Unique permission identifier
permission_name	varchar(100)	No		Permission name
permission_code	varchar(50)	No		Unique permission code
module	varchar(50)	Yes	NULL	System module
description	text	Yes	NULL	Permission description
created_at	timestamp	No	CURRENT_TIMESTAMP	Record creation time
updated_at	timestamp	No	CURRENT_TIMESTAMP	Last update time

Description:

Maintains the complete list of system permissions and access rights. This table defines granular permissions for different modules and functionalities, which can be assigned to roles to control what actions users can perform in the system

Table 17. Payment Methods

Column Name	Type	Null	Default	Comments
id (Primary)	bigint	No		
name	varchar(100)	No		
is_health_card	tinyint(1)	No	0	
deleted_at	timestamp	Yes	NULL	

Description:

Catalogs all available payment methods accepted by the healthcare facility. This table manages payment options such as cash, credit card, bank transfer, and insurance, including their configuration, reference requirements, and display order.

Table 18. Billing Settings

Column	Type	Null	Default	Comments
--------	------	------	---------	----------

billing_setting _id (Primary)	int	No	AUTO_INCR EMENT	Unique setting identifier
setting_name	varchar(100)	No		Setting name
setting_key	varchar(50)	No		Unique setting key
setting_value	text	Yes	NULL	Setting value
data_type	enum('text','nu mber','boolean' , 'decimal','json')	No	'text'	Value data type
description	varchar(255)	Yes	NULL	Setting description

is_active	tinyint(1)	No	1	Setting active status
created_at	timestamp	No	CURRENT_TIMESTAMP	Record creation time
updated_at	timestamp	No	CURRENT_TIMESTAMP	Last update time

Description:

Stores configuration settings specific to the billing and financial modules. This table manages billing-related parameters such as tax rates, default payment terms, invoice numbering schemes, and other financial system configurations.

Table 19. Discounts

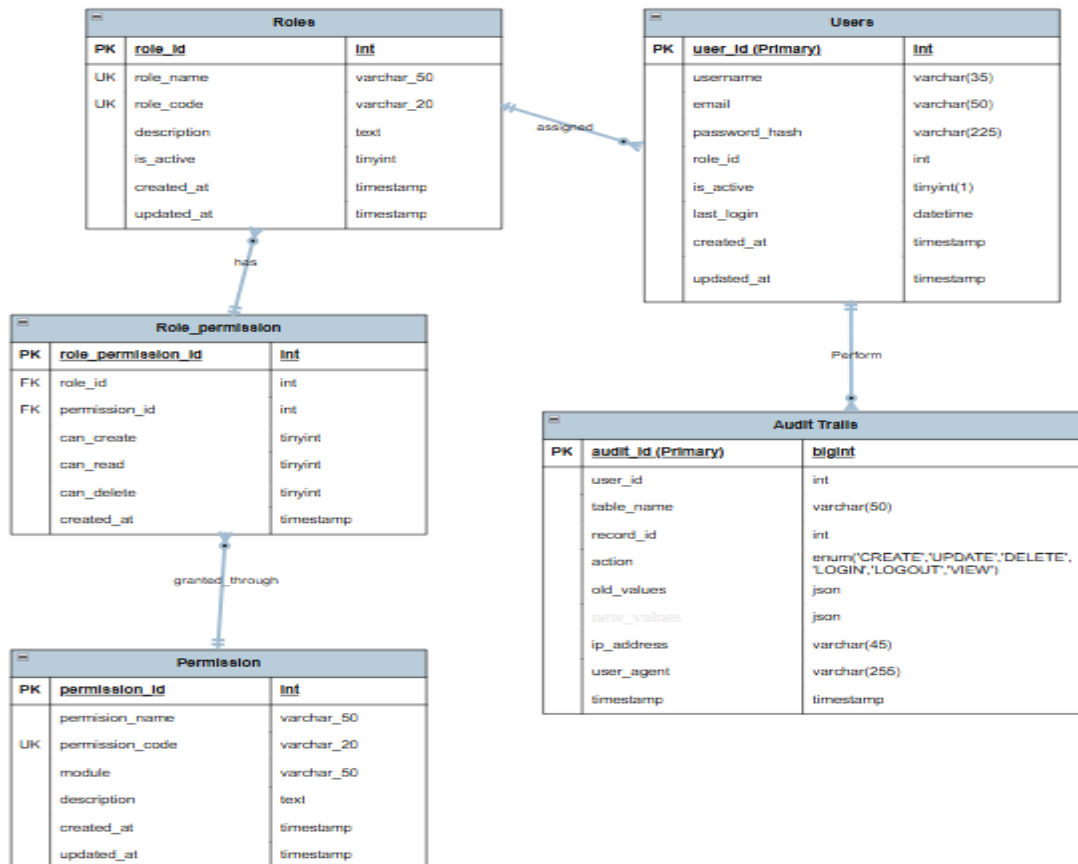
Column Name	Type	Null	Default	Comments
id (Primary)	bigint	No		
name	varchar(100)	No		
percentage	varchar(10)	No		
deleted_at	timestamp	Yes	NULL	

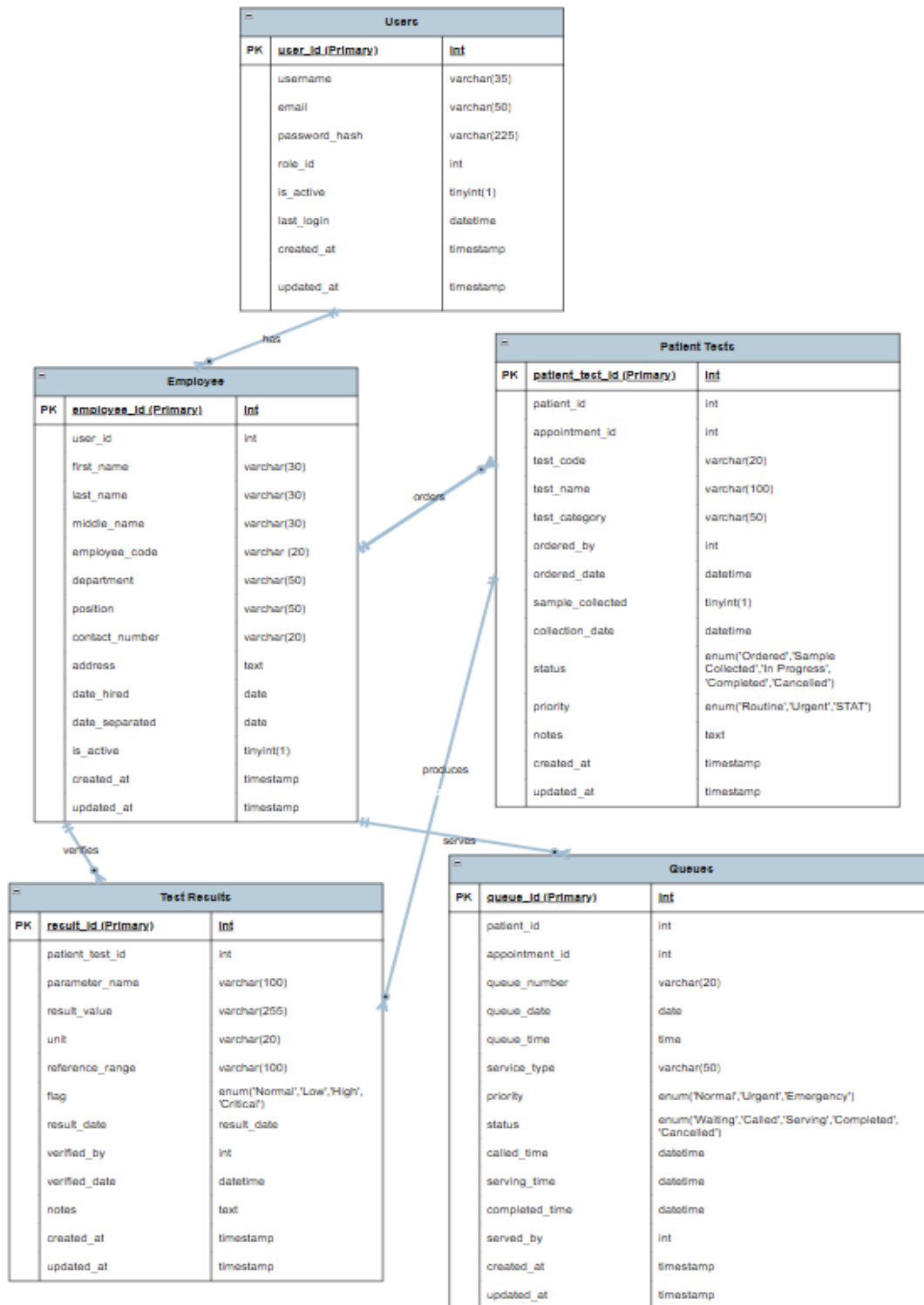
Description:

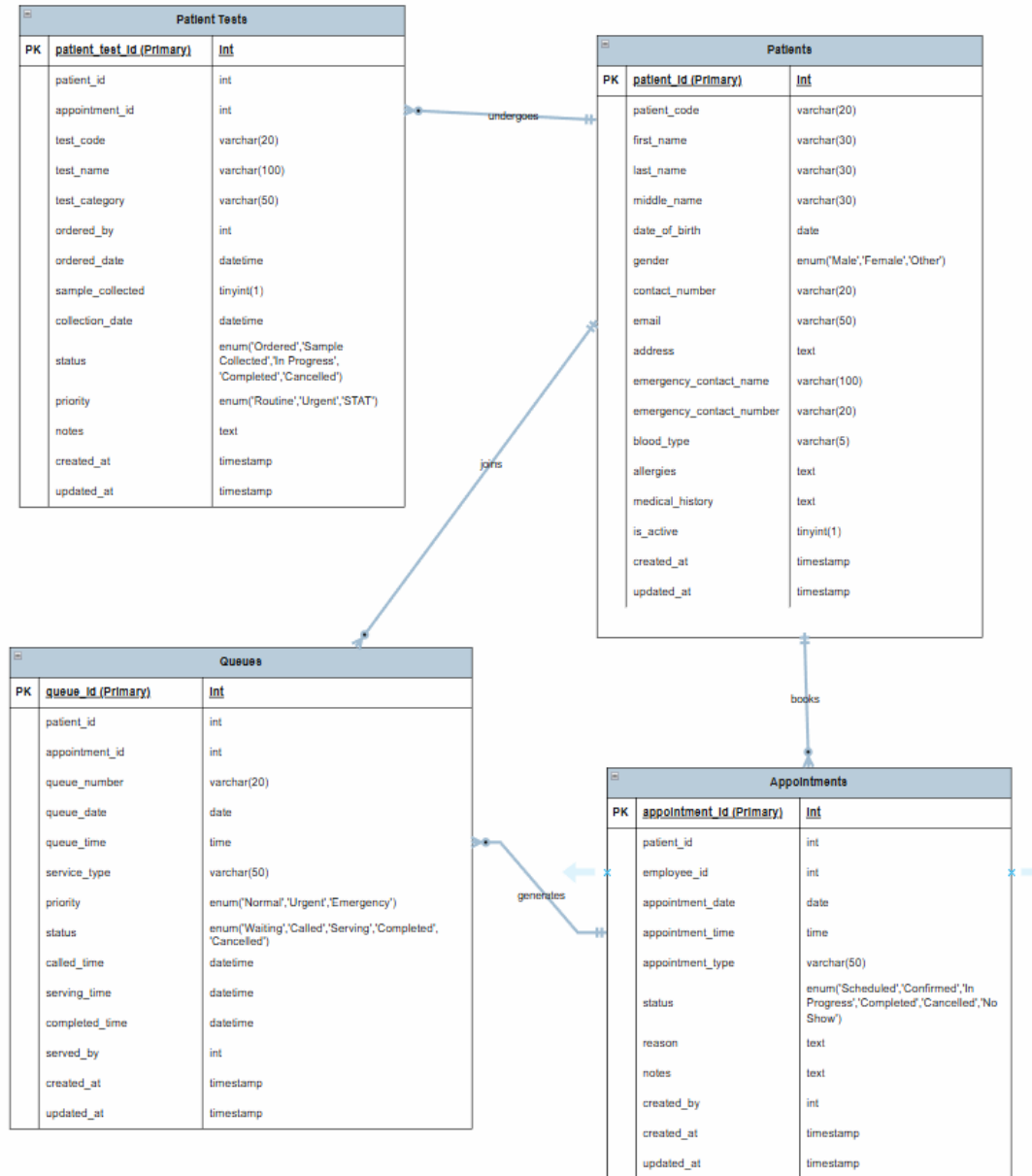
Manages discount programs and promotional offers available to patients. This table defines discount rates, validity periods, and status to support various pricing strategies such as senior citizen discounts, promotional offers, or insurance-based reductions.

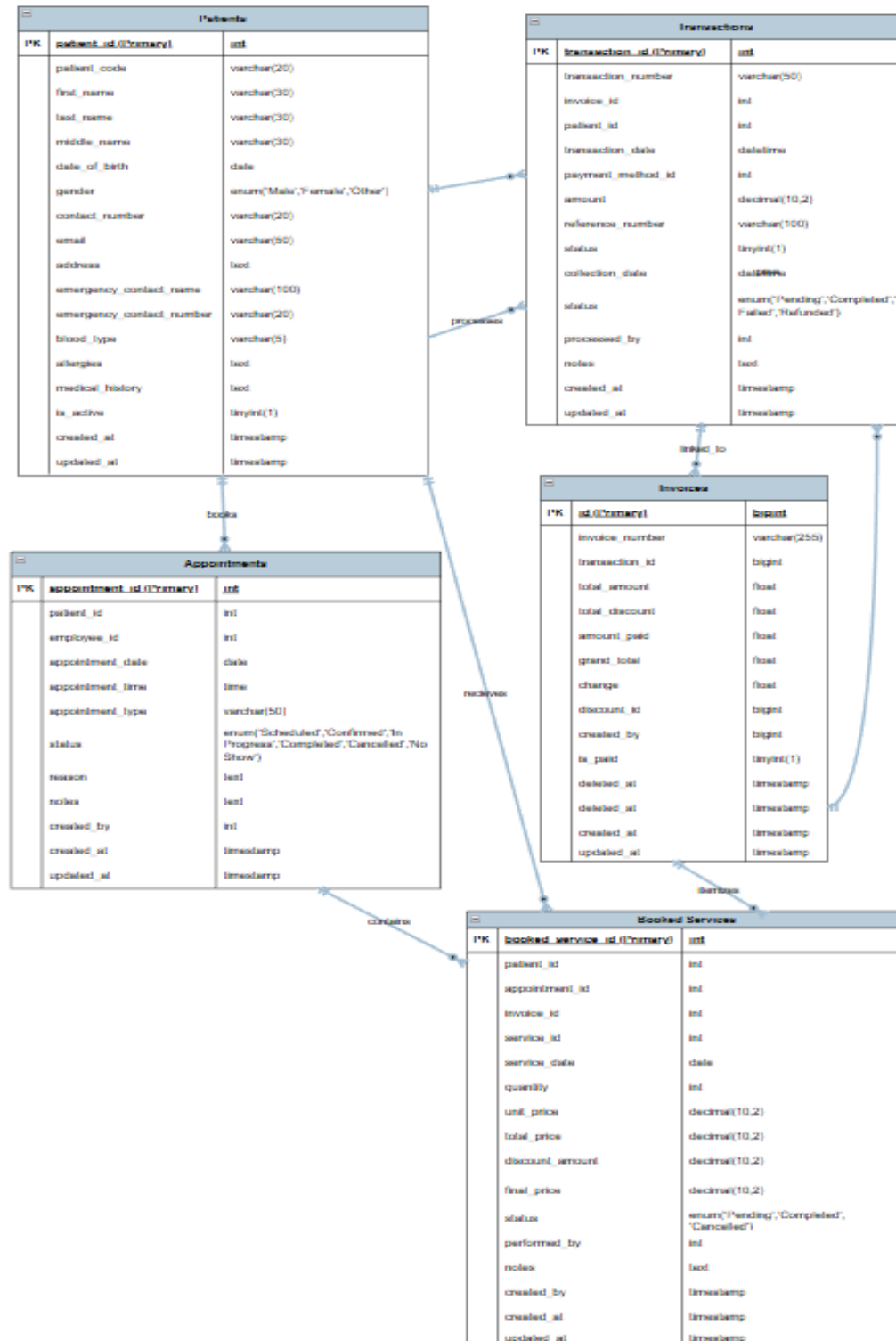
Entity Relationship Diagram

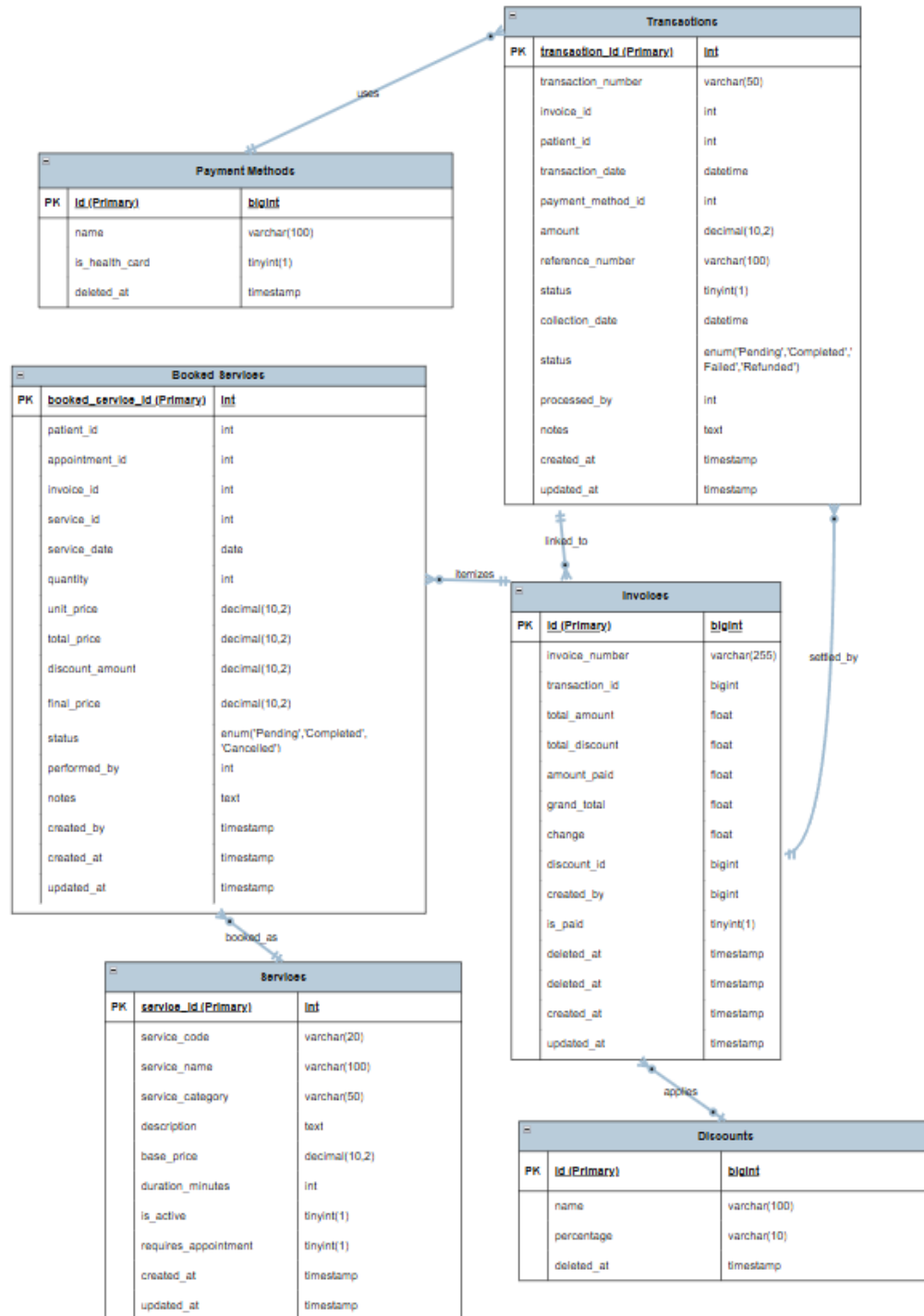
Figure 50. Entity Relationship Diagram











3.2 Conceptual Framework of the Study

The conceptual framework contains the methods and requirements for developing the Cloud-based Integrated Management System for ABR Diagnostic Center.

Figure 51. Conceptual Framework of the Study

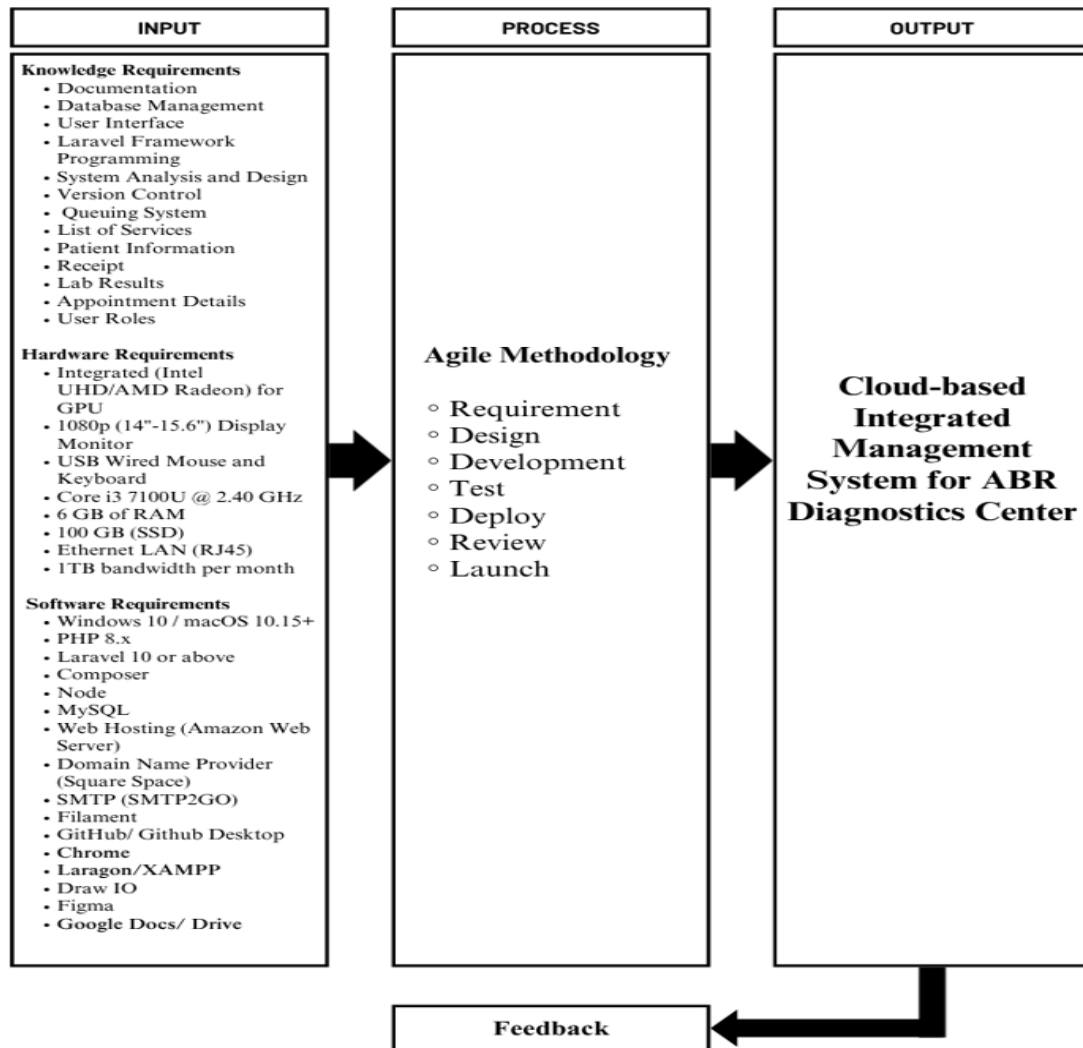


Figure 51 illustrates how the system will be constructed. The conceptual framework is divided into four components: input, process, output, and feedback. The input phase

consists of knowledge, hardware, a web server, software, and data requirements. These are essential for developing the ABR Diagnostic Integrated Management System.

The knowledge requirements define the foundational information necessary for system development, including the principles of documentation, database management, user interface, Laravel framework programming, system analysis and design, version control, and queueing system. Additionally, this phase assesses the technical expertise required to design and implement the system. The hardware requirements consist of the physical devices needed to operate the system, such as a GPU, monitor, mouse, and keyboard, while the software requirements include the development tools and platform used to create the system, such as Laravel, Laragon, and GitHub. The web server is used in hosting websites, processing backend logic, communicating with databases, load balancing and scaling, and for security and optimization. The web server requirements are a set of minimum specifications required to run a web server consistently and smoothly. Furthermore, the data requirements consist of the client's provided information being integrated into the system. The process phase outlines the software development method used in the system. The proponents used the Agile methodology, which allows flexibility, teamwork, and step-by-step development of the system. In this method, the system is built in parts called sprints, and after each sprint, feedback from the client is used to improve the next phase. To make sure the system works well; the process includes system implementation, system testing, and system maintenance. After these steps, the system will be ready for deployment and actual use.

As for the output, ABR Integrated Management System delivers comprehensive outputs across four areas. A platform that includes patient records, making appointments, processing test results and reports, billing with receipt generation, and safe access controls based on roles. These features measure outcomes to digitize workflows while boosting efficiency, security, and service quality.

The system's feedback creates a continuous improvement cycle through multiple user surveys, technical monitoring, and performance tracking. Issues are categorized by type, priority, and whether they need immediate fixes, short-term updates, or long-term updates. Transparent communication and validation with users ensure changes effectively address needs, creating an adaptive system that evolves with the clinic's demands while maintaining its reliability and user satisfaction standards through its optimization cycles.

Hardware and Software Requirements

Table 20. Hardware Requirements

Hardware Components	Minimum Requirements	Recommended Requirements
CPU	Dual-core 2.0GHz (Intel i3 / Ryzen 3)	Quad-core 2.5GHz+ (Intel i5 / Ryzen 5)
RAM	8 GB	16 GB

Storage	256 GB HDD	512 GB HDD
GPU	Integrated (Intel UHD/AMD Radeon)	Dedicated (NVIDIA MX/AMD Radeon)
Display	1080p (14"-15.6")	1080p IPS (15.6"-16")

Table 20 outlines the hardware requirements that are crucial for ensuring optimal performance during website development and deployment. For the CPU, a dual-core processor like Intel i3 or Ryzen 3 meets the minimum requirements for basic coding and testing, while a quad-core processor such as Intel i5 or Ryzen 5 is recommended for more tasks like handling large databases. When it comes to RAM, 8GB is sufficient for running a local development environment and a browser, but 16 GB is better for smoother multitasking, especially when using tools like XAMPP. For storage, a 256 GB HDD provides enough space for essential files, though a 512 GB HDD is preferable for larger projects and databases. For the GPU, integrated graphics like Intel UHD or AMD Radeon are adequate for general development, but a dedicated GPU such as NVIDIA MX or AMD Radeon is better for tasks involving UI/UX design. Finally, for the display, a standard 1080p screen is enough for basic use, but an IPS panel offers improved color accuracy and viewing angles, which is beneficial for design and debugging.

Table 21. Software Requirements

SOFTWARE COMPONENTS	USED SOFTWARE
Operating System	Windows 11
Documentation	Google Docs, Google Drive
Browser	Chrome
Web Hosting	Amazon Web Server
Domain Name Provider	Square Space
Database Designer	MySQL, Draw IO
Local Environment Development	Laragon/XAMPP
User Interface Design	Figma
Web Development	PHP 8.x, Laravel, Filament
Version Control	GitHub/GitHub Desktop
Package Manager	Composer, Node
Email Log Storing	SMTP2GO

Table 2 outlines the essential software tools and requirements for website development, covering everything from coding to deployment. The operating system used is Windows

11, though there are alternatives that can be used depending on the developer's preference. For documentation and collaboration, Google Docs and Google Drive provide cloud-based solutions for real-time editing and file storage, while Chrome serves as the primary browser for testing and research. Web hosting is handled through Amazon Web Server, offering an integrated solution for deploying and managing the website online. Database design utilizes MySQL for rational database management, complemented by Draw.io for creating visual database schemas and diagrams. The local development environment is supported by Laragon/XAMPP, which streamlines server setup for testing PHP and database applications locally. For user interface design, Figma is used for collaborative design prototyping. The web development stack includes PHP 8.x as the backend language, with Laravel providing an MVC framework (model-view-controller framework) and Filament serving as an admission panel builder. Version control is managed through GitHub or GitHub Desktop, facilitating code collaborations and tracking. For package management, it is handled by Composer for PHP packages and Node.js for the development approach. Finally, SMTP2GO replaces generic SMTP for email log storage, offering a more specialized and reliable solution for handling transactional emails.

3.3 Data Gathering Tools and Procedures

The data used in this study primarily focused on the operational workflows of the ABR Diagnostic Center. The researchers gathered information regarding patient registration, diagnostic test queuing, appointment scheduling, billing, and the generation of medical test results. These details served as the foundation for identifying the inefficiencies in the

current manual or semi-automated processes and for determining the necessary features for the proposed ABR Diagnostic Integrated Management System. The researchers also examined the existing methods of record-keeping and diagnostic reporting to understand the practical challenges that needed to be addressed through a centralized digital solution.

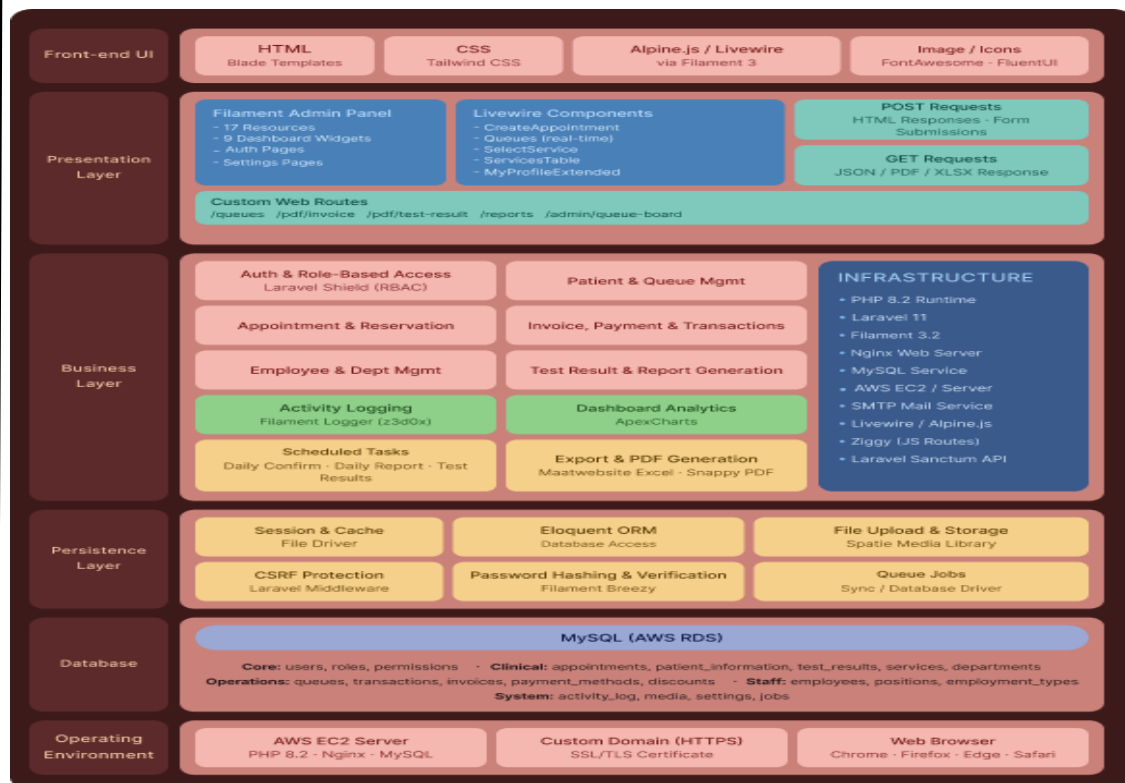
The key individuals involved in the data gathering process were the management and clinical staff of the diagnostic center, including medical technologists and administrative personnel. Their participation was vital in providing accurate information about the daily challenges encountered in managing patient queues and diagnostic data. The conduct of the study was coordinated with the center's administration to ensure that the data collected was aligned with the facility's standards and operational requirements.

To collect the necessary information, the researchers utilized the interview method. These sessions were conducted to gather detailed insights into the existing processes, common bottlenecks in patient flow, and specific administrative needs. Through these interviews, the researchers were able to define the system requirements and identify how the digital platform could improve overall efficiency. Following the development of the system, a survey questionnaire was used as an evaluation tool to assess the quality and acceptability of the ABR Diagnostic Integrated Management System. The questionnaire was based on the ISO/IEC 25010 software quality model, which served as the framework for evaluating the system's performance and usability.

The procedure began with securing the necessary permissions from the diagnostic center. Once granted, the researchers conducted scheduled interviews with identified staff members to map out the current workflow and technical issues. After analyzing the gathered information, the results were used to design and develop the system. When the application was completed, it was presented to the intended users for evaluation through a survey. This systematic approach ensured that the ABR Diagnostic Integrated Management System was grounded in actual user needs and was professionally evaluated for its effectiveness.

3.4 Project Design

Figure 52. Project Design



The project design of the ABR Diagnostic Integrated Management System follows a layered architecture where each layer performs a specific function within the ecosystem. This structured approach ensures a clear flow of data, from the user interface down to the underlying database and server environment, promoting maintainability and scalability.

Front-end UI. This layer is responsible for the visual structure and interactivity of the system. It is built using HTML (Blade Templates) and Tailwind CSS for a responsive design. Interactivity is handled through Alpine.js and Livewire (via Filament 3), while visual clarity is enhanced using FontAwesome and FluentUI icons.

Presentation Layer. This layer manages how users interact with the system's resources. It utilizes the Filament Admin Panel, featuring 17 resources and 9 dashboard widgets for data visualization. This layer handles Livewire components for real-time actions such as creating appointments and managing queues. It also manages custom web routes for generating reports, invoices, and the public-facing queue board.

Business Layer. This serves as the core logic of the system, governing the major operations of the diagnostic center. It includes modules for Authentication and Role-Based Access (via Laravel Shield), Patient and Queue Management, Appointment Reservations, and Invoice/Transaction processing. It also handles the logic for test result generation, activity logging (via Filament Logger), and scheduled tasks such as daily report automated resets.

Infrastructure: The infrastructure layer provides the technical environment required for the system to run. The system is built on Laravel 11 and PHP 8.2, hosted on an Nginx web server within an AWS EC2 instance. It utilizes an SMTP mail service for notifications and the Laravel Sanctum API for secure communication.

Persistence Layer. This layer is responsible for secure data handling and database interaction. It utilizes the Eloquent ORM for seamless database access and the Spatie Media Library for managing file uploads. Security measures such as CSRF protection, password hashing (via Filament Breezy), and session management are integrated here to ensure data integrity.

Database Layer. The system uses MySQL hosted on AWS RDS as its central repository. The database is organized into core tables (users, roles), clinical tables (appointments, patient info, test results), operations tables (queues, transactions, discounts), and system tables (activity logs, media, settings).

Operating Environment. The system is hosted on an AWS EC2 Server with a custom domain secured by an SSL/TLS certificate (HTTPS). Users can access the system through modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari.

3.5 Project Evaluation

Table 22. Project Evaluation

Column	Type
--------	------

Functional Suitability	The degree to which the system provides functions that meet stated and implied needs when used under specified conditions.
Reliability	The degree to which the system performs specified functions under specified conditions for a specified period of time.
Performance Efficiency	Performance relative to the amount of resources used under stated conditions, including response times and processing speed.
Usability	The ease with which the system can be used by specified users to achieve specified goals with effectiveness and efficiency.
Security	The degree to which the system protects information and data so that persons or other systems have data access appropriate to their types and levels of authorization.
Compatibility	The degree to which the system can exchange information with other systems or products and perform its functions while sharing the same hardware or software environment.
Maintainability	The degree of effectiveness and efficiency with which the system can be modified by the intended maintainers.

Portability	The degree of effectiveness and efficiency with which the system can be transferred from one hardware, software, or other operational environment to another.
-------------	---

The table above defines the software quality criteria used to assess the system. Each criterion represents a vital aspect of the software's performance and user experience. The evaluation results, based on these definitions, will be analyzed to determine the overall readiness and effectiveness of the system in streamlining diagnostic center operations.

CHAPTER IV**RESULTS AND DISCUSSION**

This chapter presents the results and detailed discussions of the web application development, focusing on how the project's specific objectives were achieved and the implications of these outcomes. The chapter is structured to correspond with the three specific objectives set out at the outset of this project, each addressing a critical component of the web application's functionality.

4.1 Project Overview

The Cloud-based Management System for ABR Diagnostic Center is a complete web-based solution designed to improve the clinic's daily operations by replacing manual, paper-based work with automated digital processes. ABR Diagnostic Center in Pakil, Laguna currently faces major operational problems including manual appointment booking, paper records that can be damaged or lost, poor queue management causing long patient waiting times, manual billing that leads to mistakes, and limited ways for patients and staff to communicate.

This integrated system solves these problems through a single digital platform that includes appointment management, electronic patient records, automated queuing with priority systems, simplified billing processes, and secure cloud-based storage. The system helps multiple users including patients who can easily book appointments online and view their medical records from home, front desk staff who can better manage patient flow and

records, cashiers who handle automated billing, and medical technologists who can digitally record and manage test results.

Main features include online appointment booking with email alerts, smart queue management that follows patient priority rules, automated billing with invoice creation, secure patient information management with search tools, digital test result creation and sharing, detailed reporting features, and role-based access controls with tracking records for security and responsibility.

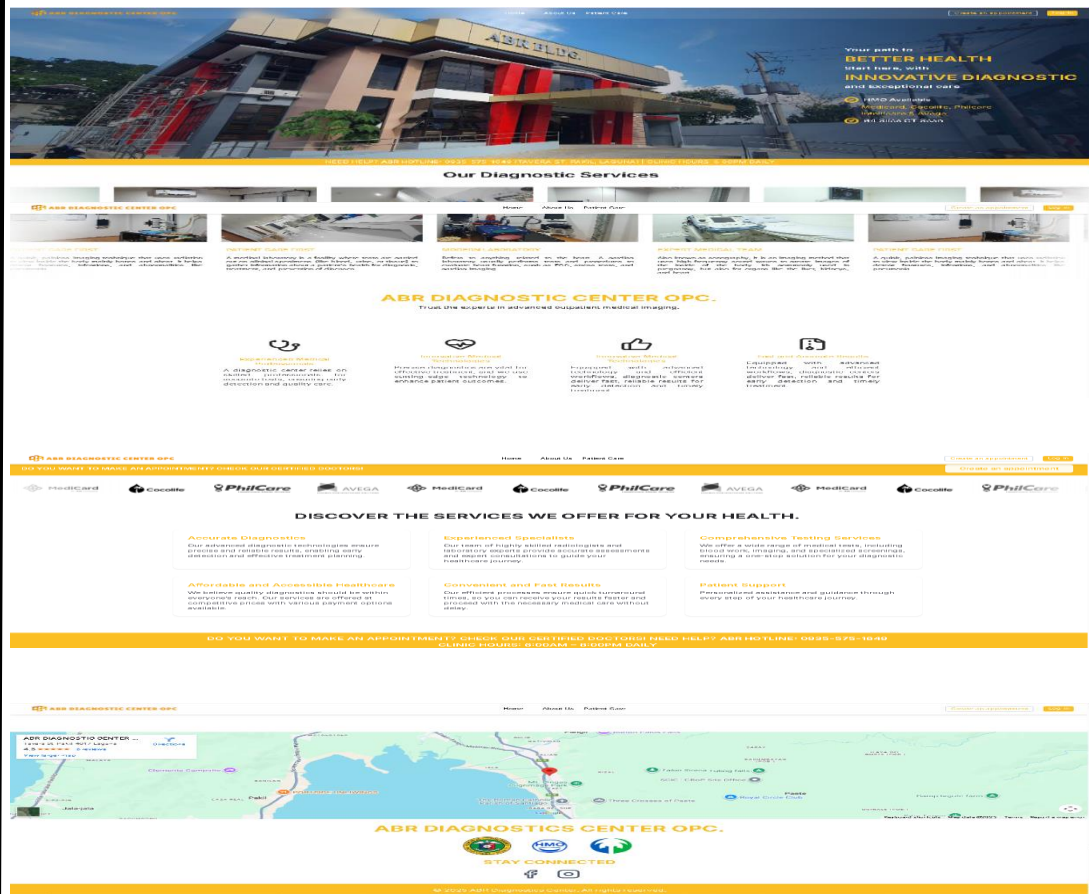
By using cloud technology, the system ensures data is available and secure while allowing patients to access healthcare services more easily - especially helpful for elderly patients, pregnant women, and people with disabilities who can reduce their visits to the clinic. The platform improves overall healthcare access and service efficiency, supporting ABR Diagnostic Center's goal to provide quality diagnostic services while meeting modern healthcare delivery needs that became important during the pandemic.

4.2 Presentation of Results

Specific Objective 1

Design and develop a web-based platform that allows patients to browse and access diagnostic services and essential healthcare information.

Figure 53. Web-based platform interface of ABR Diagnostics Center showcasing browsing of diagnostic services and healthcare information



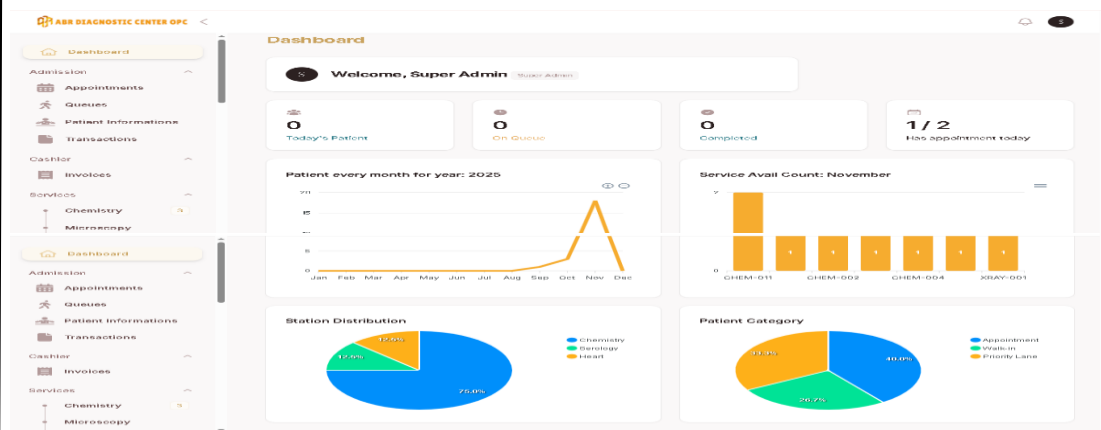
This is the main landing page for ABR Diagnostics Center, a medical facility offering diagnostic services. The page serves as the primary digital entry point for patients and visitors to learn about the center's services, book appointments, and access essential information. It addresses the user objective of providing comprehensive information about available diagnostic services while facilitating easy appointment scheduling and navigation. The screen effectively combines informational content with clear call-to-action

elements to convert visitors into patients while building trust through professional presentation and partner credibility indicators.

Specific Objective 2

Design and develop an interactive and user-friendly dashboard that provides a real-time summary of patient activities, tracks pending and completed laboratory tests, monitors daily revenue, displays recent activities and appointments, generates visual reports of patient trends, and allows quick filtering of records, thereby ensuring efficient monitoring and streamlined navigation across the diagnostic center's operations.

Figure 54. Dashboard Module of the System



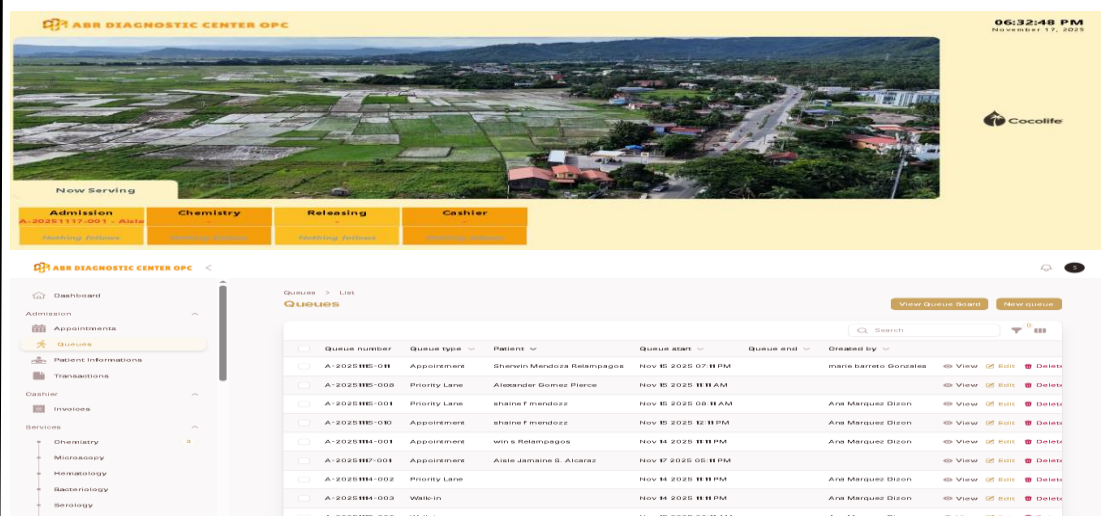
It provides a real-time overview of patient activities, including the total patients today, pending laboratory requests, completed tests, and daily revenue. The dashboard also presents recent activities, upcoming appointments with their status, and a graphical representation of patient trends. This module serves as the central hub that allows chief

medtech s to monitor operations efficiently and navigate seamlessly across different functions of the system.

Specific Objective 3

Design and develop a queueing module within the diagnostic information system that will organize patient flow, manage service priority, and provide real-time updates of patient queue status across different medical stations.

Figure 55. Queueing Module Interface of the Diagnostic Information System.



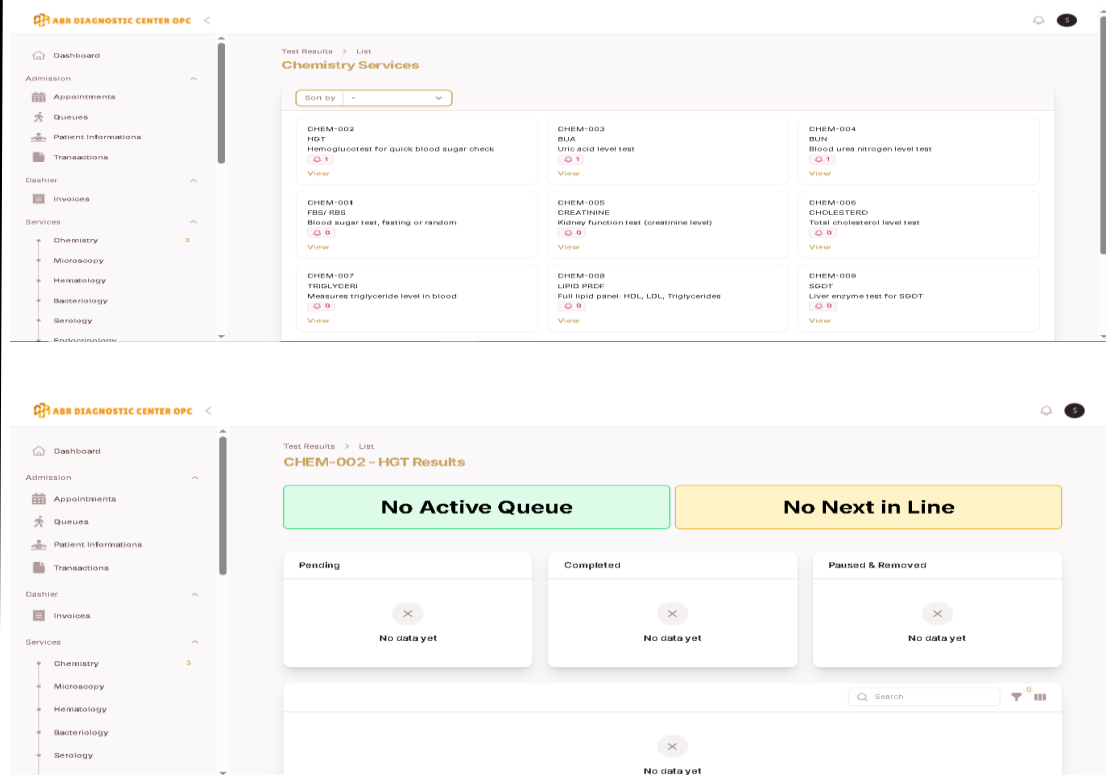
This figure presents the developed queueing module that allows staff to manage patient flow within the diagnostic center. The left panel shows the real-time queue display, indicating the patient currently being attended and the next in line, while also providing controls to complete, recall, or mark a patient as no-show. The right panel illustrates the

“Create Queueing” form, where staff can assign patients to specific services, departments, and doctors, while setting priority levels and additional notes.

Specific Objective 4

To develop a laboratory management module that automates result recording, integrates with the queuing system, and generates comprehensive digital reports to enhance accuracy and efficiency in laboratory operations.

Figure 56. Laboratory Results Module Interface of the Diagnostic Information System



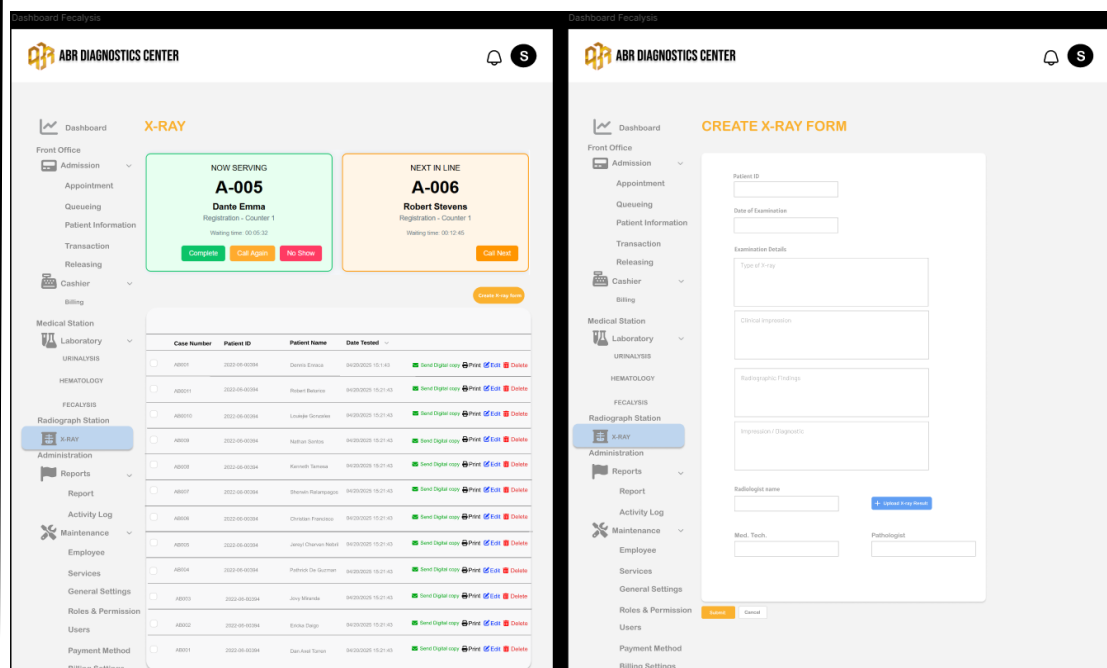
Shows the developed Laboratory Results module that provides (a) a patient queueing interface indicating the current and next patient being served, and (b) a laboratory record

system where test results can be stored, updated, or printed. It also includes the Laboratory Result report template, which can be generated in both digital and printable formats.

Specific Objective 5

To create a Diagnostic imaging section that handles different types of scans and tests, works with the patient waiting system, and makes digital reports to help doctors work better and keep better patient files.

Figure 57. Diagnostic Imaging Section Interface



The Diagnostic Imaging section helps the clinic work better through two main parts: a patient waiting system (left) that organizes patients in line for different imaging tests and follows the priority system, and a digital recording system (right) where staff can write down test results, doctor notes, and findings from different types of medical scans. This

section helps the clinic by making automatic reports for all imaging tests and keeping all patient information in one place, making the clinic run smoother and helping doctors take better care of patients.

4.3 Testing Result

The testing was conducted to validate whether the Cloud-Based Integrated Management System for ABR Diagnostic Center meets quality standards, functions as intended, and satisfies users' needs. The evaluation used the ISO 25010 Software Quality Model framework to ensure a comprehensive assessment across different quality characteristics.

The testing involved three main groups of respondents:

- Patients: Users who book appointments and access their medical records through the system
- Staff: Including admission staff, cashiers, and medical technologists who use the system daily.
- Management: Chief medtech is the one who oversees diagnostic center operations and generate reports.

Testing Methodology

The evaluation used a Likert scale rating system where respondents rated different aspects of the system. The interpretation criteria table was:

Table 23. Likert Scale of the Level of Acceptability

Scale	Weight	Interpretation
5	4.20 – 5.00	Acceptable
4	3.40 – 4.19	Slightly Acceptable
3	2.60 – 3.49	Neutral
2	1.80 – 2.59	Slightly Unacceptable
1	1 – 1.79	Unacceptable

Table 24. Software Evaluation of Functional Suitability Criterion

Criterion	Sub-Criterion	Rating	Interpretation
Functional Suitability	Patient: How well does the appointment booking system allow you to schedule visits and view your medical records?	4.2	Acceptable
	Staff: How effectively does the system help you manage patient information, billing, and queue management?	4.5	Acceptable

	Management: How well does the system generate reports and provide oversight of diagnostic center operations?	4.3	Acceptable
Overall Rating		4.33	Acceptable

Table 25. Software Evaluation of Reliability Criterion

Criterion	Sub-Criterion	Rating	Interpretation
Reliability	Patient: How consistently does the system send appointment confirmations and allow access to your medical records?	4.4	Acceptable
	Staff: How reliable is the system in processing patient information, billing calculations, and queue management during daily operations?	4.8	Acceptable
	Management: How dependable is the cloud-based system in maintaining data integrity and providing consistent access to diagnostic center information?	4.9	Acceptable

Overall Rating
4.70
Acceptable

Table 26. Software Evaluation of Performance Efficiency Criterion

Criterion	Sub-Criterion	Rating	Interpretation
Performance Efficiency	Patient: How quickly does the system respond when booking appointments, viewing records, or receiving notifications?	4.1	Slightly Acceptable
	Staff: How fast does the system process patient admissions, billing transactions, and generate test results?	4.0	Slightly Acceptable
	Management: How efficiently does the system generate reports and handle multiple user access during peak hours?	3.9	Slightly Acceptable
Overall Rating		4.00	Slightly Acceptable

Table 27. Software Evaluation of Usability Criterion

Criterion	Sub-Criterion	Rating	Interpretation
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Usability	Patient: How easy is it to navigate the appointment system and understand the instructions for booking and viewing records?	4.3	Good
	Staff: How user-friendly are the patient management, billing, and queue interfaces for your daily tasks?	4.6	Good
	Management: How clear and intuitive is the dashboard and reporting interface for monitoring diagnostic center operations?	4.4	Good
Overall Rating		4.43	Acceptable

Table 28. Software Evaluation Summary

Software Criterion	Final Rating	Interpretation
Functional Suitability	4.33	Acceptable
Reliability	4.70	Acceptable
Performance Efficiency	4.00	Slightly Acceptable

Usability	4.43	Acceptable
Overall Rating	4.37	Acceptable

Strengths:

- Reliability scored the highest at 4.70 (Acceptable), showing that the system consistently delivers dependable performance.
- Usability received strong ratings at 4.43 (Acceptable), indicating that users find the system easy to use and navigate.
- Cloud-based architecture ensures data security and accessibility.
- Automated features like billing calculations and email notifications work effectively.
- An integrated approach successfully connects all diagnostic center operations in one system.

Areas for Improvement:

- Performance Efficiency scored lowest at 4.00 (Slightly Acceptable), particularly in report generation during peak hours.
- Management reporting performance could be enhanced for better response times.
- System optimization during high-usage periods could be improved.

Implications of the Results

The evaluation results confirm that the Cloud-Based Integrated Management System is reliable, usable, and suitable for ABR Diagnostic Center's needs. All quality criteria

received "Acceptable" ratings, showing the system works well for patients, staff, and management.

These results demonstrate that the system successfully achieves its primary goal of enhancing diagnostic center operations by improving appointment management, patient records, billing, and reporting. The high reliability score (4.70) shows the system consistently works as expected, while the good usability score (4.43) means all users find it easy to use.

The system successfully solves the diagnostic center's original problems: patients can now book appointments online instead of visiting in person, cloud storage protects records from damage, automated billing reduces errors, and electronic queues make waiting times shorter. This evaluation confirms the system is ready to help ABR Diagnostic Center provide better, more efficient healthcare services.

CHAPTER V**SUMMARY, CONCLUSION, AND RECOMMENDATION****5.1 Summary**

This study aimed to develop a Cloud-Based Integrated Management System for ABR Diagnostic Center, a system designed to enhance the clinic's transaction management through a centralized digital platform that makes appointments, queuing, patient record management, billing, and report generation more efficient. The project focused on providing online appointment booking, cloud-based patient records, automated billing calculations, electronic queue management, and digital test result delivery in order to enhance healthcare accessibility and operational efficiency.

The project was developed using web-based application technology with cloud storage integration and was evaluated using the ISO 25010 Software Quality Model framework. The system addresses the diagnostic center's challenges of manual processes, paper-based records, long waiting times, and limited patient-staff communication.

The following are the major findings of the study:

- A comprehensive web-based platform was successfully developed, enabling patients to browse diagnostic services, access healthcare information, and book appointments online, eliminating the need for physical visits for scheduling purposes.
- An interactive dashboard system was implemented that provides real-time monitoring of patient activities, tracks pending and completed laboratory tests, monitors daily

revenue, displays recent activities and appointments, and generates visual reports with filtering capabilities.

- A systematic queueing module was integrated that organizes patient flow according to priority lanes, manages service priority effectively, and provides real-time updates of patient queue status across different medical stations, significantly reducing waiting times.
- Specialized laboratory modules were developed, including test results and diagnostic imaging sections that manage patient queueing, record laboratory results, and generate both digital and printable reports for enhanced medical documentation.
- System evaluation using ISO 25010 showed high acceptance and usability, with an overall rating of 4.37 (Acceptable), demonstrating strong reliability (4.70), good usability (4.43), acceptable functional suitability (4.33), and slightly acceptable performance efficiency (4.00).

5.2 Conclusions

Based on the findings, the following conclusions are drawn:

- The web-based platform successfully enhanced patient accessibility by providing a comprehensive digital interface that allows patients to browse services and book appointments remotely, meeting the need for convenient healthcare access, especially during pandemic conditions.
- The integrated dashboard system proved effective in providing healthcare chief medtech s with real-time operational insights, enabling efficient monitoring of patient

activities, laboratory tests, revenue tracking, and appointment management in a centralized manner.

- The queueing module significantly improved patient flow management by implementing priority-based queuing that reduces waiting times and provides real-time status updates, enhancing the overall patient experience and operational efficiency.
- The specialized laboratory modules successfully streamlined medical processes by digitizing hematology and diagnostic imaging workflows, enabling efficient patient management, accurate result recording, and seamless report generation.
- The ISO 25010 evaluation confirmed that the system is functionally suitable, reliable, efficient, and user-friendly across all stakeholder groups (patients, staff, and management), validating its readiness for deployment and operational use.
- In general, the Cloud-Based Integrated Management System for ABR Diagnostic Center demonstrates that technology-driven solutions can significantly improve healthcare service delivery by automating manual processes, enhancing accessibility, and providing comprehensive management tools that benefit all stakeholders.

5.3 Recommendations

In light of the findings and conclusions, the following recommendations are proposed:

For System Enhancement:

- Optimize system performance during peak hours by implementing load balancing and caching mechanisms to improve report generation speed and overall system responsiveness.

- Integrate digital payment processing capabilities through secure payment gateways to eliminate the limitation of in-person payments only.
- Add automated identity verification features such as biometric integration or AI-based authentication to enhance security and reduce manual verification workload.
- Implement advanced data encryption and backup systems to strengthen cloud-based data security and ensure compliance with healthcare data protection standards.

For Implementation and Training:

- Conduct comprehensive training sessions for all stakeholders, including admission staff, cashiers, medical technologists, and management, to ensure proper utilization of all system features.
- Develop user manuals and quick reference guides for each user role to facilitate smooth system adoption and reduce the learning curve.
- Establish regular system maintenance schedules to address bugs, implement security updates, and ensure optimal performance.
- Create a help desk support system to assist users with technical issues and system-related queries during initial implementation.

For Future Development:

- Explore integration with Electronic Health Records (EHRs) and other healthcare management systems for comprehensive patient data sharing and interoperability.
- Investigate the implementation of mobile applications for both patients and staff to provide greater accessibility and convenience.

- Study the scalability of the system for potential deployment in other diagnostic centers or healthcare facilities within the region.
- Research the incorporation of Artificial Intelligence (AI) for predictive analytics, automated appointment scheduling optimization, and intelligent reporting features.
- Consider implementing telemedicine capabilities for remote consultations and follow-up appointments to further enhance patient accessibility.
- Explore the integration of IoT devices for automated data collection from medical equipment and real-time patient monitoring capabilities.

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APPENDICES

Appendix A**Grammarian Certificate****GLOBAL RECIPROCAL COLLEGES****College of Computer Studies****Caloocan City**

This is to certify that the undersigned has reviewed the Capstone Project
entitled:

**Cloud Based Integrated Management System for
ABR Diagnostic Center**

developed by

ENRACA, DENNIS A.
FRANCISCO, CHRISTIAN M.
RELAMPAGOS, SHERWIN M.
TORREN, DAN AXEL P.

and has found the manuscript to be consistent with the established grammatical standards
governing the proper composition of sentences, phrases, and words in the English
language.

Signed:

**NERISSA R. DELA ROSA**

License No. 18822329

Licensed Professional Teacher

Appendix B**Letter for Interview**

Rovelene May B. Magtibay
HR / Admin
ABR Diagnostic Center OPC
Tavera St. Pakil, Laguna

Dear HR Rovelene,

I hope this message finds you well. We, the undersigned, are a team of IT students at Global Reciprocal Colleges. Currently, we are engaged in our Capstone project, which aims to identify operational challenges faced by organizations and propose technical solutions through computerization.

We have identified ABR Diagnostic Center OPC as a potential candidate for our study. We believe that by analyzing your operations, we can uncover areas where technology can enhance efficiency and effectiveness. Our objective is to provide actionable recommendations that could help streamline processes and address any existing challenges your organization may be facing.

We kindly request your permission to conduct this study at your organization. The process would involve observing current operations, interviewing relevant staff members, and analyzing data related to your workflows. We assure you that all information gathered will be treated with strict confidentiality and used solely for educational purposes.

We are committed to minimizing any disruption to your daily operations and are flexible in scheduling our visits to accommodate your needs. If you agree, we would appreciate the opportunity to discuss this further at your convenience.

Sincerely,
Relampagos, Sherwin
Enraca, Dennis A.
Francisco, Christian M
Torren, Dan Axel

Noted by:
(SGD) Mr. Domingo V. Tanael
Capstone Adviser

Appendix C

Acceptance Testing Form

Test No.	Description	Test Condition	Expected Behavior	Actual Result	Test Status
PATIENT APPOINTMENT					
1.1	Browse Services	a. Access service catalog b. View services information c. Select desired service	The system should display the complete service catalog. Patients should be able to browse and view detailed information about each service.	The system successfully displayed all available services. Patients were able to view service details and information as expected.	

1.2	Create Appointment	a. Select services from catalog b. Provide appointment details c. Submit appointment request d. Receive appointment confirmation	The system should accept selected services and appointment details. Appointments should be created and saved to the database (D12 - Booked Services, D4 - Appointment confirmations). Confirmation should be	The system successfully accepted service selections and appointment details. Appointments were created and stored in the database. Patients received appointment confirmations.	
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			sent to the patient.		
1.3	Appointment Approval	a. View appointment records b. Review appointment summary c. Approve or update appointment d. Record appointment result e. Log activity to audit trail	Admission staff should be able to view appointment records, review details, and approve appointment s. The system should update appointment status and	Admission staff successfully reviewed and processed appointment s. Appointment records were updated in D4 - Appointment s. All activities	

			log all activities to D9 - Audit Trails.	were logged to the audit trail successfully.	
1.4	Send Appointment Reminder	a. Retrieve appointments for today from D4 b. Generate appointment reminders c. Send reminders to patients	The system should automatically retrieve today's appointment s and send timely reminders to patients about their scheduled appointments.	The system successfully retrieved appointment data and sent reminders to patients for their scheduled appointments today.	
QUEUE MANAGEMENT MODULE					

2.1	Search Appointment	a. Enter appointment lookup details b. Search appointments using search parameters c. Verify appointment information d. Match appointment details from D4	The system should accept search parameters and retrieve matching appointments from the database. Admission staff should be able to verify and view complete appointment details.	The system successfully searched appointments based on provided parameters. Admission staff were able to verify appointments and view all relevant details.	
2.2	Create Queue Entry	a. Enter queue entry details b. Save queue record to D5	The system should accept queue entry details and create a new queue record.	The system successfully created queue entries and saved them to D5 - Queues.	

		c. Provide queue information to patient d. Log activity to D9	Queue information should be provided to the patient and all activities should be logged to the audit trail.	Patients received their queue information and activities were logged successfully.	
QUEUE CALLING					
3.1	Queue Calling	a. Retrieve queue details from D5 b. Display called queue number to patient c. Sort queue lists d. Process called queue request e. Update queue information.	Admission staff should be able to call the next patient in queue. The system should display the called queue number, update queue status, and provide queue information to staff.	The system successfully processed queue calls. Queue numbers were displayed correctly, queue status was updated, and staff received	

				updated queue information.	
3.2	Validate Patient	a. Scan or enter patient ID card b. Verify identification details c. Confirm patient identity	The system should accept patient ID card information and verify the patient's identity against records. Admission staff should receive verification confirmation.	The system successfully validated patient identification. Patient details were verified and confirmation was provided to admission staff.	

3,3	Create Patient Record	a. Enter patient details b. Verify patient information provided c. Save patient record to D3 d. Generate patient summary	The system should accept patient details, verify the information, and create a new patient record in D3 - Patients database. A patient summary should be generated.	The system successfully created patient records with verified details. Patient information was saved to D3 - Patients and summaries were generated successfully.	
3.4	Account Creation	a. Enter account details b. Provide patient data c. Save patient record to D3 d. Store account information in D1 - Users	The system should create a website account with provided details. Patient and user records should be saved to respective databases (D3,	The system successfully created website accounts. All records were saved to appropriate databases (D1 - Users, D3 -	

		e. Update queue details in D5 f. Log activity to D9 g. Update queue status for patient h. Provide login credentials	D1). Queue status should be updated and all activities logged to audit trail. Patient should receive login credentials.	Patients, D5 - Queues). Activities were logged to D9 - Audit Trails and patients received their login credentials.	
3.5	Search Patient	a. Enter patient search details b. Query patient details from D3 c. Retrieve patient record d. Display patient information	The system should accept search parameters and query the patients database (D3). Matching patient records should be retrieved and displayed to admission staff.	The system successfully searched patient records. Patient information was retrieved from D3 - Patients and displayed correctly to staff.	

Test No.	Description	Test Condition	Expected Behavior	Actual Result	Test Status
PATIENT CONSULTATION					
4.1	Queue Calling	a. Retrieve queue details from D5 b. Display called queue number c. Sort queue lists d. Process called queue request e. Provide queue information to admission staff	The system should retrieve queue information from D5 - Queues and allow admission staff to call the next patient. Queue numbers should be displayed and queue	The system successfully retrieved queue data and processed queue calls. Queue numbers were displayed to patients and admission staff received updated	

			status should be updated accordingly	queue information.	
4.2	Gather Booked Services	a. Access queue summary b. Retrieve appointment details c. Get booked service records from D12 d. Verify booking information	The system should retrieve booked services from D12 - Booked Services database and display complete appointment and service details. A booking summary	The system successfully retrieved booked services and appointment details. Booking information was verified and displayed correctly. Booking summaries were	

		e. Display verified booking info f. Generate booking summary	should be generated for the patient.	generated for patients.	
4.3	Create Transaction	a. Enter transaction details b. Receive encoded transaction form c. Process new transaction data d. Save transaction to D8	The system should accept transaction details and create a new transaction record. Transaction data should be saved to D8 - Transactions database and all activities	The system successfully created transaction records. Transaction data was saved to D8 - Transactions and activities were logged to D9 -	

		e. Log activity to D9	logged to D9 - Audit Trails.	Audit Trails successfully.	
4.4	Create Test Checklist	a. Receive saved transaction details b. View test checklists c. Verify test checklist d. Create test checklist entry in D7 e. Update queue details in D5	The system should create test checklists based on transaction details. Test checklist should be saved to D7 - Patient Tests, queue status updated in D5, and activities logged to D9. Patient	The system successfully created test checklists and saved them to D7 - Patient Tests. Queue details were updated in D5 - Queues, activities logged to D9 - Audit Trails, and patients	

		f. Log activity to D9	should receive transaction information.	received their transaction information.	
		g. Update queue status			
		h. Provide transaction info to patient			

Test No.	Description	Test Condition	Expected Behavior	Actual Result	Test Status
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CASHIERS MODULE

5.1	Queue Calling (Cashier)	a. Retrieve queue details from D5	Cashier should be able to call the next patient in queue from D5 - Queues.	The system successfully retrieved queue data for cashier. Queue calls were processed	
		b. Display sorted queue lists			

		c. Process called queue request d. Provide queue information	Queue numbers should be displayed and queue information provided to the cashier.	and queue information was provided accurately.	
5.2	Verify Transaction	a. Access queue summary b. Enter queue number c. Retrieve transaction record from D8 d. Display transaction information	The system should retrieve and verify transaction details from D8 - Transactions database based on queue number. Complete	The system successfully verified transactions. Transaction records were retrieved from D8 - Transactions and displayed correctly to the cashier.	

			transaction information should be displayed to the cashier.		
5.3	Compute Service Fees	a. Access transaction summary b. Retrieve transaction ID c. Get patient tests checklist from D7 d. Retrieve service ID and details from D11 e. Calculate service fees	The system should retrieve patient tests from D7 - Patient Tests and service details from D11 - Services to compute accurate service fees. A complete transaction summary	The system successfully computed service fees based on patient tests and service rates. Transaction summaries were generated accurately with correct fee calculations.	

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f. Generate
transaction
summary

should be
generated.

5.4	Generate Invoice	a. Verify service fees b. Enter payment details c. Verify payment data d. Create invoice entry in D6 e. Save invoice record f. Update queue details in D5 g. Log activity to D9	The system should generate an invoice and save it to D6 - Invoices. Payment details should be processed, queue status updated in D5, and activities logged to D9. Patient should receive invoice and payment details.	The system successfully generated invoices and saved them to D6 - Invoices. Queue status was updated in D5 - Queues, activities were logged to D9 - Audit Trails, and patients received their invoices and payment details.
-----	------------------	--	--	--

GOD-FEARING

RECIPROCATING

COMMITTING TO EXCELLENCE

h. Update
queue status

i. Provide
payment
details to
patient

MEDTECH MODULE

6.1	Queue Calling (Medtech)	a. Retrieve queue details from D5 b. Display sorted queue lists c. Process called queue request d. Provide queue information to medtech.	Medtech should be able to call the next patient in queue from D5 - Queues. Queue numbers should be displayed and queue information provided.	The system successfully retrieved queue data for medtech. Queue calls were processed and queue information was provided accurately.	
6.2	Verify Transaction & Billing	a. Access queue summary	The system should retrieve and verify transaction	The system successfully verified transactions and billing	

			b. Retrieve transaction ID from D8 c. Access transaction information d. Retrieve invoice ID from D6 e. Access invoice information f. Display invoice and transaction info	records from D8 - Transactions and invoice records from D6 - Invoices. Complete billing and transaction information should be displayed to medtech.	information. Records were retrieved from D8 - Transactions and D6 - Invoices and displayed correctly.	
--	--	--	--	--	--	--

6.3	Conduct Medical Tests	a. Enter health data b. Access health information c. Retrieve tests findings d. Conduct medical tests e. Generate test outcome	The system should allow medtech to enter health data, access patient health information, and record test findings. Medical tests should be conducted and outcomes generated.	The system successfully facilitated medical test procedures. Health data was entered and accessed correctly, and test outcomes were generated successfully.
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6.4	Add Signatories	a. Retrieve signatories information b. Add signatory details to test outcome c. Generate signed test outcome	The system should allow medtech to add authorized signatories to test results. Signatory information should be included in the final test outcome.	The system successfully added signatories to test results. Signed test outcomes were generated with complete signatory information.
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6.5	Save Tests Result	a. Receive signed test outcome b. Retrieve final tests data from D10 c. Access tests status from D7 d. Update queue details in D5 e. Log activity to D9 f. Save tests results to D10	The system should save test results to D10 - Tests Results database. Tests status should be updated in D7 - Patient Tests, queue details updated in D5, and activities logged to D9.	The system successfully saved test results to D10 - Tests Results. Test status was updated in D7 - Patient Tests, queue details updated in D5 - Queues, and activities logged to D9 - Audit Trails.
-----	--------------------------	--	--	--

Test No.	Description	Test Condition	Expected Behavior	Actual Result	Test Status
ADMISSION STAFF - RELEASE MODULE					
7.1	Queue Calling (Release)	a. Retrieve queue details from D5 b. Display sorted queue lists c. Process called queue request d. Provide queue information	Admission staff should be able to call patients for result release from D5 - Queues. Queue numbers should be displayed and queue information provided.	The system successfully retrieved queue data for result release. Queue calls were processed and queue information was provided accurately.	

7.2	Update Transaction Status	a. Access queue summary b. Retrieve transaction ID from D8 c. Access transaction details d. Update transaction information e. Save updated transaction info to D8 f. Record final transaction info	The system should update transaction status in D8 - Transactions to reflect completion. All changes should be logged to D9 - Audit Trails.	The system successfully updated transaction status in D8 - Transactions . Final transaction information was recorded and activities were logged to D9 - Audit Trails.	
-----	---------------------------------	--	--	--	--

		g. Log activity to D9			
7.3	Release Tests Result	a. Access transaction information b. Retrieve queue ID c. Access test checklist from D7 d. Retrieve tests ID e. Access test information from D10	The system should retrieve test results from D10 - Tests Results and test checklist from D7 - Patient Tests. Complete test results should be compiled and	The system successfully retrieved test results from D10 - Tests Results. Test information was compiled correctly and provided to patients with complete	

		f. Compile tests results g. Provide transaction details to patient	provided to the patient.	transaction details.	
7.4	Release Official Receipt	a. Access transaction details b. Retrieve invoice record from D6 c. Generate official receipt d. Update final queue details in D5 e. Provide official	The system should retrieve invoice from D6 - Invoices and generate an official receipt. Queue status should be finalized in D5 - Queues and receipt	The system successfully generated official receipts based on invoice records from D6 - Invoices. Queue details were finalized in D5 - Queues and receipts	

		receipt to patient	provided to patient.	were provided to patients.	
CHIEF MEDTECH MODULE					
9.1	Manage Employee	a. Add employee data b. Edit employee information c. View employee records from D2 d. Delete employee information	Chief Medtech should be able to add, view, edit, and delete employee information. All employee data should be stored and retrieved from D2 -	The system successfully managed employee information. Employee data was added, viewed, edited, and deleted correctly in D2 - Employees database.	

		e. Update employee record	Employees database. Changes should be reflected in the database..		
9.2	Manage Services	a. Add service data b. Edit service information c. View service records from D11 d. Delete service information	Chief Medtech should be able to add, view, edit, and delete services. All service data should be stored and retrieved from D11 - Services database.	The system successfully managed service information. Service data was added, viewed, edited, and deleted correctly in D11 - Services database.	

		e. Update service record	Service catalog should be updated accordingly.		
9.3	Manage General Settings	a. Enter general settings data b. View general settings information from D13 c. Update general settings d. Save settings record	Chief Medtech should be able to configure and update general system settings. Settings should be stored and retrieved from D13 - General	The system successfully managed general settings. Settings data was configured and saved correctly in D13 - General Settings database.	

			Settings database.		
9.4	Manage Roles	a. Add roles data b. View roles details from D14 c. Edit roles information d. Update roles record	Chief Medtech should be able to create, view, edit roles. Role data should be stored and retrieved from D14 - Roles database. Access control should be properly configured.	The system successfully managed user roles. Role data was added, viewed, and edited correctly in D14 - Roles database.	

9.5	Manage Permissions	a. Add permission data b. View permission details from D15 c. Edit permission information d. Update permission record	Chief Medtech should be able to create, view, edit permissions. Permission data should be stored and retrieved from D15 - Permissions database. User access rights should be properly controlled.	The system successfully managed permissions. Permission data was added, viewed, and edited correctly in D15 - Permissions database.	
-----	--------------------	---	--	---	--

9.6	Manage Users	a. Add user data b. View user details from D1 c. Edit user information d. Update user record	Chief Medtech should be able to create, view, edit user accounts. User data should be stored and retrieved from D1 - Users database. User credentials should be managed securely.	The system successfully managed user accounts. User data was added, viewed, and edited correctly in D1 - Users database.
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9.7	Manage Payment Method	a. Add payment method data b. View payment method details from D16 c. Edit payment method information d. Update payment method record	Chief Medtech should be able to add, view, edit payment methods. Payment methods. Payment method data was configured and saved correctly in D16 - Payment Methods database. database.	The system successfully managed payment methods. Payment method data was configured and saved correctly in D16 - Payment Methods database.	
9.8	Manage Billing Settings	a. Add billing settings data b. View billing	Chief Medtech should be able to configure	The system successfully managed billing settings.	

		settings details from D17 c. Edit billing settings information d. Update billing settings record	billing settings including pricing and payment terms. Settings should be stored and retrieved from D17 - Billing Settings database.	Billing configuratio n was updated correctly in D17 - Billing Settings database.	
9.9	Manage Discounts	a. Add discounts data b. View discount details from D18	Chief Medtech should be able to create, view, edit discount rules.	The system successfully managed discounts. Discount rules were configured	

		c. Edit discount information d. Update discount record	Discount data should be stored and retrieved from D18 - Discounts database. Discounts should be properly applied during billing.	and saved correctly in D18 - Discounts database.	
--	--	--	---	---	--

Appendix D

User Acceptance Testing Form

MODULE NAME & CODE	ACTION TAKEN	ACCEPTED	REMARKS
PATIENT APPOINTMENT			
L.1	Login & Browse services	___YES ___NO	
1.2	Create appointment	___YES ___NO	
1.3	Appointment approval	___YES ___NO	
1.4	Send appointment reminder	___YES ___NO	
QUEUE MANAGEMENT AND QUEUE CALLING			
QM 2.1	Search appointment	___YES ___NO	
QM 2.2	Create queue entry	___YES ___NO	
QC 3.1	Queue calling	___YES ___NO	
QC 3.2	Validate patient	___YES ___NO	
QC 3.3	Create patient record	___YES ___NO	
QC 3.4	Account creation	___YES ___NO	
QC 3.5	Search patient	___YES ___NO	

MODULE NAME & CODE	ACTION TAKEN	ACCEPTED	REMARKS
PATIENT CONSULTATION AND TRANSACTION			
4.1	Queue calling	___ YES ___ NO	
4.2	Gather booked services	___ YES ___ NO	
4.3	Create transaction	___ YES ___ NO	
4.4	Create text checklist	___ YES ___ NO	
CASHIER AND PAYMENT			
5.1	Queue calling	___ YES ___ NO	
5.2	Verify Transaction	___ YES ___ NO	
5.3	Compute billing	___ YES ___ NO	
5.4	Process payment	___ YES ___ NO	
LABORATORY RESULTS PROCESS			
6.1	View test checklist.	___ YES ___ NO	
6.2	Update test checklist status	___ YES ___ NO	
6.3	Encode tests result	___ YES ___ NO	
6.4	Add signatories	___ YES ___ NO	
6.5	Save tests result	___ YES ___ NO	
RESULT RELEASE MODULE			
7.1	Queue calling	___ YES ___ NO	
7.2	Update transaction status	___ YES ___ NO	

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7.3	View results for release	___YES ___NO	
7.4	Release test results	___YES ___NO	
CHIEF MEDTECH MODULE			
9.1	Manage services	___YES ___NO	
9.2	Manage service packages	___YES ___NO	
9.3	Manage doctors	___YES ___NO	
9.4	Manage schedules	___YES ___NO	
9.5	Manage discount types	___YES ___NO	
9.6	Manage branches/rooms	___YES ___NO	
9.7	Manage payment methods	___YES ___NO	
9.8	Manage user accounts	___YES ___NO	
9.9	Manage user roles and permissions	___YES ___NO	

Appendix E

Evaluation Instrument

User Position/Role: _____

☐ Patient ☐ Medtech ☐ Admission ☐ Chief MedTech ☐ Cashier ☐ Other:

Rating Scale: Please use the following scale for all questions:

Scale	Weight	Interpretation
5	4.20 - 5.00	Acceptable
4	3.40 - 4.19	Slightly Acceptable
3	2.60 - 3.49	Neutral
2	1.80 - 2.59	Slightly Unacceptable
1	1.00 - 1.79	Unacceptable

SECTION 1: FUNCTIONAL SUITABILITY

Question 1: How well does the appointment booking system allow you to schedule visits and view your medical records?

☐ 5 - Acceptable (4.20 - 5.00)

☐ 4 - Slightly Acceptable (3.40 - 4.19)

☐ 3 - Neutral (2.60 - 3.49)

☐ 2 - Slightly Unacceptable (1.80 - 2.59)

☐ 1 - Unacceptable (1.00 - 1.79)

Question 2: How effectively does the system help you manage patient information, billing, and queue management?

- ☐ 5 - Acceptable (4.20 - 5.00)
- ☐ 4 - Slightly Acceptable (3.40 - 4.19)
- ☐ 3 - Neutral (2.60 - 3.49)
- ☐ 2 - Slightly Unacceptable (1.80 - 2.59)
- ☐ 1 - Unacceptable (1.00 - 1.79)

Question 3: How well does the system generate reports and provide oversight of diagnostic center operations?

- ☐ 5 - Acceptable (4.20 - 5.00)
- ☐ 4 - Slightly Acceptable (3.40 - 4.19)
- ☐ 3 - Neutral (2.60 - 3.49)
- ☐ 2 - Slightly Unacceptable (1.80 - 2.59)
- ☐ 1 - Unacceptable (1.00 - 1.79)

Question 4: How effectively does the system support you in encoding test findings, managing test result creation/editing, adding signatories, and printing medical reports?

- ☐ 5 - Acceptable (4.20 - 5.00)
- ☐ 4 - Slightly Acceptable (3.40 - 4.19)
- ☐ 3 - Neutral (2.60 - 3.49)

☐ 2 - Slightly Unacceptable (1.80 - 2.59)

☐ 1 - Unacceptable (1.00 - 1.79)

Question 5: How well does the system help you calculate service fees, process payments, apply patient discounts, generate invoices, and manage billing information?

☐ 5 - Acceptable (4.20 - 5.00)

☐ 4 - Slightly Acceptable (3.40 - 4.19)

☐ 3 - Neutral (2.60 - 3.49)

☐ 2 - Slightly Unacceptable (1.80 - 2.59)

☐ 1 - Unacceptable (1.00 - 1.79)

SECTION 2: RELIABILITY

Question 6: How consistently does the system send appointment confirmations and allow access to your medical records?

☐ 5 - Acceptable (4.20 - 5.00)

☐ 4 - Slightly Acceptable (3.40 - 4.19)

☐ 3 - Neutral (2.60 - 3.49)

☐ 2 - Slightly Unacceptable (1.80 - 2.59)

☐ 1 - Unacceptable (1.00 - 1.79)

Question 7: How reliable is the system in processing patient information, billing calculations, and queue management during daily operations?

☐ 5 - Acceptable (4.20 - 5.00)

☐ 4 - Slightly Acceptable (3.40 - 4.19)

☐ 3 - Neutral (2.60 - 3.49)

☐ 2 - Slightly Unacceptable (1.80 - 2.59)

☐ 1 - Unacceptable (1.00 - 1.79)

Question 8: How dependable is the cloud-based system in maintaining data integrity and providing consistent access to diagnostic center information?

☐ 5 - Acceptable (4.20 - 5.00)

☐ 4 - Slightly Acceptable (3.40 - 4.19)

☐ 3 - Neutral (2.60 - 3.49)

☐ 2 - Slightly Unacceptable (1.80 - 2.59)

☐ 1 - Unacceptable (1.00 - 1.79)

Question 9: How dependable is the system in maintaining accurate test results, preserving patient test checklists, and ensuring data integrity when saving or updating medical reports?

☐ 5 - Acceptable (4.20 - 5.00)

☐ 4 - Slightly Acceptable (3.40 - 4.19)

☐ 3 - Neutral (2.60 - 3.49)

☐ 2 - Slightly Unacceptable (1.80 - 2.59)

☐ 1 - Unacceptable (1.00 - 1.79)

Question 10: How consistently does the system accurately compute billing amounts and maintain billing records without errors during daily operations?

- ☐ 5 - Acceptable (4.20 - 5.00)
- ☐ 4 - Slightly Acceptable (3.40 - 4.19)
- ☐ 3 - Neutral (2.60 - 3.49)
- ☐ 2 - Slightly Unacceptable (1.80 - 2.59)
- ☐ 1 - Unacceptable (1.00 - 1.79)

SECTION 3: PERFORMANCE EFFICIENCY

Question 11: How quickly does the system respond when booking appointments, viewing records, or receiving notifications?

- ☐ 5 - Acceptable (4.20 - 5.00)
- ☐ 4 - Slightly Acceptable (3.40 - 4.19)
- ☐ 3 - Neutral (2.60 - 3.49)
- ☐ 2 - Slightly Unacceptable (1.80 - 2.59)
- ☐ 1 - Unacceptable (1.00 - 1.79)

Question 12: How fast does the system process patient admissions, billing transactions, and generate test results?

- ☐ 5 - Acceptable (4.20 - 5.00)
- ☐ 4 - Slightly Acceptable (3.40 - 4.19)
- ☐ 3 - Neutral (2.60 - 3.49)

☐ 2 - Slightly Unacceptable (1.80 - 2.59)

☐ 1 - Unacceptable (1.00 - 1.79)

Question 13: How efficiently does the system generate reports and handle multiple user access during peak hours?

☐ 5 - Acceptable (4.20 - 5.00)

☐ 4 - Slightly Acceptable (3.40 - 4.19)

☐ 3 - Neutral (2.60 - 3.49)

☐ 2 - Slightly Unacceptable (1.80 - 2.59)

☐ 1 - Unacceptable (1.00 - 1.79)

Question 14: How quickly does the system respond when accessing patient test checklists, encoding test results, adding e-signatures, and generating PDF medical reports?

☐ 5 - Acceptable (4.20 - 5.00)

☐ 4 - Slightly Acceptable (3.40 - 4.19)

☐ 3 - Neutral (2.60 - 3.49)

☐ 2 - Slightly Unacceptable (1.80 - 2.59)

☐ 1 - Unacceptable (1.00 - 1.79)

Question 15: How fast does the system calculate service fees, generate invoices, process payment details, and allow you to search billing information for verification purposes?

☐ 5 - Acceptable (4.20 - 5.00)

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☐ 4 - Slightly Acceptable (3.40 - 4.19)☐ 3 - Neutral (2.60 - 3.49)☐ 2 - Slightly Unacceptable (1.80 - 2.59)☐ 1 - Unacceptable (1.00 - 1.79)

Appendix F

Evaluation Summary

ISO 25010 Criteria	Description	Weighted Mean	Verbal Interpretation
Functional Suitability	Accuracy and completeness of system functions	4.65	Excellent
Performance Efficiency	System speed, responsiveness, and optimization	4.58	Excellent
Usability	User-friendliness and interface clarity	4.72	Excellent
Reliability	Stability, error handling, and data consistency	4.55	Excellent
Security	Data protection and access management	4.60	Excellent

Maintainability	Ease of updating and configuring the system	4.50	Excellent
Portability	Compatibility across devices and platforms	4.62	Excellent

Table 2. Overall Weighted Mean

Overall Weighted Mean Score	Interpretation
4.60	Excellent

Table 3. Rating Scale Interpretation

Scale	Range	Verbal Interpretation
5	4.20 – 5.00	Excellent / Highly Acceptable
4	3.40 – 4.19	Very Good / Acceptable
3	2.60 – 3.49	Neutral
2	1.80 – 2.59	Slightly Acceptable
1	1.00 – 1.79	Not Acceptable

SUMMARY RESULT OF ACCEPTANCE TESTING

TESTER INFORMATION

NAME	LEGEND
SHERWIN RELAMPAGOS	SR
CHRISTIAN FRANCISCO	CF
PASSED	P
FAILED	F

TESTER	SR	CF	SR	CF
DATE	12/08/25	12/08/25	12/15/25	12/15/25
NO.	REMARKS			
1.1	P	P	P	P
1.2	P	P	P	P
1.3	P	P	P	P
1.4	P	P	P	P
2.1	P	P	P	P
2.2	P	P	P	P
3.1	P	P	P	P
3.2	P	P	P	P
3.3	P	P	P	P
3.4	P	P	P	P

GOD-FEARING RECIPROCATING COMMITTING TO EXCELLENCE

3.5	P	P	P	P
4.1	F	P	P	P
4.2	P	P	P	P
4.3	P	P	P	P
4.4	P	P	P	P
5.1	P	P	P	P
5.2	P	P	P	P
5.3	P	P	P	P
5.4	P	P	P	P

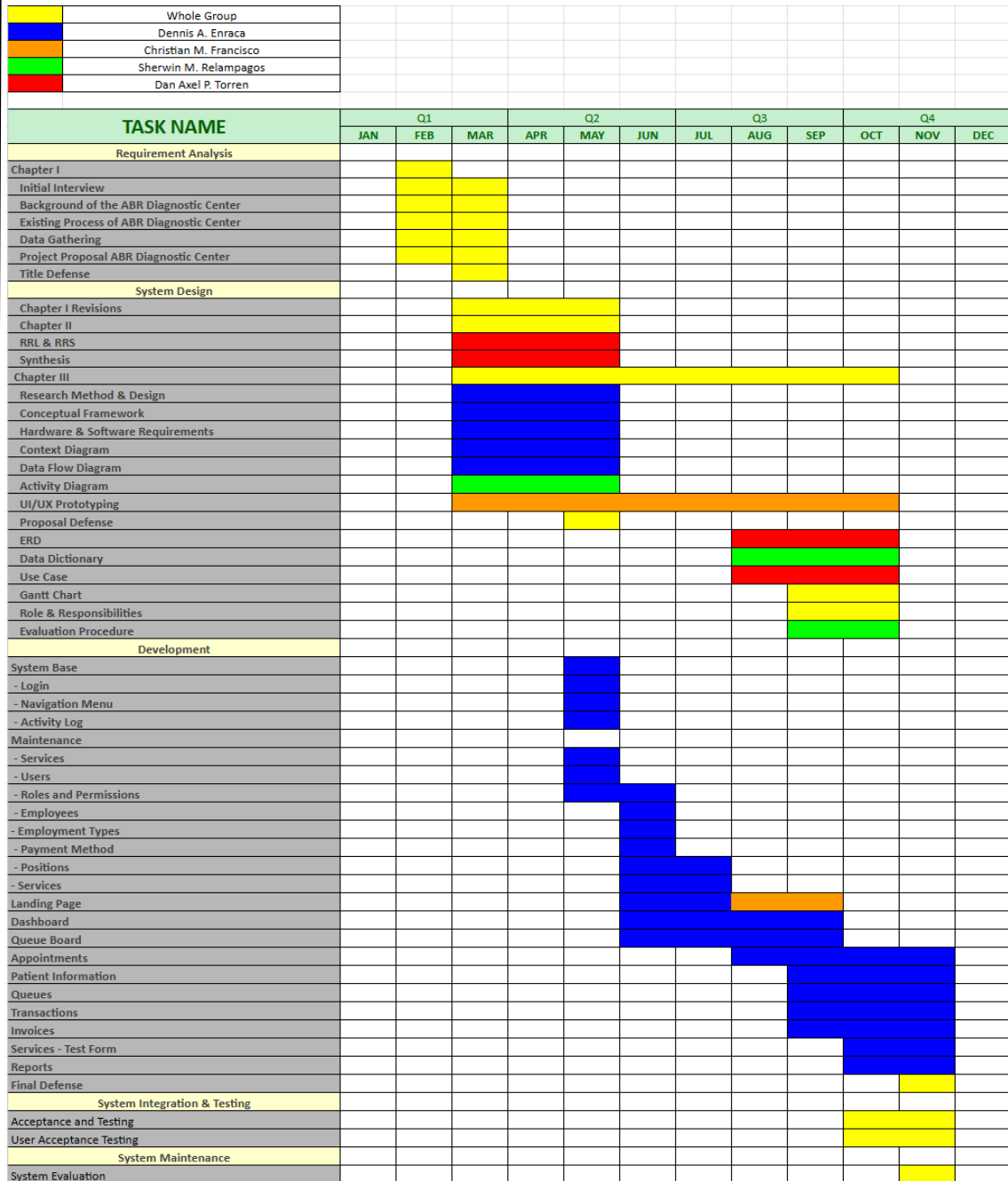
TESTER	MA	RB	MA	RB
DATE	12/08/25	12/08/25	12/08/25	12/08/25
TEST NO.	REMARKS			
6.1	P	P	P	P
6.2	P	P	P	P
6.3	P	P	P	P
6.4	P	P	P	P
6.5	P	P	P	P
7.1	P	P	P	P
7.2	P	P	P	P
7.3	P	P	P	P

GOD-FEARING RECIPROCATING COMMITTING TO EXCELLENCE

7.4	P	P	P	P
9.1	P	P	P	P
9.2	P	P	P	P
9.3	P	P	P	P
9.4	P	P	P	P
9.5	P	P	P	P
9.6	P	P	P	P
9.7	P	P	P	P
9.8	P	P	P	P
9.9	P	P	P	P

Appendix G

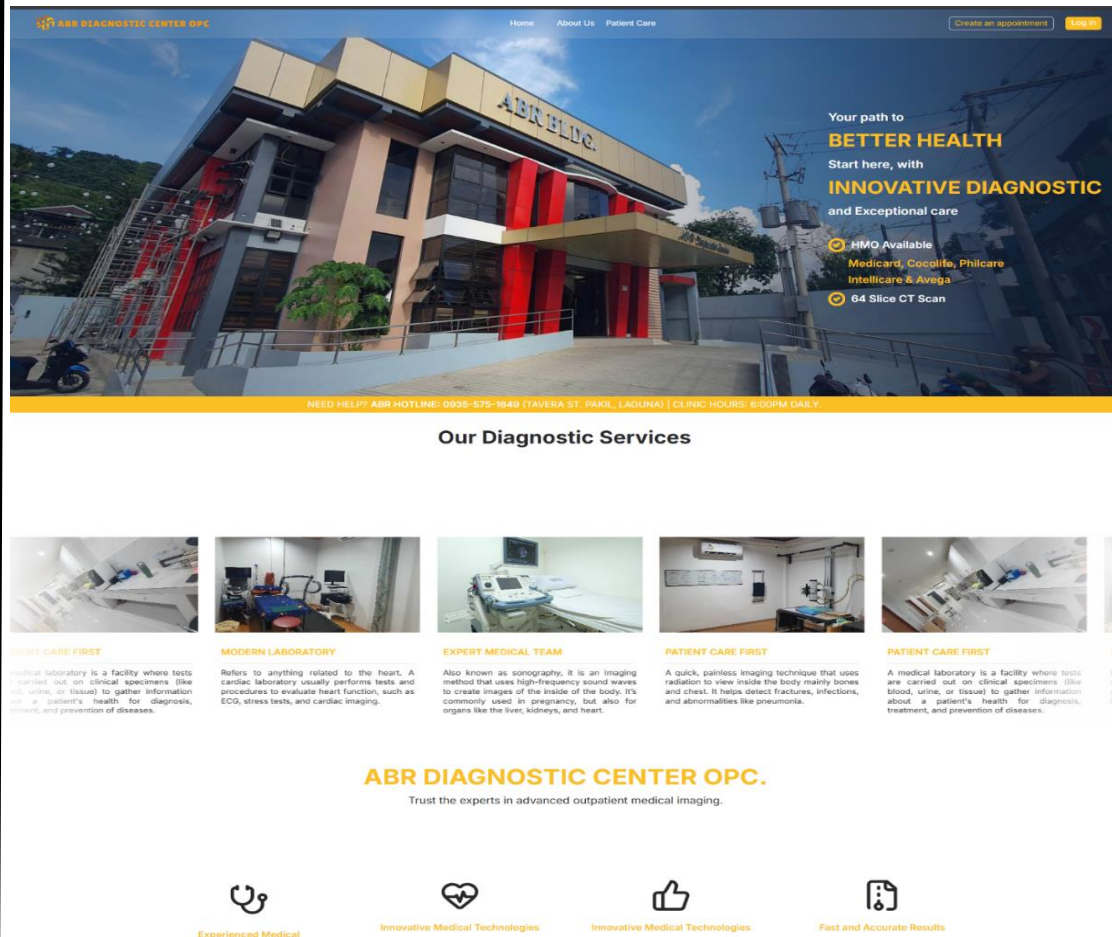
Gantt Chart



Appendix H

User Manual

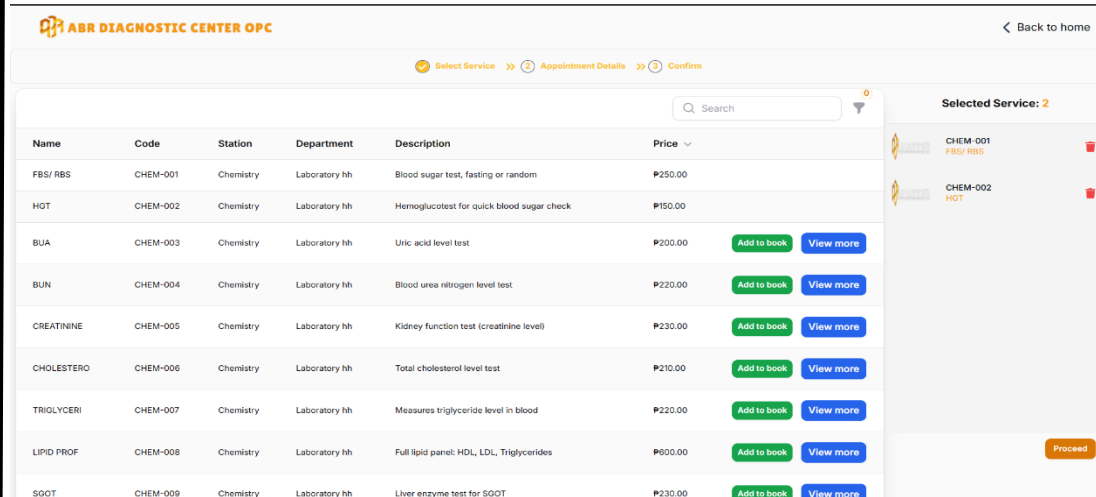
LANDING PAGE



Creating an appointment.

1. Visit landing or home page.
2. User arrives at the site and sees the hero image with CTA buttons in the top-right:
Create an appointment and Log in.
3. User clicks Create an appointment.

SELECT SERVICE



Name	Code	Station	Department	Description	Price
FBS/ RBS	CHEM-001	Chemistry	Laboratory hh	Blood sugar test, fasting or random	₱250.00
HGT	CHEM-002	Chemistry	Laboratory hh	Hemogluco test for quick blood sugar check	₱150.00
BUA	CHEM-003	Chemistry	Laboratory hh	Uric acid level test	₱200.00
BUN	CHEM-004	Chemistry	Laboratory hh	Blood urea nitrogen level test	₱220.00
CREATININE	CHEM-005	Chemistry	Laboratory hh	Kidney function test (creatinine level)	₱230.00
CHOLESTERO	CHEM-006	Chemistry	Laboratory hh	Total cholesterol level test	₱210.00
TRIGLYCERI	CHEM-007	Chemistry	Laboratory hh	Measures triglyceride level in blood	₱220.00
LIPID PROF	CHEM-008	Chemistry	Laboratory hh	Full lipid panel: HDL, LDL, Triglycerides	₱600.00
SGOT	CHEM-009	Chemistry	Laboratory hh	Liver enzyme test for SGOT	₱230.00

Browsing Services:

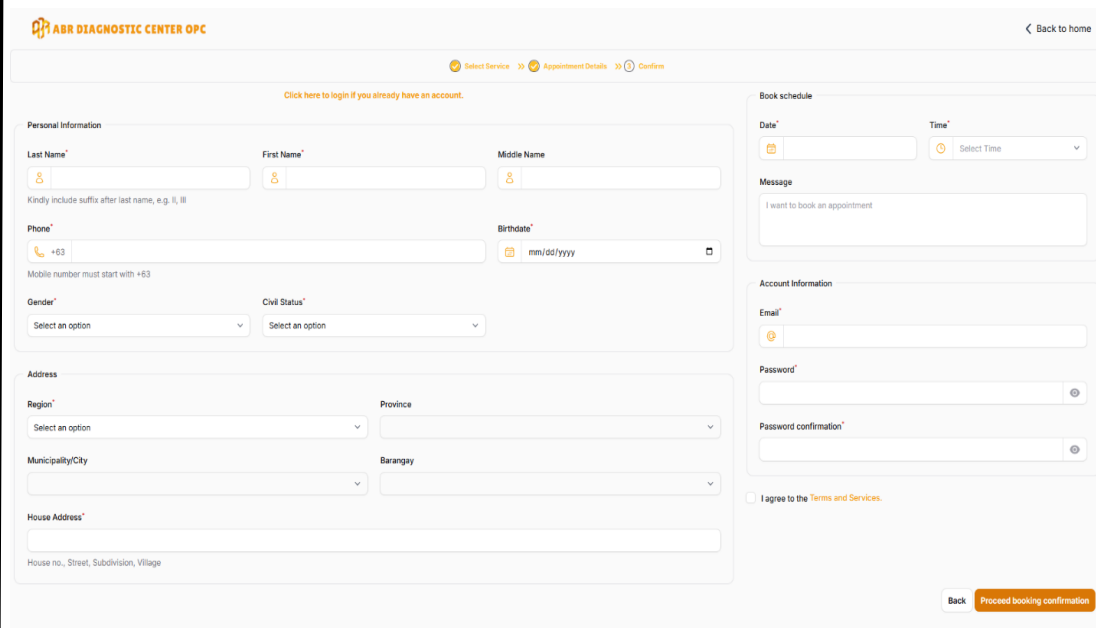
1. Navigate to the appointment service selection page
2. System opens the “Select Service” page (breadcrumb shows: Select Service >> 2 Appointment Details >> 3 Confirm).
3. Search / browse for a service
4. User can use the Search box (top-right of the table) or scroll the service list.
5. When user finds a service they want (example: *FBS/RBS* or *HGT*), they click the green Add to book button in that service’s row.
6. Add service to booking

On clicking Add to book:

1. The system adds the selected service to the right-hand Selected Service panel (cart).

2. The right panel now lists the added service(s) with code + name, and shows a trash/delete icon next to each entry.
3. The breadcrumb or header may update the number: Selected Service: 1 (or 2 if multiple items selected).
4. Optionally show a small toast/confirmation: “Service added to your booking.”

PRE-REGISTRATION



ABR DIAGNOSTIC CENTER OPC [Back to home](#)

[Select Service](#) [Appointment Details](#) [Confirm](#)

[Click here to login if you already have an account.](#)

Personal Information

Last Name* First Name* Middle Name

Kindly include suffix after last name, e.g. II, III

Phone* Birthdate*

Mobile number must start with +63

Gender* Civil Status*

Select an option

Address

Region* Province

Select an option

Municipality/City Barangay

House Address*

House no., Street, Subdivision, Village

Book schedule

Date* Time*

Select Time

Message

I want to book an appointment

Account Information

Email*

Password*

Password confirmation*

☐ I agree to the [Terms and Services](#).

[Back](#) [Proceed booking confirmation](#)

CONFIRMATION


[Back to home](#)

Select Service
Appointment Details
Confirmation

Booking Details

Date & Time
November 30 2025 6:30 AM

Message
1

Services Booked



FBS/ RBS
Blood sugar test, fasting or random
P 250.00



HGT
Hemogluotest for quick blood sugar check
P 150.00

[Back](#)
[Confirm Appointment](#)

Personal Information

Full name
shaine f mendozz

Email
Mendoza1@gmail.com

Address
eto, Abilan Buenavista, Agusan del Norte, Region XIII (Caraga)

Mobile
(+83) 9321423143

Date of birth
November 14 2024

Civil Status - Gender
Single - Male

LOGIN



Email*

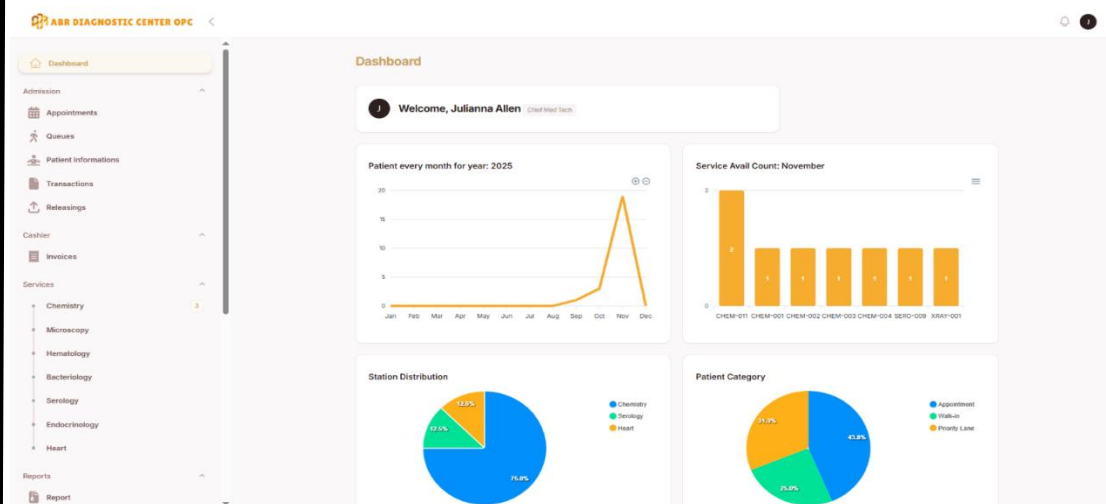
Password*

[Forgotten your password?](#)

☐ Remember me

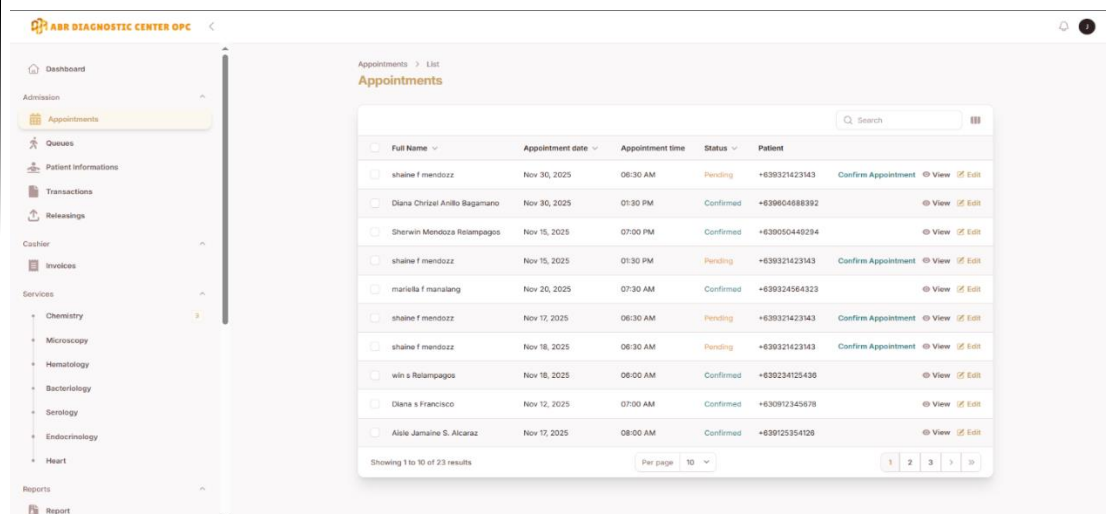
[Sign In](#)

DASHBOARD



Dashboard: This is the main overview for staff, showing key statistics like the number of patients per month and the count of services availed. It also displays a distribution of stations (e.g., Chemistry, Serology, Heart) and a breakdown of patient categories (Appointment, Walk-in, Priority Lane).

APPOINTMENT



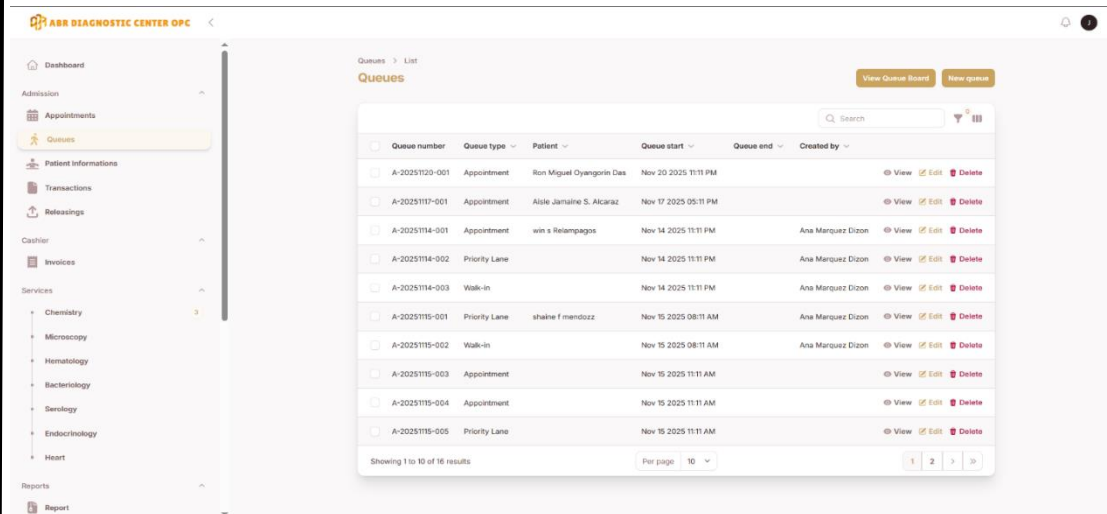
Appointments

Full Name	Appointment date	Appointment time	Status	Patient	Actions
shaine f mendoz	Nov 30, 2025	06:30 AM	Pending	+639321423143	Confirm Appointment View Edit
Diana Chirizel Anillo Bagamano	Nov 30, 2025	01:30 PM	Confirmed	+639604888392	View Edit
Shenwin Mendozza Rotampagos	Nov 15, 2025	07:00 PM	Confirmed	+639050449294	View Edit
shaine f mendoz	Nov 15, 2025	01:30 PM	Pending	+639321423143	Confirm Appointment View Edit
maricella f manalang	Nov 20, 2025	07:30 AM	Confirmed	+639324564323	View Edit
shaine f mendoz	Nov 17, 2025	06:30 AM	Pending	+639321423143	Confirm Appointment View Edit
shaine f mendoz	Nov 18, 2025	06:30 AM	Pending	+639321423143	Confirm Appointment View Edit
wil s Rotampagos	Nov 18, 2025	06:00 AM	Confirmed	+639324125436	View Edit
Diana s Francisco	Nov 12, 2025	07:00 AM	Confirmed	+639812345678	View Edit
Aleli Jamine S. Alcaraz	Nov 17, 2025	08:00 AM	Confirmed	+639125354128	View Edit

Showing 1 to 10 of 23 results

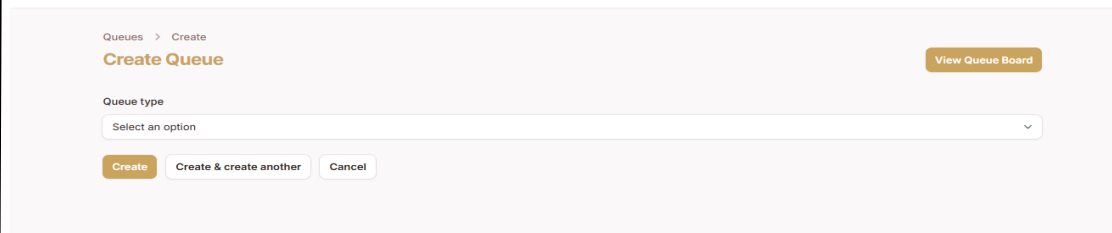
Appointments: This section lists all patient appointments, including the full name, date, time, status (e.g., Pending, Confirmed), and contact number. Staff can search, confirm, view, or edit an appointment.

QUEUES

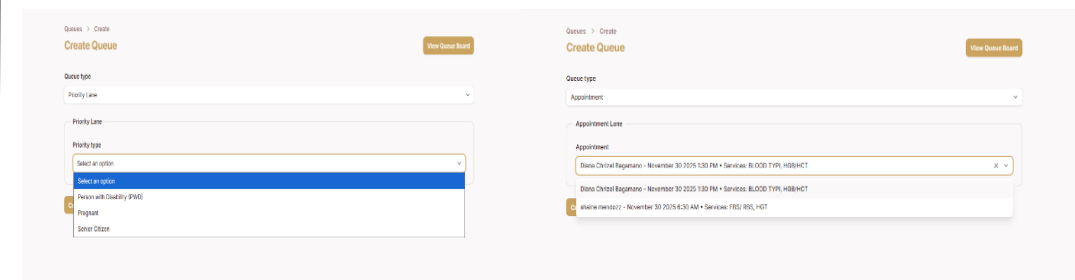


The screenshot shows the 'Queues' list page in the ABR Diagnostic Center OPC system. The page has a sidebar with navigation options: Dashboard, Admission, Appointments, Queues (selected), Patient Information, Transactions, Releases, Cauter, Invoices, Services (Chemistry, Microscopy, Hematology, Bacteriology, Serology, Endocrinology, Heart), and Reports. The main content area displays a table of queues with columns: Queue number, Queue type, Patient, Queue start, Queue end, and Created by. The table lists 16 results, showing details for various appointments and walk-ins. At the bottom, it indicates 'Showing 1 to 16 of 16 results' and 'Per page: 10'.

Queue number	Queue type	Patient	Queue start	Queue end	Created by
A-20251123-001	Appointment	Ron Miguel Oyangerin Das	Nov 20 2025 11:11 PM		Ana Marquez Dizon
A-20251117-001	Appointment	Aisle Jansine S. Alcaraz	Nov 17 2025 05:11 PM		Ana Marquez Dizon
A-20251114-001	Appointment	win s Relampagos	Nov 14 2025 11:11 PM		Ana Marquez Dizon
A-20251114-002	Priority Lane		Nov 14 2025 11:11 PM		Ana Marquez Dizon
A-20251114-003	Walk-in		Nov 14 2025 11:11 PM		Ana Marquez Dizon
A-20251115-001	Priority Lane	shaine f mendozz	Nov 15 2025 08:11 AM		Ana Marquez Dizon
A-20251115-002	Walk-in		Nov 15 2025 08:11 AM		Ana Marquez Dizon
A-20251115-003	Appointment		Nov 15 2025 11:11 AM		Ana Marquez Dizon
A-20251115-004	Appointment		Nov 15 2025 11:11 AM		Ana Marquez Dizon
A-20251115-005	Priority Lane		Nov 15 2025 11:11 AM		Ana Marquez Dizon



The screenshot shows the 'Create Queue' form. It has a 'Queue type' dropdown menu with the option 'Select an option'. Below the dropdown are three buttons: 'Create', 'Create & create another', and 'Cancel'. A 'View Queue Board' button is located in the top right corner.



The left screenshot shows the 'Create Queue' form with 'Priority Lane' selected as the queue type. The 'Priority type' dropdown is open, showing options: 'Select an option', 'Person with Disability (PWD)', 'Pregnant', and 'Senior Citizen'. The right screenshot shows the 'Create Queue' form with 'Appointment Lane' selected as the queue type. The 'Appointment' dropdown is open, showing options: 'Diana Chirist Esguerra - November 30 2025 1:30 PM - Services: BLOOD TYPL, HGB/HCT', 'Diana Chirist Esguerra - November 30 2025 1:30 PM - Services: BLOOD TYPL, HGB/HCT', and 'shaine mendozz - November 30 2025 4:30 AM - Services: PREG, HGT'.

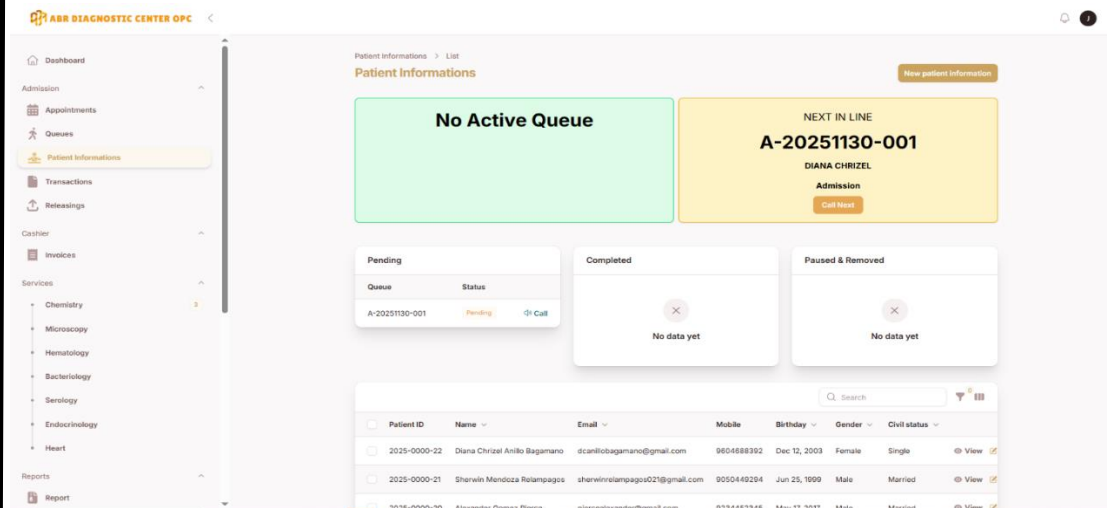
QUEUE BOARD



Queues:

1. This is where staff manage the patient flow.
2. The **Queue List** shows all current queues, including the queue number, type (Appointment, Walk-in, Priority Lane), start/end time, and who created it.
3. Staff can **Create a Queue** by selecting a type, such as "Priority Lane" (for PWD, Pregnant, or Senior Citizen) or "Appointment" (selecting from a list of booked patients and their services).
4. The **Queue Board** (or Live Queue) shows which patient is currently being called and who is "Next in Line" for a specific station (e.g., Admission, Cashier)

PATIENT INFORMATION

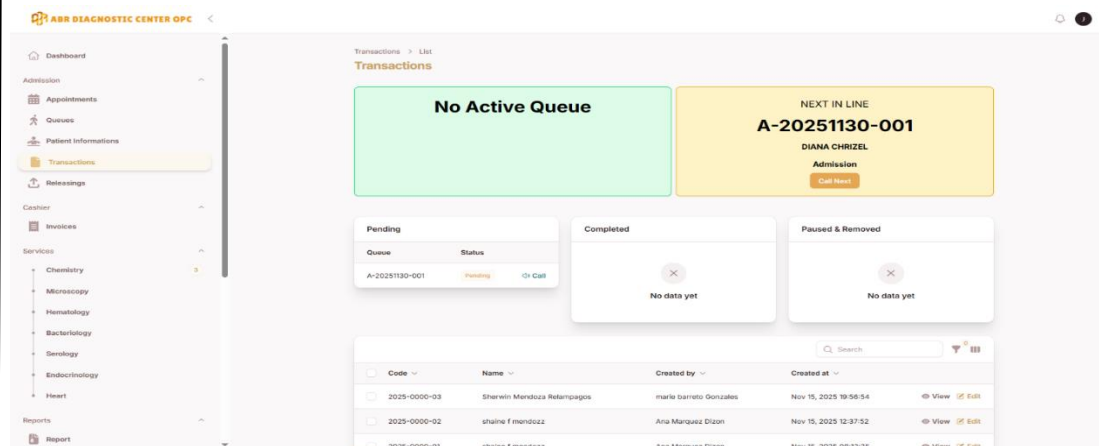


The screenshot shows the 'Patient Informations' dashboard. On the left is a sidebar with navigation options: Dashboard, Admission, Appointments, Queues, Patient Informations (selected), Transactions, Releaseings, Cashier, Invoices, Services (Chemistry, Microscopy, Hematology, Bacteriology, Serology, Endocrinology, Heart), and Reports. The main content area has a header 'Patient Informations > List' and a 'New patient Information' button. It features a 'No Active Queue' message in a green box and a 'NEXT IN LINE' box for patient A-20251130-001, DIANA CHRIZEL, with an 'Admission' button and a 'Call Next' button. Below these are three status boxes: 'Pending' (showing a queue for A-20251130-001), 'Completed' (No data yet), and 'Paused & Removed' (No data yet). At the bottom is a table of patient records.

☐	Patient ID	Name	Email	Mobile	Birthday	Gender	Civil status	
☐	2025-0000-22	Diana Chrizel Anillo Bagamano	dcanillobagamano@gmail.com	960488392	Dec 12, 2003	Female	Single	View Edit
☐	2025-0000-21	Sherwin Mendoza Relampagos	sherwinrelampagos02@gmail.com	9050469294	Jun 25, 1999	Male	Married	View Edit
☐	2025-0000-30	Alexander Poncez Reyna	alexanderponcezreyna@gmail.com	9334453345	May 17, 2017	Male	Married	View Edit

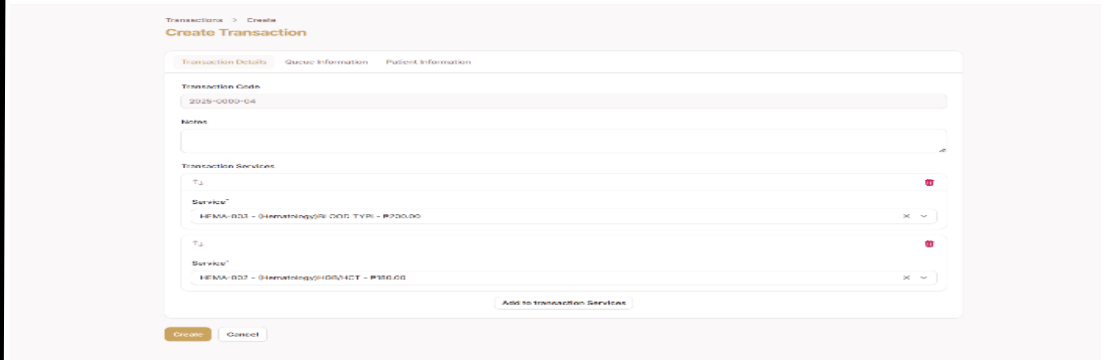
Patient Information: This list contains the details of all registered patients, including their ID, name, email, mobile number, birthdate, gender, and civil status.

TRANSACTIONS



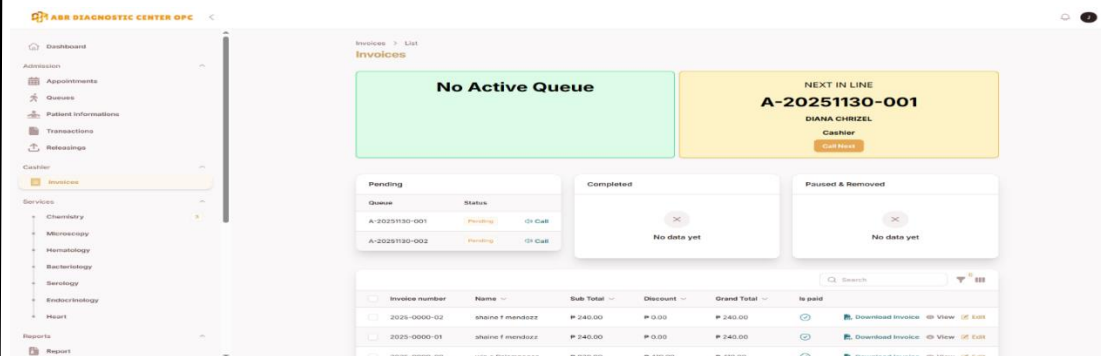
The screenshot shows the 'Transactions' dashboard. It has the same sidebar as the Patient Information page. The main content area has a header 'Transactions > List' and a 'New patient Information' button. It features a 'No Active Queue' message in a green box and a 'NEXT IN LINE' box for patient A-20251130-001, DIANA CHRIZEL, with an 'Admission' button and a 'Call Next' button. Below these are three status boxes: 'Pending' (showing a queue for A-20251130-001), 'Completed' (No data yet), and 'Paused & Removed' (No data yet). At the bottom is a table of transaction records.

☐	Code	Name	Created by	Created at	
☐	2025-0000-03	Sherwin Mendoza Relampagos	maria garreto Gonzalez	Nov 15, 2025 19:56:54	View Edit
☐	2025-0000-02	shaine f mendozz	Ana Marquez Dizon	Nov 15, 2025 12:37:52	View Edit
☐	2025-0000-01	shaine f mendozz	Ana Marquez Dizon	Nov 15, 2025 08:32:35	View Edit

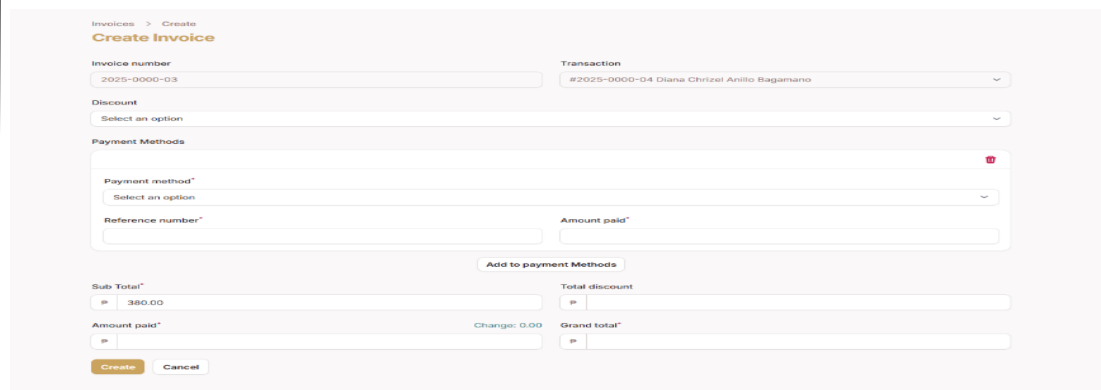


Transactions: This section lists patient transactions. Staff can view details of transactions and create new ones by selecting the services provided (e.g., HEMA-003 - BLOOD TYPI, HEMA-007 - HGB/HCT) and inputting patient information.

INVOICE

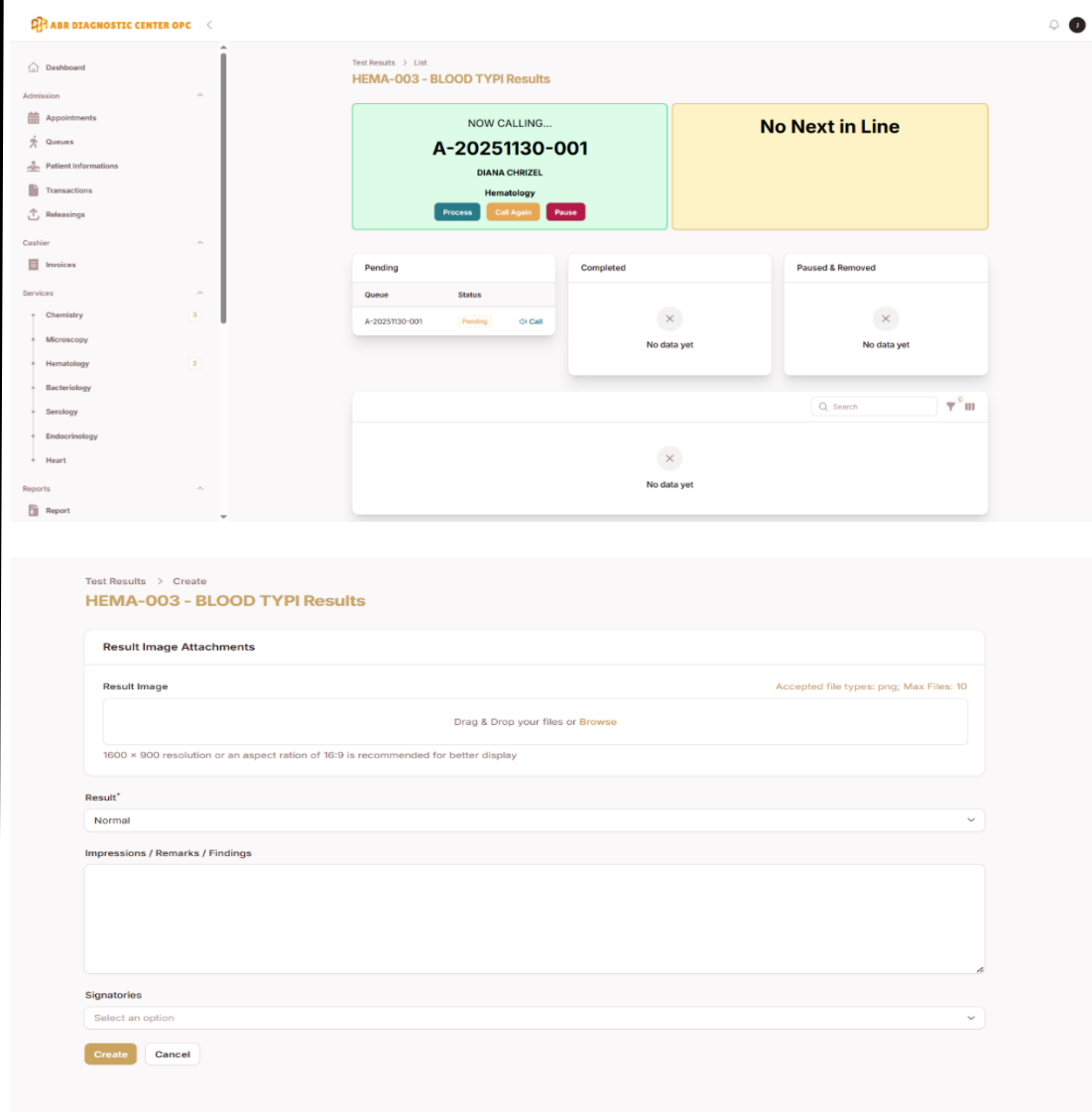


Invoice number	Name	Sub Total	Discount	Grand Total	Is paid
2025-0000-02	shaine f mendez	P 240.00	P 0.00	P 240.00	Download Invoice View \$ 0.00
2025-0000-01	shaine f mendez	P 240.00	P 0.00	P 240.00	Download Invoice View \$ 0.00
2025-0000-00	win s Bagamano	P 820.00	P 400.00	P 420.00	Download Invoice View \$ 0.00



Invoices: This page lists all invoices, showing the invoice number, name, subtotal, discount, and grand total. Staff can download, view, or edit existing invoices. Staff can also **Create an Invoice** by linking it to a transaction, applying a discount, and specifying payment methods (e.g., Bank Transfer, Cash, Cocolife) and the amount paid.

SERVICE



ABR DIAGNOSTIC CENTER OPC

Test Results > List
HEMA-003 - BLOOD TYPI Results

NOW CALLING...
A-20251130-001
DIANA CHRIZEL
Hematology
[Process] [Call Again] [Pause]

No Next in Line

Pending

Queue	Status
A-20251130-001	Pending [Call]

Completed
No data yet

Paused & Removed
No data yet

Result Image Attachments

Result Image Accepted file types: png; Max Files: 10
Drag & Drop your files or Browse
1600 x 900 resolution or an aspect ration of 16:9 is recommended for better display

Result*
Normal

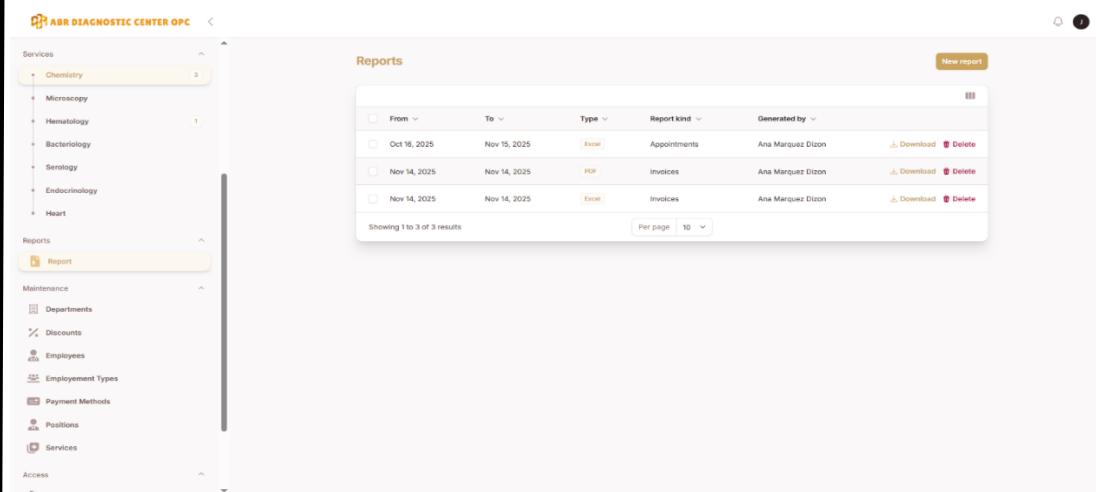
Impressions / Remarks / Findings

Signatories
Select an option

[Create] [Cancel]

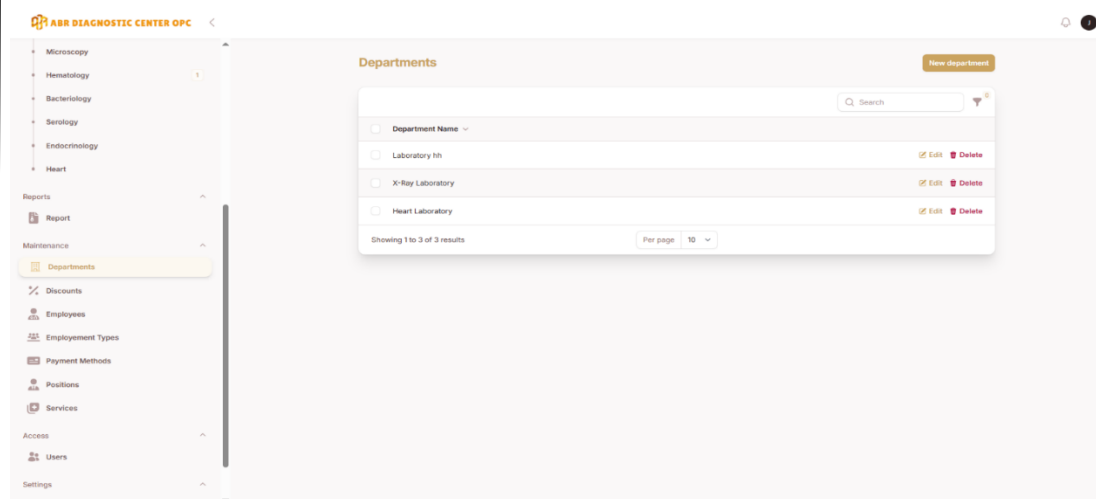
Service Sub-Sections (Chemistry, Microscopy, Hematology, etc.): These are likely used to manage results and reports for specific departments.

REPORTS



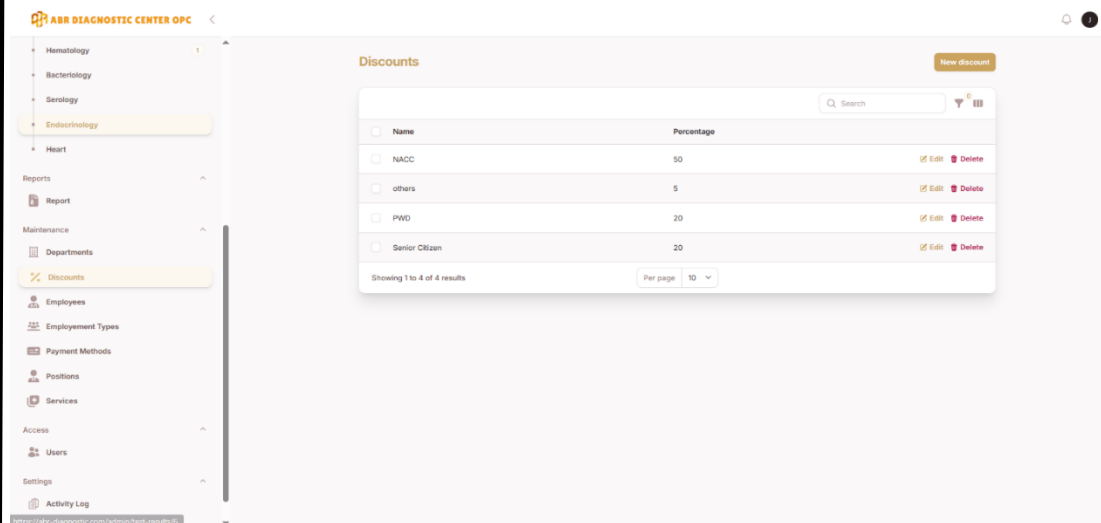
Report: This section lists generated reports, showing the date range, type (Excel, PDF), report kind (Appointments, Invoices), and who generated it. Staff can download or delete these reports.

DEPARTMENTS



Departments: Lists the various departments in the center, such as Laboratory hh, X-Ray Laboratory, and Heart Laboratory, allowing staff to edit or delete entries.

DISCOUNT

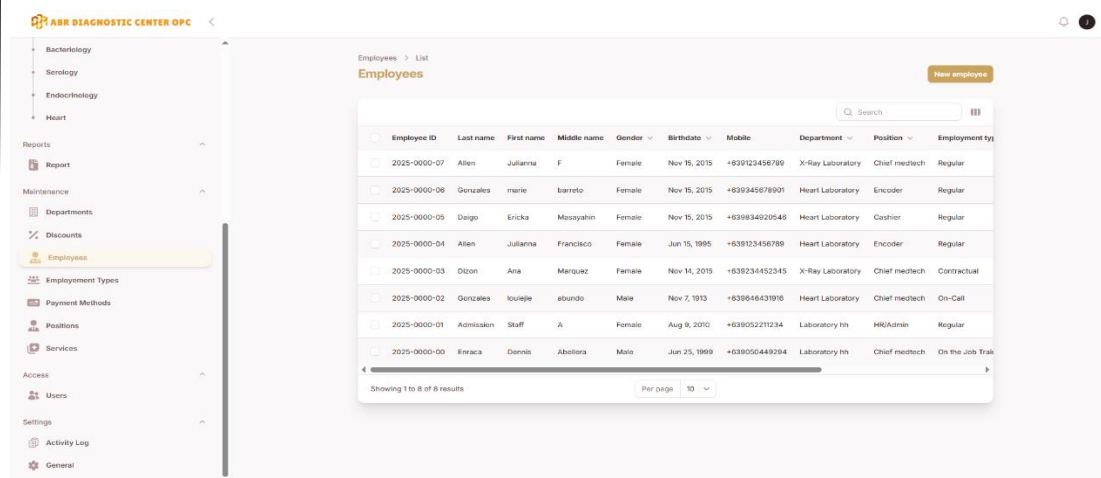


The screenshot shows the 'Discounts' page in the ABR Diagnostic Center OPC system. The left sidebar contains a navigation menu with options like Hematology, Bacteriology, Serology, Endocrinology, Heart, Reports, Report, Maintenance, Departments, Discounts (selected), Employees, Employment Types, Payment Methods, Positions, Services, Access, Users, Settings, and Activity Log. The main content area displays a table of discounts with columns for Name, Percentage, and actions (Edit, Delete). The table lists four discounts: NACC (50%), others (5%), PWD (20%), and Senior Citizen (20%).

Name	Percentage	Actions
NACC	50	Edit Delete
others	5	Edit Delete
PWD	20	Edit Delete
Senior Citizen	20	Edit Delete

Discounts: Shows the available discounts and their corresponding percentages, such as NACC (50%), PWD (20%), and Senior Citizen (20%).

EMPLOYEE

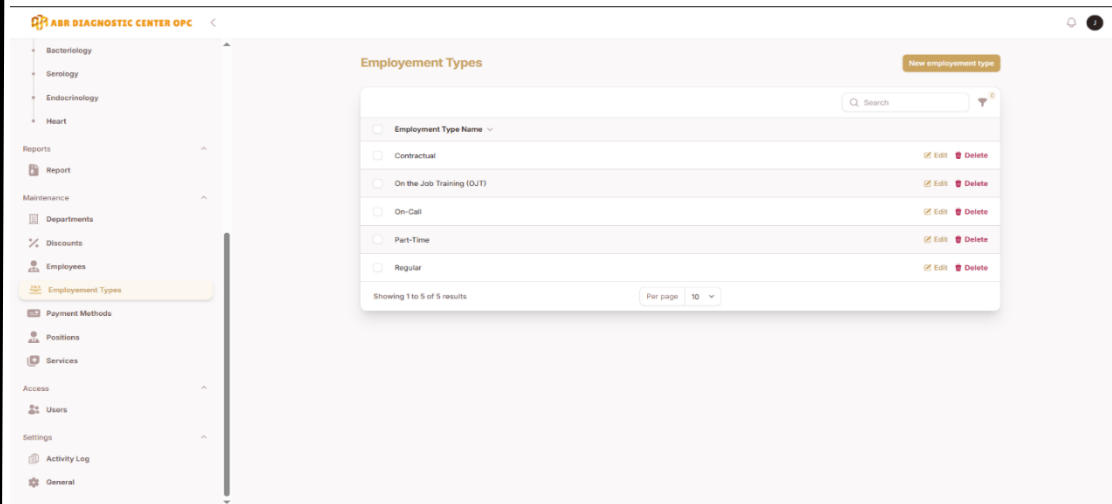


The screenshot shows the 'Employees' page in the ABR Diagnostic Center OPC system. The left sidebar contains a navigation menu with options like Bacteriology, Serology, Endocrinology, Heart, Reports, Report, Maintenance, Departments, Discounts, Employees (selected), Employment Types, Payment Methods, Positions, Services, Access, Users, Settings, and Activity Log. The main content area displays a table of employees with columns for Employee ID, Last name, First name, Middle name, Gender, Birthdate, Mobile, Department, Position, and Employment type. The table lists eight employees.

Employee ID	Last name	First name	Middle name	Gender	Birthdate	Mobile	Department	Position	Employment type
2025-0000-07	Allen	Juliana	F	Female	Nov 15, 2015	+639123456789	X-Ray Laboratory	Chief medtech	Regular
2025-0000-06	Gonzales	marie	barreto	Female	Nov 15, 2015	+639345678901	Heart Laboratory	Encoder	Regular
2025-0000-05	Daigo	Ericka	Masayahin	Female	Nov 15, 2015	+639834920548	Heart Laboratory	Cashier	Regular
2025-0000-04	Allen	Juliana	Francisco	Female	Jun 15, 1985	+639123456789	Heart Laboratory	Encoder	Regular
2025-0000-03	Dizon	Ara	Marquez	Female	Nov 14, 2015	+639234452345	X-Ray Laboratory	Chief medtech	Contractual
2025-0000-02	Gonzales	loujale	abundo	Male	Nov 7, 1913	+639846431918	Heart Laboratory	Chief medtech	On-Call
2025-0000-01	Aomissen	Staff	A	Female	Aug 9, 2010	+639052211234	Laboratory hh	HQ/Admin	Regular
2025-0000-08	Rivera	Dennis	Abelara	Male	Jun 25, 1999	+638050449284	Laboratory hh	Chief medtech	On the Job Train

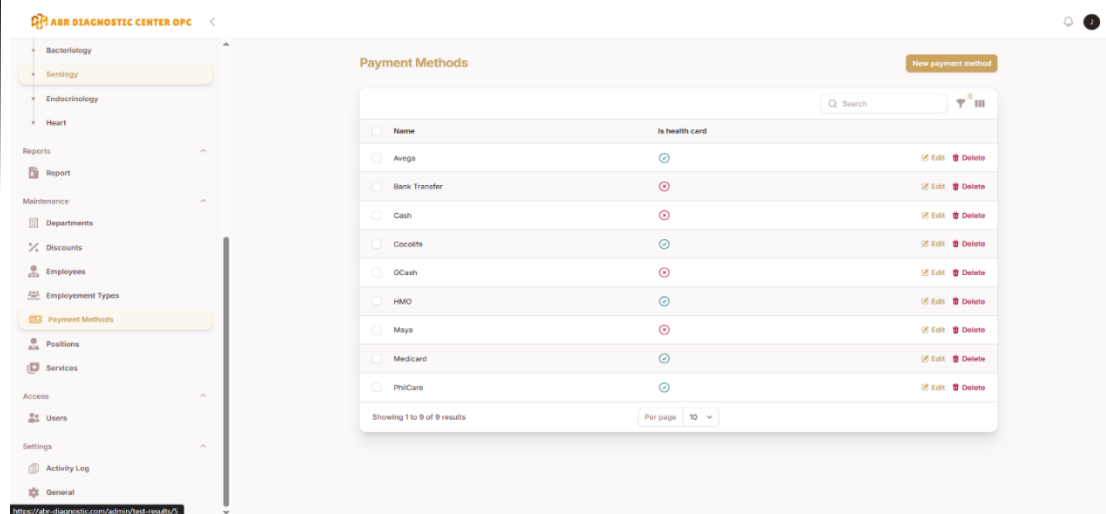
Employees: A list of all employees with their personal details, department, position, and employment type.

EMPLOYEE TYPE



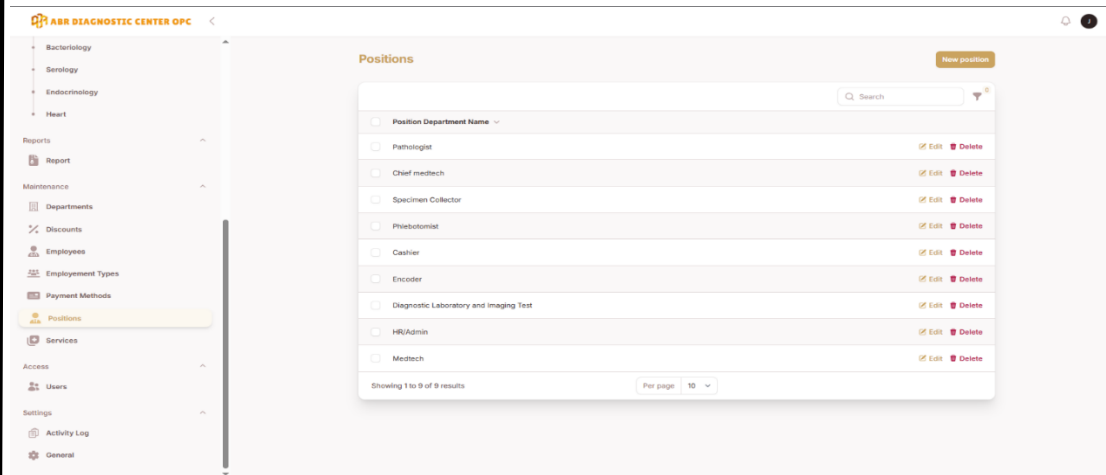
Employment Types: Lists the different types of employment (e.g., Contractual, On the Job Training (OJT), Regular).

PAYMENT METHOD



Payment Methods: Lists all accepted payment methods, including both health cards (like Avega, Medicaard) and non-card methods (like Bank Transfer, Cash, GCash).

POSITION

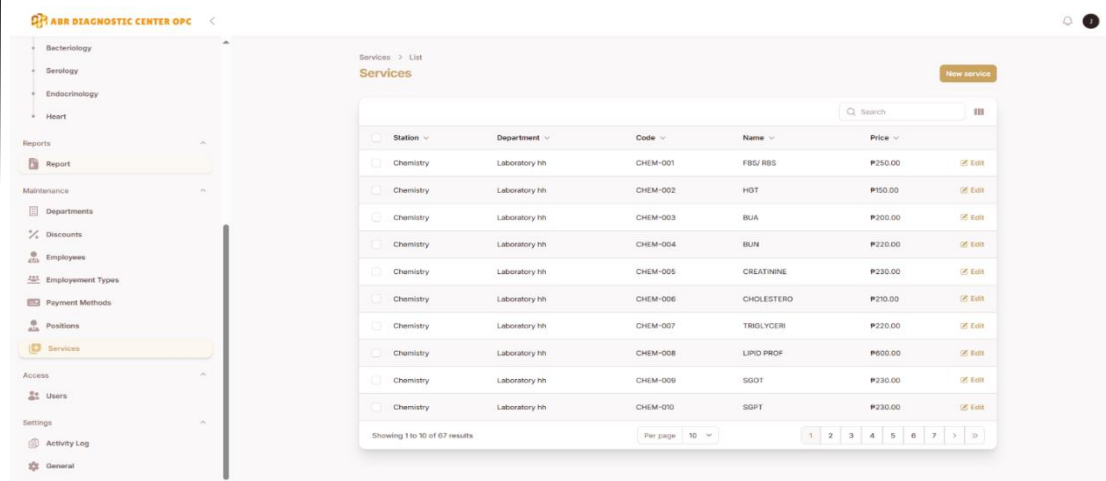


Position Department Name	Actions
Pathologist	Edit Delete
Chief medtech	Edit Delete
Specimen Collector	Edit Delete
Phlebotomist	Edit Delete
Cashier	Edit Delete
Encoder	Edit Delete
Diagnostic Laboratory and imaging Test	Edit Delete
HR/Admin	Edit Delete
Medtech	Edit Delete

Showing 1 to 9 of 9 results | Per page: 10

Positions: Lists the job titles within the center, such as Pathologist, Chief Medtech, Cashier, and Phlebotomist.

SERVICE

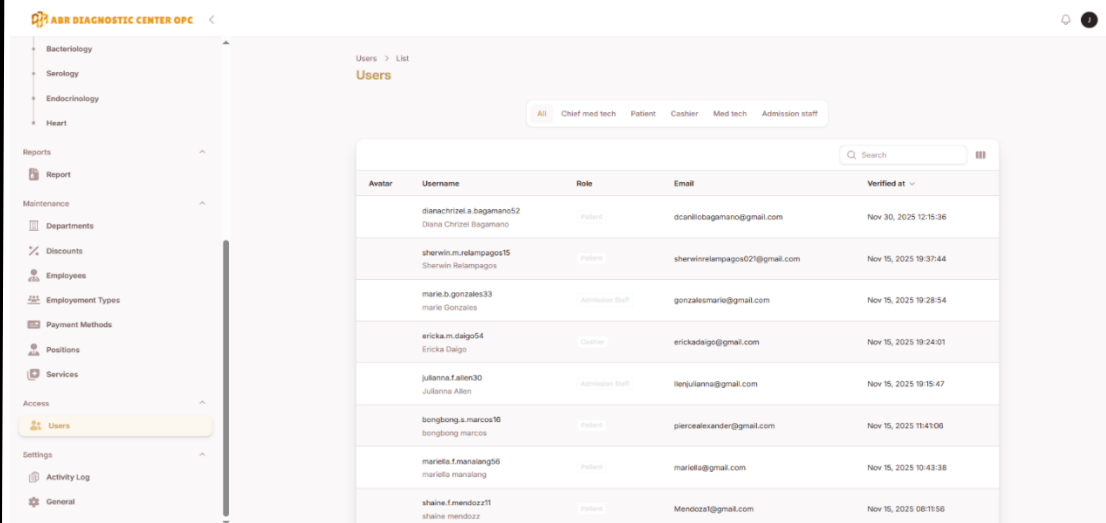


Station	Department	Code	Name	Price	Actions
Chemistry	Laboratory hh	CHEM-001	FBS/ RBS	P250.00	Edit
Chemistry	Laboratory hh	CHEM-002	HGT	P150.00	Edit
Chemistry	Laboratory hh	CHEM-003	BUA	P200.00	Edit
Chemistry	Laboratory hh	CHEM-004	BUN	P220.00	Edit
Chemistry	Laboratory hh	CHEM-005	CREATININE	P230.00	Edit
Chemistry	Laboratory hh	CHEM-006	CHOLESTERO	P210.00	Edit
Chemistry	Laboratory hh	CHEM-007	TRIGLYCER	P220.00	Edit
Chemistry	Laboratory hh	CHEM-008	LIPID PROF	P600.00	Edit
Chemistry	Laboratory hh	CHEM-009	SGOT	P230.00	Edit
Chemistry	Laboratory hh	CHEM-010	SGPT	P230.00	Edit

Showing 1 to 10 of 67 results | Per page: 10 | 1 2 3 4 5 6 7 8 9 10

Services: The master list of all diagnostic services offered, including the station, department, code, name, and price.

USERS

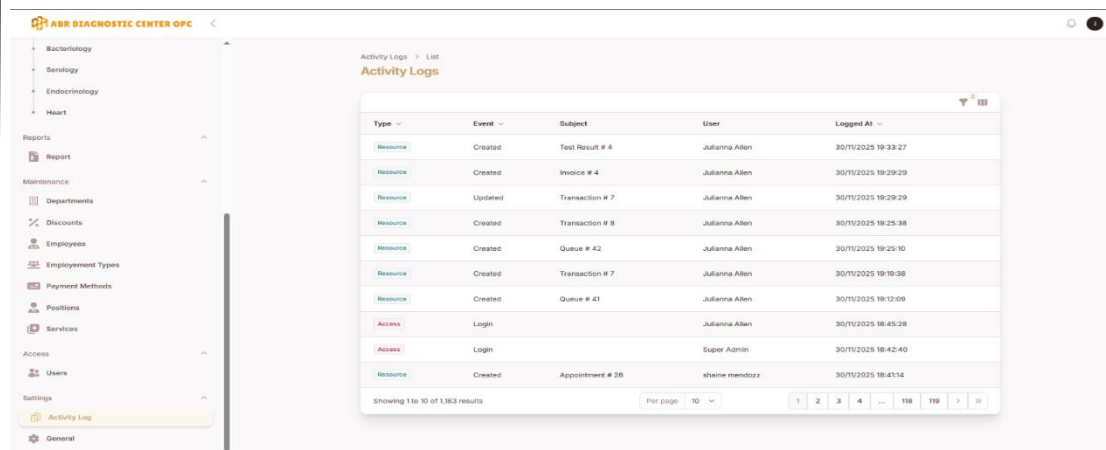


The screenshot shows the 'Users' list interface. On the left is a sidebar with navigation options: Reports, Maintenance, Access, and Settings. The 'Users' option is highlighted under the 'Access' section. The main content area displays a table of users with columns: Avatar, Username, Role, Email, and Verified at. The table lists 10 users with their respective details.

Avatar	Username	Role	Email	Verified at
	dianachrizel.a.bagamano52 Diana Chrizel Bagamano	Patient	dcanilicbagamano@gmail.com	Nov 30, 2025 12:15:36
	sherevin.mustampagos15 Sherevin Relampagos	Patient	sherevinrelampagos021@gmail.com	Nov 15, 2025 19:37:44
	marie.b.gonzales33 marie Gonzales	Admission Staff	gonzalesmarie@gmail.com	Nov 15, 2025 19:28:54
	ericka.m.dalgo54 Ericka Dalgo	Cashier	erickadalgo@gmail.com	Nov 15, 2025 19:24:01
	julianna.f.allen30 Julianna Allen	Admission Staff	llenjulianna@gmail.com	Nov 15, 2025 19:15:47
	bongbong.s.marco510 Bongbong marcos	Patient	pierealexander@gmail.com	Nov 15, 2025 11:41:06
	mariele.f.mansiang56 mariele mansiang	Patient	mariele@gmail.com	Nov 15, 2025 10:43:38
	shaine.f.mendozaz71 shaine mendozz	Patient	Mendozaz@gmail.com	Nov 15, 2025 08:11:56

Users: Lists all system users, their role (Patient, Admission Staff, Cashier, Chief Med tech, etc.), email, and verification status.

ACTIVITY LOG

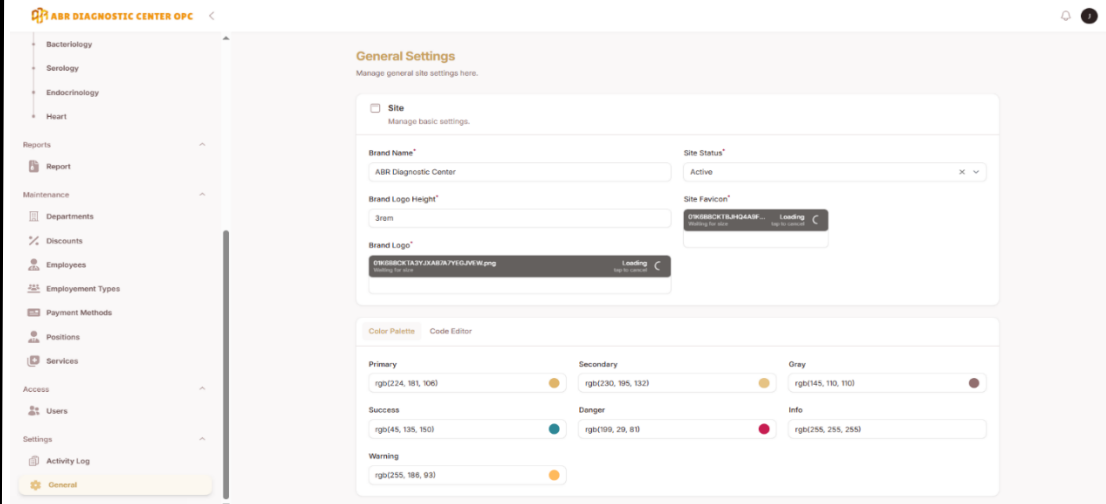


The screenshot shows the 'Activity Logs' interface. On the left is a sidebar with navigation options: Reports, Maintenance, Access, and Settings. The 'Activity Log' option is highlighted under the 'Settings' section. The main content area displays a table of activity logs with columns: Type, Event, Subject, User, and Logged At. The table lists 10 activity log entries.

Type	Event	Subject	User	Logged At
Resource	Created	Test Result # 4	Julianna Allen	30/11/2025 19:33:27
Resource	Created	Invoice # 4	Julianna Allen	30/11/2025 19:29:29
Resource	Updated	Transaction # 7	Julianna Allen	30/11/2025 19:29:29
Resource	Created	Transaction # 8	Julianna Allen	30/11/2025 19:25:36
Resource	Created	Queue # 42	Julianna Allen	30/11/2025 19:25:10
Resource	Created	Transaction # 7	Julianna Allen	30/11/2025 19:19:38
Resource	Created	Queue # 41	Julianna Allen	30/11/2025 19:12:09
Access	Login		Julianna Allen	30/11/2025 18:45:28
Access	Login		Super Admin	30/11/2025 18:42:40
Resource	Created	Appointment # 28	shaine mendozz	30/11/2025 18:41:14

Activity Log: Tracks all actions performed in the system, including the type of resource/access, the event (Created, Updated, Login), the subject the user, and the time the action was logged.

GENERAL SETTINGS



General: Manages basic site settings like the Brand Name, Site Status, Logo, and the primary/secondary color palette of the system.

CURRICULUM VITAE



Dennis Enraca

Full Stack Developer



About Me

I will serve as the Full Stack Developer responsible for designing and implementing both the front-end and back-end components of the queuing system. My role includes:

- Developing the Flutter-based queuing app for kiosks and customer-facing interfaces
- Creating and managing the Laravel back-end API for queue logic, user management, and reporting
- Integrating with the in-house accounting system via secure REST API endpoints
- Ensuring seamless data flow between the queuing system and the internal accounting APIs

Working closely with stakeholders to align system features with operational requirements
Performing testing, deployment, and post-launch support



Contact

+63 905 044 9294

dennisenraca25@gmail.com

Bocaue bulacan



Language

- English
- Filipino



Education



Approx. 2026

GLOBAL RECIPROCAL COLLEGE

Bachelor of Science Information Technology



Experience



(February 2025 – Present)

**SOFTWARE DEVELOPER – THERMOZONE
PHILIPPINES CORPORATION**

- Provides part-time web development support, handling selected internal projects and backend tasks
- Developed API endpoints for reporting features integrated with Flutter-based kiosks
- Built a lightweight landing page CMS for dynamic content management and internal use



(November 2023 – February 2025)

**SENIOR WEB DEVELOPER – MARVILL WEB
DEVELOPMENT**

- Led a team of junior developers across multiple client projects, ensuring quality and timely delivery. Providing mentorship to junior devs through peer programming, architecture planning, and debugging sessions
- Managed task delegation and project tracking using the ClickUp platform for agile workflows.



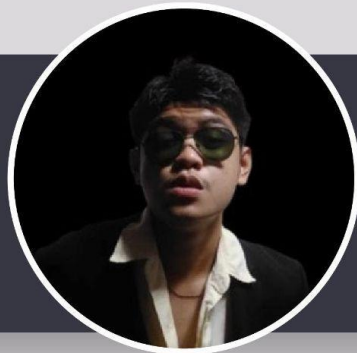
(March 2023 – November 2023)

WEB DEVELOPER – MARVILL WEB DEVELOPMENT



(Nov. 2021 – March 2023)

**JUNIOR WEB DEVELOPER – PPPR WEB
PROGRAMMING SERVICES**



Christian Francisco

UX/UI Designer

About Me

As a UI/UX Designer, I focus on building intuitive interfaces that prioritize user experience through thoughtful wireframing, prototyping, and usability testing. I have hands-on experience using design tools such as Figma, Adobe XD, and Sketch to bring concepts to life.

Contact

 +63 926 265 7265

 christianfrancisco040603@gmail.com

 6-2 J Sebastian St.
Balangkas, Valenzuela city.

Skills

- Web Design
- Graphic Design

Language

- English
- Filipino

Education

- **(2022-2026)**
GLOBAL RECIPROCAL COLLEGE
Bachelor of Science Information Technology
- **(2019-2021)**
OBANDO MONTESSORI INC.
Accounting Business and Management
- **(2015-2019)**
POLO NATIONAL HIGH SCHOOL
- **(2009-2015)**
PIO VALENZUELA ELEMENTARY SCHOOL

Experience

- **(2015 -2021)**
DAC COMPUTER RENTALS AND PRINTING SHOP.
Polo Valenzuela city.
 - Assisted customers with printing, scanning, and basic computer use
 - Managed computer rental time logs and billing
 - Maintained workstation cleanliness and functionality



Sherwin Relampagos

Front-end Developer

About Me

Creative front-end developer specializing in crafting pixel-perfect, accessible, and mobile-first web experiences. Skilled in React, Tailwind CSS, and modern UI workflows. I love solving design challenges and turning complex requirements into intuitive digital solutions.

Contact

 +63 994 334 8966

 sherwinrelampagos021@gmail.com

 Jasmin st. Balangkas
Valenzuela City.

Language

- English
- Filipino

Education

- **(2022-2026)**
GLOBAL RECIPROCAL COLLEGES, CALOOCAN
Bachelor of Science Information Technology
- **(2020-2022)**
POLO NATIONAL SENIOR HIGH SCHOOL
Science, Technology, Engineering, and
Mathematics (STEM)

Skills

- Front-End Development: Proficient in HTML, CSS, and basic JavaScript for building user interfaces (based on programming certifications).
- Programming: Intermediate knowledge of C# and Python, with experience in best coding practices.
- Network Security: Familiar with essentials like shielding connections and information security concepts.
- Documentation and Tools: Skilled in managing documentation and using productivity software for reports and logs.
- Technical Support: Hands-on experience with computer assembly, troubleshooting, and customer assistance.
- Soft Skills: Strong attention to detail, time management, and teamwork from retail and educational settings.



Dan Axel Torren

Data Analysis

About Me

Detail-oriented Data Analyst with strong skills in SQL, Excel, Python, and data visualization tools. Experienced in transforming complex datasets into clear insights that support strategic decision-making. Proven ability to improve reporting efficiency, automate workflows, and communicate findings to both technical and nontechnical teams.

Contact

 +63 939 175 0919

 danaxeltorren@gmail.com

 808 P. Lopez Cmpd. Karuhatan
Valenzuela City, 1441

Language

- English
- Filipino

Education

- **(July 2023 - Present)**
GLOBAL RECIPROCAL COLLEGES, CALOOCAN
Bachelor of Science Information Technology
- **(August 2022 - May 2023)**
STI COLLEGE, MEYCAUAYAN
Bachelor of Science Information Technology

Skills

- Programming & Development
- Languages: Python, Java, C#, C++
- Database Management: Basic SQL Database Design & Querying
- Software Development Lifecycle: Understanding of requirements analysis, design, coding, and testing Technical & Hard Skills
- Data Management & Analysis: Data Encoding & Entry, Data Interpretation, Data Validation
- Digital Design & Multimedia:
- Graphic Design (Canva)
- Photo Editing & Retouching
- Professional PowerPoint Presentation Design
- Basic Video Editing
- Software & Tools: Microsoft Office Suite, Basic Software Troubleshooting Projects & Practical Application Kitchen Supply Shop Inventory Management System
- Designed and developed a console-based application in Java/C# to automate inventory tracking for a small business.
- Implemented core features including product registration, stock level monitoring, sales recording, and low-stock alerts.
- Utilized SQL for backend data storage to persistently manage product information, supplier details, and transaction history. Gym Booking System
- Created a web-based application using Python to manage gym class schedules and member bookings.
- Developed functionality for users to view available time slots, reserve sessions, and cancel bookings.
- Designed an intuitive user interface and an SQL database to store member profiles, class details, and attendance records. Academic Presentation & Data Visualization
- Developed a comprehensive PowerPoint presentation on a core IT concept, incorporating custom graphics and clear data visualizations to effectively communicate complex information. Personalized Souvenir Design Project
- Applied photo editing and graphic design skills in Canva to create a series of personalized souvenirs, managing the project from concept to final, print-ready design.